

# Genetic response to supplements (999 supplements)

Genetic response predicted to 999 popular dietary supplements, by category  
Report generated in October 2 of 2026 · revised the 3 of October 2026 (aligned with the current protocol)

**Genetic basis:** DNA crude genotyping chip (GRCh38, 684 967 SNPs) + local statistical imputation (public reference panel of genomes complete to 30x; only genotypes with DR2 ≥ 0.8). Genetic data did not leave the local computer.

**Evidence base:** published gene-nutrient interactions (nutrigenomics), safety genes (liver, G6PD, HLA, TERT), plant pharmacogenetics (CYP3A4/P-gp, CYP2C9, CYP2D6, UGT1A1) with the same profile of the 632 drug report, LiverTox (NIH), NIH ODS and NCCIH chips, upper limits EFSA and CPIC guidelines used by analogy.

**List of supplements:** 999 products (substances, chemical forms, strains and combined formulas), constructed from NIH ODS and NCCIH chips and completed by curation of the usual commercial categories (vitamins and forms, minerals, amino acids, lipids, polyphenols, Western plants, Ayurvedic, Chinese/Kampo, Americas and Africas, mushrooms, probiotics, enzymes, nootropics, sport, hormones, longevity, joints/skin, metabolism) + current stack. Duplicates removed.

**Important:** This is a genetic reading of chip data, not a diagnosis or a prescription. It uses only genetics (without analysis). All indications are suggestion (not applied) and should be decided with the doctor. The evidence of nutrient is, in general, weaker than that of pharmacogenetics.

**red** avoid/pause **orange** adjust dose, shape or time **yellow** monitoring/confirming **green** normal (relevant gene evaluated) **gray** indeterminate **compact list** with no known genetic relevance ★ stack ➕ favourable genetic indication

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# 1. Executive summary

| Class                             | Number of supplements | %           |
|-----------------------------------|-----------------------|-------------|
| <b>Avoid / pause</b>              | 53                    | 5.3%        |
| <b>Adjust dose/form/time</b>      | 66                    | 6.6%        |
| <b>Monitoring / confirm</b>       | 111                   | 11.1%       |
| <b>Undetermined</b>               | 11                    | 1.1%        |
| <b>Normal (gene evaluated)</b>    | 221                   | 22.1%       |
| <b>No known genetic relevance</b> | 537                   | 53.8%       |
| <b>Total analyzed</b>             | <b>999</b>            | <b>100%</b> |

**Fast reading:** 230 Supplements require attention (avoid, adjust or monitor) and 67 have a favourable genetic indication (⊕). Almost everything is explained by eight axes: (1) genetic liver (PNPLA3 I148M + MBOAT7 + Pi\*MZ probable) → avoid hepatotoxic plants (green tea in extract, kava, ashwagandha, red rice yeast, comfrey, He Shou Wu... (Complete abstract click electronic access below) (2) UGT1A1 \*1/\*28 (Gilbert) + inhibition of CYP3A4/P-gp → curcumin, quercetin, luteolin and apigenin (pause while ALT is high, current protocol); (3) COMT Val/Val + MTHFR compound (homocysteine p92 score) → high dose catechols, SAMe/methionine, L-dopa; prefer 5-MTHF, TMG, glycine and creatine; (4) DHFR CC → folic acid substituted by 5-MTHF; (5) HFE H63D → without iron by routine and vitamin C away from meals; (6) normal to intermediate CYP1A2 + MTNR1B G → caffeine ≤ 200 mg in the morning and melatonin 0.3–1 mg; (7) TERT risk + nevi p88 → avoid telomerase activators; (8) CYP3A5 expresser + sirolimus → nothing that induces or inhibits CYP3A4 in the window of rapamycin (St John's wort, schisandra, berberine, piperine...). Current stack (44 items): 4 to pause, 11 to adjust, normal 22, indeterminate 1 and 6 without specific gene — all consistent with the genetic plan and its coordination note. Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.

## Avoid / pause (53)

| Supplement                             | Category           | Genes (state)                                                                   | Predicted response      | What to do (summer)                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|--------------------|---------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vitamin A (retinyl acetate)            | Vitamins and forms | <b>BCO1:</b> slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable | avoid preformed retinol | The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied). |
| vitamin A (retinol, retinyl palmitate) | Vitamins and forms | <b>BCO1:</b> slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable | avoid preformed retinol | The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied). |
| vitamin A + D (oil)                    | Vitamins and forms | <b>BCO1:</b> slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable | avoid preformed retinol | The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied). |

| Supplement                           | Category                             | Genes (state)                                                                                                                                          | Predicted response         | What to do (summer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cod liver oil                        | Fatty acids                          | <b>BCO1</b> : slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                      | avoid preformed retinol    | The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).                                                                                                                                                                                                                                                                                                        |
| EGCG / green tea extract             | Polyphenols and other phytochemicals | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>COMT</b> : Val/Val (high)<br><b>HLA-B*35:01</b> : probable negative<br><b>UGT1A1</b> : IM (*1/*28)    | avoid (liver)              | Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                   |
| ★ apigenin                           | Polyphenols and other phytochemicals | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP3A5:expressor (*1/*3)</b><br><b>CYP1A2</b> : normal to intermediate<br><b>CYP2C9</b> : IM (*1/*3)                 | avoid / pause (test-pause) | Current stack (100 mg) — plan S28: PAUSE (test-pause). Test-pause (current protocol). Test pause for the same reasons of luteolin: inhibits UGT1A1 (*1/*28: up to bilirubin and confuses liver reading) and CYP3A4/P-gp (sirolimus; CYP3A5 expressor). It is not catechol, but also inhibits CYP1A2 (already normal to intermediate: *1F het + rs2472297 C/C), which rises melatonin and caffeine, and CYP2C9 (*1/*3). Pause; pause while ALT is high (current protocol). Coordination note: in the current protocol, the pause lasts while ALT is high; if it returns, 25-50 mg at night, outside the rapamycin window (suggestion, not applied). |
| curcumin (95% extract)               | Polyphenols and other phytochemicals | <b>UGT1A1</b> : IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>HLA-B*35:01</b> : probable negative<br><b>CYP3A5:expressor (*1/*3)</b> | avoid/pause                | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                            |
| curcumin + piperine                  | Polyphenols and other phytochemicals | <b>UGT1A1</b> : IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>HLA-B*35:01</b> : probable negative<br><b>CYP3A5:expressor (*1/*3)</b> | avoid/pause                | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                            |
| colloidal curcumin (nanoparticles)   | Polyphenols and other phytochemicals | <b>UGT1A1</b> : IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>HLA-B*35:01</b> : probable negative<br><b>CYP3A5:expressor (*1/*3)</b> | avoid/pause                | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                            |
| curcumin with curcuman essential oil | Polyphenols and other phytochemicals | <b>UGT1A1</b> : IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>HLA-B*35:01</b> : probable negative<br><b>CYP3A5:expressor (*1/*3)</b> | avoid/pause                | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                            |

| Supplement                           | Category                             | Genes (state)                                                                                                                                        | Predicted response | What to do (summer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| curcumin phytosome                   | Polyphenols and other phytochemicals | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative<br><b>CYP3A5:</b> expressor (*1/*3) | avoid/pause        | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                     |
| ★ solid lipid curcumin (SLCP)        | Polyphenols and other phytochemicals | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative<br><b>CYP3A5:</b> expressor (*1/*3) | avoid/pause        | Current stack (1 g) — plan S26: PAUSE. Pause (current protocol). UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                                                                                                                    |
| isoquercytrin enzymatically modified | Polyphenols and other phytochemicals | <b>UGT1A1:</b> IM (*1/*28)<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>CYP3A5:</b> expressor (*1/*3)                          | avoid/pause        | UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.                                                                                                                                                                                                                                           |
| ★ luteolin                           | Polyphenols and other phytochemicals | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>UGT1A1:</b> IM (*1/*28)<br><b>CYP2C9:</b> IM (*1/*3)                                 | avoid/pause        | Current stack (100 mg) — plan S27: PAUSE. Pause (current protocol). Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| luteolin + PEA                       | Polyphenols and other phytochemicals | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>UGT1A1:</b> IM (*1/*28)<br><b>CYP2C9:</b> IM (*1/*3)                                 | avoid/pause        | Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ★ quercetin                          | Polyphenols and other phytochemicals | <b>UGT1A1:</b> IM (*1/*28)<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>CYP3A5:</b> expressor (*1/*3)                          | avoid/pause        | Current stack (250 mg (3×/month)) — plan S29: PAUSE (suggestion, not applied). It is suggested to pause as an isolated supplement (genetic verdict); in the current protocol enters the monthly pulse with fisetin, only with normal ALT. UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. |
| quercetin phytosome                  | Polyphenols and other phytochemicals | <b>UGT1A1:</b> IM (*1/*28)<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>CYP3A5:</b> expressor (*1/*3)                          | avoid/pause        | UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.                                                                                                                                                                                                                                           |

| Supplement                               | Category                             | Genes (state)                                                                                                               | Predicted response                 | What to do (summer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| resveratrol + quercetin (combined)       | Polyphenols and other phytochemicals | <b>UGT1A1:</b> IM (*1/*28)<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>CYP3A5:</b> expressor (*1/*3) | avoid/pause                        | UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. |
| greater celandine                        | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                     |
| comfrey (conferred)                      | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (pyrrolizidine alkaloids)    | Pyrrolizidine alkaloids cause veno-occlusive liver disease; oral use is prohibited in several countries. Genetic liver reinforces.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| germander (Teucrium)                     | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                     |
| St. John's wort                          | Western medicinal plants             | <b>CYP3A4:</b> normal<br><b>CYP3A5:</b> expressor (*1/*3)<br><b>CYP2C19:</b> NM<br><b>TERT/melanoma:</b> unfavourable       | avoid (strong inducer)             | Strong inducer of CYP3A4, P-gp and CYP2C19: low cyclosporine/tacrolimus 30–60% → predictable descent of sirolimus, which has already reduced exposure to CYP3A5 expresser; also low tadalafil. Photosensitizing hypericin (p88 + isotretinoin) (suggestion, not applied).                                                                                                                                                                                                                                                                                         |
| pore (oil)                               | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                     |
| tussiling                                | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (pyrrolizidine alkaloids)    | Pyrrolizidine alkaloids cause veno-occlusive liver disease; oral use is prohibited in several countries. Genetic liver reinforces.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ashwagandha (full spectrum root extract) | Ayurvedic plants                     | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).                                                                                                                                                                                                                                                                                 |
| ashwagandha (root and leaf extract)      | Ayurvedic plants                     | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).                                                                                                                                                                                                                                                                                 |
| ashwagandha (root)                       | Ayurvedic plants                     | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).                                                                                                                                                                                                                                                                                 |
| babchi (Psoralea corylyfolia)            | Ayurvedic plants                     | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>TERT/melanoma:</b> unfavourable                                             | avoid (liver + photosensitization) | Babchi: photosensitizing psoralens (p88 + isotretinoin) and hepatitis (LiverTox) cases.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| brahmi + ashwagdha (combined)            | Ayurvedic plants                     | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025).                                                                                                                                                                                                                                                                                 |

| Supplement                                            | Category                                      | Genes (state)                                                                                              | Predicted response           | What to do (summer)                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| giloy / guduchi (Tinospora cordifolia)                | Ayurvedic plants                              | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                  | avoid (liver)                | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                                                             |
| He Shou Wu (Polygonum multiflorum, fo-ti)             | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative                         | avoid (liver)                | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Polygonum multiflorum: HLA-B*35:01-linked lesion (probable negative), but intrinsic hepatotoxicity remains. |
| Jin Bu Huan (tetra-hydropalmatin)                     | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                  | avoid (liver)                | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                                                             |
| astragalosid IV                                       | Traditional Chinese medicine plants and Kampo | <b>TERT/melanoma:</b> unfavourable                                                                         | avoid (telomerase activator) | Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma has low polygenic risk: consensus p18), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).              |
| astralis (root)                                       | Traditional Chinese medicine plants and Kampo | <b>TERT/melanoma:</b> unfavourable                                                                         | avoid (telomerase activator) | Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma has low polygenic risk: consensus p18), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).              |
| cycloastragenol                                       | Traditional Chinese medicine plants and Kampo | <b>TERT/melanoma:</b> unfavourable                                                                         | avoid (telomerase activator) | Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma has low polygenic risk: consensus p18), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).              |
| purified extract of astrogalum (telomerase activator) | Traditional Chinese medicine plants and Kampo | <b>TERT/melanoma:</b> unfavourable                                                                         | avoid (telomerase activator) | Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma has low polygenic risk: consensus p18), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).              |
| ma huang (ephedra)                                    | Traditional Chinese medicine plants and Kampo | <b>CETP/9p21/APOB:</b> genetic vascular risk<br><b>NOS3:</b> -786CT + 298GT                                | avoid                        | Prohibited sympathomimetics (UE/EUA): Stroke/farce; 9p21 risk homozygous and NOS3 reinforce.                                                                                                                                                                                                                                              |
| tripterygium (thunder Godvine)                        | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                  | avoid                        | Tripterygium: hepatic, renal and reproductive toxicity; immunosuppressant.                                                                                                                                                                                                                                                                |
| pharmacaria (Larrea)                                  | Plants of the Americas and Africa; superfood  | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                  | avoid (liver)                | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                                                             |
| kava (Piper methylicum)                               | Plants of the Americas and Africa; superfood  | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>CYP2D6:</b> Probable NM (without CNV)                      | avoid (liver)                | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Kava: severe hepatitis and transplants; removal in several countries.                                       |
| kratom (Mitragyna speciosa)                           | Plants of the Americas and Africa; superfood  | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>CYP2D6:</b> Probable NM (without CNV)<br><b>OPRM1:</b> A/G | avoid                        | Atypical opioid (dependence), cholestatic liver injury (LiverTox); CYP3A4/2D6 (CNV not evaluated) and OPRM1 A/G.                                                                                                                                                                                                                          |

| Supplement                                 | Category              | Genes (state)                                                                                                                                           | Predicted response          | What to do (summer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DMHA / DMAA (prohibited stimuli)           | Sport and performance | <b>CETP/9p21/APOB</b> : genetic vascular risk<br><b>NOS3</b> : –786CT + 298GT                                                                           | avoid                       | Prohibited sympathomimetics (UE/EUA): Stroke/farce; 9p21 risk homozygous and NOS3 reinforce.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| prohormones/SARMs (sold as supplement)     | Sport and performance | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                                                              | avoid                       | Synthesis steroids and SARMs: severe cholestatic lesion (LiverTox) and hormonal axis; genetic liver reinforces. They are not legal supplements in UE.                                                                                                                                                                                                                                                                                                                                                                                                             |
| desiccated liver (capsules)                | Hormones              | <b>HFE</b> : H63D het<br><b>BCO1</b> : slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                              | avoid (retinol + iron)      | Desiccated liver concentrates retinol (isotretinoin) and iron heme (HFE H63D).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| organ glandular (mixture)                  | Hormones              | <b>HFE</b> : H63D het<br><b>BCO1</b> : slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                              | avoid (retinol + iron)      | Desiccated liver concentrates retinol (isotretinoin) and iron heme (HFE H63D).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| apigenin + magnesium (sleep)               | Longevity             | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP3A5</b> :expressor (*1/*3)<br><b>TRPM6</b> : typical                                                               | avoid / pause (by apigenin) | Magnesium does not raise a problem (typical TRPM6), but for apigenin a test break (UGT1A1 *28) is suggested. CYP3A4/P-gp with sirolimus: pause while ALT is high (current protocol). Prefer magnesium without apigenin (suggestion, not applied).                                                                                                                                                                                                                                                                                                                 |
| senolytic combined (fisetin + quercetin)   | Longevity             | <b>UGT1A1</b> : IM (*1/*28)<br><b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC<br><b>CYP3A5</b> :expressor (*1/*3)                          | avoid/pause                 | UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. |
| turmeric + boswellia (articular)           | Articulations         | <b>UGT1A1</b> : IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>HLA-B*35:01</b> : probable negative<br><b>CYP3A5</b> :expressor (*1/*3) | avoid/pause                 | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                           |
| caffeine + green tea + garcinia (combined) | Metabolism            | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>COMT</b> : Val/Val (high)<br><b>HLA-B*35:01</b> : probable negative<br><b>UGT1A1</b> : IM (*1/*28)     | avoid (liver)               | Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).                                                                                                                                                                                                                                                                  |
| thermogenic green tea (extract)            | Metabolism            | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>COMT</b> : Val/Val (high)<br><b>HLA-B*35:01</b> : probable negative<br><b>UGT1A1</b> : IM (*1/*28)     | avoid (liver)               | Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).                                                                                                                                                                                                                                                                  |
| red rice yeast (monacoline K)              | Metabolism            | <b>SLCO1B1</b> : increased function<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>CYP3A5</b> :expressor (*1/*3)                                   | avoid (unregulated statin)  | Monacholine K = lovastatin with non-standardized dose and possible citrinine. SLCO1B1 *1/*14 is favourable, but it already takes pitavastatin (statin duplication) and there are genetic livers and cases of hepatitis (suggestion, not applied).                                                                                                                                                                                                                                                                                                                 |

| Supplement                   | Category   | Genes (state)                             | Predicted response | What to do (summer)                                                                                                                                                                                                           |
|------------------------------|------------|-------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| multi-ingredient thermogenic | Metabolism | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable | avoid (liver)      | Multi-ingredient thermogenic formulas were associated with acute hepatitis (LiverTox); genetic liver.                                                                                                                         |
| Usnic acid (fat burner)      | Metabolism | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable | avoid (liver)      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). |

## Adjust dose, shape or time (66)

| Supplement                             | Category           | Genes (state)                                                                                               | Predicted response                              | What to do (summer)                                                                                                                                                                                                                                                                                            |
|----------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| acerola (natural vitamin C)            | Vitamins and forms | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical                           | drinking between meals (hour)                   | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                            |
| alpha-GPC (glycerophosphocholine)      | Vitamins and forms | <b>PEMT:</b> V175M het<br><b>FM03:</b> typical                                                              | prefer citicoline or phosphatidylcholine (form) | PEMT V175M het: a slightly smaller endogenous choline synthesis (rational to choline). Typical FM03: bitartrate and alpha-GPC generate more TMA/TMAO than phosphatidylcholine/citicoline; with genetic vascular priority (9p21, CETP), preferring citicoline 250–500 mg or PC 1–2 g (suggestion, not applied). |
| calcium ascorbate (buffered vitamin C) | Vitamins and forms | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical                           | drinking between meals (hour)                   | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                            |
| sodium ascorbate                       | Vitamins and forms | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical                           | drinking between meals (hour)                   | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                            |
| choline (bitartrate)                   | Vitamins and forms | <b>PEMT:</b> V175M het<br><b>FM03:</b> typical                                                              | prefer citicoline or phosphatidylcholine (form) | PEMT V175M het: a slightly smaller endogenous choline synthesis (rational to choline). Typical FM03: bitartrate and alpha-GPC generate more TMA/TMAO than phosphatidylcholine/citicoline; with genetic vascular priority (9p21, CETP), preferring citicoline 250–500 mg or PC 1–2 g (suggestion, not applied). |
| B complex                              | Vitamins and forms | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high<br><b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC) | choose formula (low B6, without folic acid)     | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                           |
| B complex in gums                      | Vitamins and forms | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high<br><b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC) | choose formula (low B6, without folic acid)     | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                           |
| Methylated B complex                   | Vitamins and forms | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high<br><b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC) | choose formula (low B6, without folic acid)     | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                           |
| iron + vitamin C (combined)            | Vitamins and forms | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                     | only with confirmed deficit                     | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                            |

| Supplement                    | Category           | Genes (state)                                                                                                                                                             | Predicted response            | What to do (summer)                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| multivitamin 50+              | Vitamins and forms | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)<br><b>HFE:</b> H63D het<br><b>BCO1:</b> slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6):</b> typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  |
| multivitamin in gums          | Vitamins and forms | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)<br><b>HFE:</b> H63D het<br><b>BCO1:</b> slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6):</b> typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  |
| female multivitamin           | Vitamins and forms | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)<br><b>HFE:</b> H63D het<br><b>BCO1:</b> slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6):</b> typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  |
| General multivitamin          | Vitamins and forms | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)<br><b>HFE:</b> H63D het<br><b>BCO1:</b> slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6):</b> typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  |
| Multivitaminic male           | Vitamins and forms | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)<br><b>HFE:</b> H63D het<br><b>BCO1:</b> slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6):</b> typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  |
| multivitamin prenatal         | Vitamins and forms | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)<br><b>HFE:</b> H63D het<br><b>BCO1:</b> slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6):</b> typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  |
| ascorbyl palmitate            | Vitamins and forms | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical                                                                                         | drinking between meals (hour) | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                                                                                                              |
| ★ pyridoxal-5-phosphate (P5P) | Vitamins and forms | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high                                                                                                                                 | lowering dose (≤ 5 mg/day)    | Current stack (15 mg) — plan S6: ADJUST (descend). ≤ 5 mg/day or stop (above the upper limit of ingestion, without genetic indication). PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. ≤ 5 mg/day or pause (suggestion, not applied). |

| Supplement                                         | Category                    | Genes (state)                                                                                               | Predicted response                          | What to do (summer)                                                                                                                                                                                                                                                                 |
|----------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vitamin B6 (pyridoxine hydrochloride)              | Vitamins and forms          | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high                                                                   | lowering dose ( $\leq 5$ mg/day)            | PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. $\leq 5$ mg/day or pause (suggestion, not applied).                            |
| vitamin C (ascorbic acid)                          | Vitamins and forms          | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical                           | drinking between meals (hour)               | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied). |
| vitamin C + zinc                                   | Vitamins and forms          | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical                           | drinking between meals (hour)               | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied). |
| vitamin C with metabolites (ascorbate + threonate) | Vitamins and forms          | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical                           | drinking between meals (hour)               | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied). |
| ★ vitamin C liposomal                              | Vitamins and forms          | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical                           | <b>drinking between meals (hour)</b>        | Current stack (500 mg) — plan S16: ADJUST (hour). Always $\geq 2$ h away from meals. HFE H63D het: vitamin C with food increases the absorption of non-heme iron. 500 mg always $\geq 2$ h away from meals. Normal G6PD; Typical SLC23A1/2 (suggestion, not applied).               |
| folic acid (synthetic)                             | Vitamins and forms          | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)                                              | prefer 5-MTHF (form)                        | MTHFR 677CT + 1298AC (activity $\sim 50\text{--}60\%$ ) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).                           |
| folic acid + B12 (combined)                        | Vitamins and forms          | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)                                              | prefer 5-MTHF (form)                        | MTHFR 677CT + 1298AC (activity $\sim 50\text{--}60\%$ ) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).                           |
| ZMA (zinc, magnesium, B6)                          | Minerals and trace elements | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high<br><b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC) | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                |
| iron bisglycinate                                  | Minerals and trace elements | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                 |
| iron-polysaccharide complex                        | Minerals and trace elements | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                 |
| iron carbonyl                                      | Minerals and trace elements | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                 |
| iron heme (polypeptide)                            | Minerals and trace elements | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                 |

| Supplement                                    | Category                             | Genes (state)                                                                                                             | Predicted response          | What to do (summer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| liposomal / sucrosomal iron                   | Minerals and trace elements          | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                                   | only with confirmed deficit | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                     |
| ferrous fumarate                              | Minerals and trace elements          | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                                   | only with confirmed deficit | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                     |
| ferrous gluconate                             | Minerals and trace elements          | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                                   | only with confirmed deficit | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                     |
| ★ trace minerals (multineural) ⊕              | Minerals and trace elements          | <b>SLC30A8:</b> C/C (usual zinc)<br><b>GPX1:</b> Pro198Leu het<br><b>SLC39A8:</b> normal<br><b>AOC1 (DAO):</b> slightly ↓ | up zinc and copper          | Current stack (Zn 11, Cu 0.9, I 0.05, Se 0.055, Mn 2.3, Cr 0.035, Mo 0.045, B 1 mg) — plan S12: ADJUST (Zn and Cu). Total zinc 20–25 mg + total copper 1.5–2 mg; remaining equal. SLC30A8 C/C + rs11558471 A/A (genetically useful zinc); GPX1 Pro198Leu (selenium already suitable, without Brazil nuts); SLC39A8 C/C (manganese without adjustment); DAO is a copper enzyme (AOC1 het). Total zinc 20–25 mg + total copper 1.5–2 mg, ≥ 2 h away from NAC, Ca-AMG and fiber (suggestion, not applied). |
| ferrous sulfate                               | Minerals and trace elements          | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect                                                                   | only with confirmed deficit | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                     |
| ★ TMG (trimethylglycine, betaine anhydrous) ⊕ | Amino acids                          | <b>BHMT:</b> R239Q het<br><b>GNMT:</b> TT<br><b>MTHFR:</b> 677CT/1298AC                                                   | adjust dose (plan)          | Current stack (1500 mg) — plan S7: ADJUST (subir). Up to 3 g/day in 2 tomas; do not exceed 3 g (LDL). BHMT R239Q het, GNMT TT and MTHFR compound: the BHMT pathway compensates for folate. 3 g/day (≥ 4 g up to LDL) (suggestion, not applied).                                                                                                                                                                                                                                                         |
| ★ glycine ⊕                                   | Amino acids                          | <b>CPS1:</b> low glycine<br><b>GSTM1:</b> probable null<br><b>GNMT:</b> TT                                                | adjust dose (plan)          | Current stack (3 g) — plan S8: ADJUST (subir). Up to 5 g/day (3 g at bedtime + 2 g in the morning). CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).                                                                                                                                                                                                                    |
| ★ fish oil (EPA+DHA, TG/rTG) ⊕                | Fatty acids                          | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3                                                                   | adjust dose (plan)          | Current stack (EPA 1125 + DHA 875 mg) — plan S1: ADJUST. Up to 3.2–4 g/day of EPA+DHA (Coordination Note), form rTG, with the meal fatter and far from psyllium. FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).                                                               |
| tea catechins (drink/matcha)                  | Polyphenols and other phytochemicals | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C<br><b>COMT:</b> Val/Val (high)                               | count caffeine              | Tea/matcha in drink does not have the liver risk of the extract; counts for the caffeine ceiling (≤ 200 mg/day, only in the morning; Normal to intermediate CYP1A2).                                                                                                                                                                                                                                                                                                                                    |

| Supplement                               | Category                                     | Genes (state)                                                                                 | Predicted response          | What to do (summer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ fisetin                                | Polyphenols and other phytochemicals         | COMT: Val/Val (high)<br>UGT1A1: IM (*1/*28)<br>CYP3A5:expressor (*1/*3)                       | isolated monthly pulse      | Current stack (2.5 g (3×/month)) — plan S29: ADJUST. Monthly pulse (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI; current protocol: with quercetin, only with normal ALT. Catechol (COMT Val/Val + MTHFR) and inhibitor of CYP3A4/P-gp and UGT1A1. 9p21 homozygous (CDKN2A/B region, senescence) gives senolytic rationale. Genetic reading: 1 pulse/month (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. |
| ★ lycopene                               | Carotenoids and pigments                     | BCO1: slightly ↓ conversion                                                                   | lower dose (≤ 10 mg)        | Current stack (35 mg) — plan S23: ADJUST (descend). 10 mg or stop. Genetic levels of lycopene tendentially high: no genetic indication for 35 mg. 10 mg or tomato cooked with olive oil (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| green coffee (extract)                   | Plants of the Americas and Africa; superfood | CYP1A2: normal to intermediate<br>ADORA2A: C/C<br>COMT: Val/Val (high)<br>MTHFR: 677CT/1298AC | caffeine + chlorogenic acid | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). The concentrated chlorogenic acid can rise to homocysteine (COMT/MTHFR).                                                                                                                                                                                                                                                                        |
| yerba mate                               | Plants of the Americas and Africa; superfood | CYP1A2: normal to intermediate<br>ADORA2A: C/C                                                | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                 |
| guarana                                  | Plants of the Americas and Africa; superfood | CYP1A2: normal to intermediate<br>ADORA2A: C/C                                                | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                 |
| guayusa                                  | Plants of the Americas and Africa; superfood | CYP1A2: normal to intermediate<br>ADORA2A: C/C                                                | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                 |
| walnut                                   | Plants of the Americas and Africa; superfood | CYP1A2: normal to intermediate<br>ADORA2A: C/C                                                | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                 |
| yaupon                                   | Plants of the Americas and Africa; superfood | CYP1A2: normal to intermediate<br>ADORA2A: C/C                                                | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                 |
| ★ PHGG (partially hydrolyzed guar gum) ☺ | Probiotics                                   | TCF7L2/FTO/GCKR: unfavourable<br>CETP/9p21/APOB: genetic vascular risk<br>FUT2: secretor      | adjust dose (plan)          | Current stack (6 g) — plan S35: ADJUST (subir). 10 g/day. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ★ psyllium (ispaghula bark) ☺            | Probiotics                                   | TCF7L2/FTO/GCKR: unfavourable<br>CETP/9p21/APOB: genetic vascular risk<br>FUT2: secretor      | adjust dose (plan)          | Current stack (10 g) — plan S35: ADJUST (subir). 15–20 g/day in 2–3 takes with water; drugs ≥ 1 h before or 2–4 h after. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Supplement                        | Category   | Genes (state)                                                                                               | Predicted response                          | What to do (summer)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| caffeine + L-theanine             | Nootropics | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                             |
| anhydrous caffeine                | Nootropics | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                             |
| prolonged release caffeine        | Nootropics | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                             |
| magnesium + B6 (cognition/stress) | Nootropics | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high<br><b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC) | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                                                                                                                                                                                                                       |
| melatonin (immediate release)     | Nootropics | <b>MTNR1B:</b> C/G (carrier)<br><b>CYP1A2:</b> normal to intermediate                                       | low dose (0.3–1 mg)                         | MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).                                                                                                                                                                                          |
| prolonged release melatonin       | Nootropics | <b>MTNR1B:</b> C/G (carrier)<br><b>CYP1A2:</b> normal to intermediate                                       | low dose (0.3–1 mg)                         | MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).                                                                                                                                                                                          |
| ★ melatonin high dose (≥ 10 mg)   | Nootropics | <b>MTNR1B:</b> C/G (carrier)<br><b>CYP1A2:</b> normal to intermediate                                       | low (0.3–1 mg)                              | Current stack (up to 100 mg) — plan S30: ADJUST (desert much). 0.3–1 mg, 30–60 min before bedtime. MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied). Doses ≥ 10 mg (up to 100 mg) have no genetic basis and prolong the effect on insulin. |
| melatonin in spray/sublingual     | Nootropics | <b>MTNR1B:</b> C/G (carrier)<br><b>CYP1A2:</b> normal to intermediate                                       | low dose (0.3–1 mg)                         | MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).                                                                                                                                                                                          |
| paraxanthine                      | Nootropics | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                             |
| tryptophan + B6 (combined)        | Nootropics | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high<br><b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC) | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                                                                                                                                                                                                                       |

| Supplement                                    | Category              | Genes (state)                                                                                                  | Predicted response                          | What to do (summer)                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| electrolytes with caffeine                    | Sport and performance | <b>CYP1A2</b> : normal to intermediate<br><b>ADORA2A</b> : C/C                                                 | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). |
| energy gel with caffeine                      | Sport and performance | <b>CYP1A2</b> : normal to intermediate<br><b>ADORA2A</b> : C/C                                                 | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). |
| pre-training with caffeine                    | Sport and performance | <b>CYP1A2</b> : normal to intermediate<br><b>ADORA2A</b> : C/C                                                 | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). |
| tauro-caffeine (energy drink powder)          | Sport and performance | <b>CYP1A2</b> : normal to intermediate<br><b>ADORA2A</b> : C/C                                                 | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). |
| energy formula (B + caffeine)                 | Others                | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC) | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                                           |
| homocysteine formula with folic acid          | Others                | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                                               | prefer 5-MTHF (form)                        | MTHFR 677CT + 1298AC (activity ~50–60%) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).                                                                                 |
| methylation formula (B12 + folate + B6 + TMG) | Others                | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC) | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                                           |
| lysine + vitamin C (Pauling protocol)         | Others                | <b>HFE</b> : H63D het<br><b>G6PD</b> : normal<br><b>SLC23A1/2 (vit. C)</b> : typical                           | drinking between meals (hour)               | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                                            |

## Monitor / confirm before using (111)

| Supplement            | Category                    | Genes (state)                                                   | Predicted response                         | What to do (summer)                                                                                                                                                                                                     |
|-----------------------|-----------------------------|-----------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula AREDS2 (eyes) | Vitamins and forms          | <b>CFH/ARMS2</b> : CFH het<br><b>SLC30A8</b> : C/C (usual zinc) | without indication; zinc × CFH controversy | CFH Y402H het and ARMS2 G/G.                                                                                                                                                                                            |
| potassium bicarbonate | Minerals and trace elements | <b>Sal (AGT/ADD1/GNB3/CYP11B2)</b> : sensitive (modest)         | potassium only for food                    | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. |
| potassium citrate     | Minerals and trace elements | <b>Sal (AGT/ADD1/GNB3/CYP11B2)</b> : sensitive (modest)         | potassium only for food                    | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. |

| Supplement                                     | Category                    | Genes (state)                                                                     | Predicted response                            | What to do (summer)                                                                                                                                                                                                     |
|------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| potassium chloride                             | Minerals and trace elements | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)                            | potassium only for food                       | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. |
| salt tablets (sodium chloride)                 | Minerals and trace elements | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)                            | avoid routine use                             | Genetic sensitivity to salt: salt tablets only for intense and prolonged sweat.                                                                                                                                         |
| potassium gluconate                            | Minerals and trace elements | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)                            | potassium only for food                       | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. |
| salt with potassium chloride (salt substitute) | Minerals and trace elements | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)                            | potassium only for food                       | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. |
| BCAA (leucine, isoleucine, valine)             | Amino acids                 | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                              | the genetic plan advises to add               | High BCAA are associated with insulin resistance;                                                                                                                                                                       |
| L-isoleucine                                   | Amino acids                 | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                              | the genetic plan advises to add               | High BCAA are associated with insulin resistance;                                                                                                                                                                       |
| L-leucine                                      | Amino acids                 | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                              | the genetic plan advises to add               | High BCAA are associated with insulin resistance;                                                                                                                                                                       |
| L-methionine                                   | Amino acids                 | <b>COMT:</b> Val/Val (high)<br><b>GNMT:</b> TT<br><b>MTHFR:</b> 677CT/1298AC      | avoiding isolated supplement                  | Extra methionine turns SAM and then homocysteine;                                                                                                                                                                       |
| L-valine                                       | Amino acids                 | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                              | the genetic plan advises to add               | High BCAA are associated with insulin resistance;                                                                                                                                                                       |
| SAMe (S-adenosylmethionine)                    | Amino acids                 | <b>COMT:</b> Val/Val (high)<br><b>GNMT:</b> TT<br><b>MTHFR:</b> 677CT/1298AC      | the genetic plan advises to add               | COMT Val/Val + GNMT TT: plus SAM tends to be transformed into homocysteine (genetic score p92).                                                                                                                         |
| CLA (conjugated linoleic acid)                 | Fatty acids                 | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>TCF7L2/FTO/GCKR:</b> unfavourable | without benefit; possible metabolic worsening | CLA t10,c12 increased insulin resistance and hepatic fat in trials.                                                                                                                                                     |
| Sewer oil (GLA)                                | Fatty acids                 | <b>FADS1/FADS2:</b> high activity<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable    | GLA→AA + alkaloids                            | High FADS1 (GLA→AA) and possible presence of hepatotoxic pyrrolizidine alkaloids (genetic liver): only certified products without PA.                                                                                   |
| coconut oil                                    | Fatty acids                 | <b>CETP/9p21/APOB:</b> genetic vascular risk                                      | saturated fat (LDL)                           | Genetic priority to low LDL/ApoB (high CETP, 9p21, APOB): coconut oil rises to LDL;                                                                                                                                     |
| safflower oil                                  | Fatty acids                 | <b>FADS1/FADS2:</b> high activity                                                 | Omega-6 unnecessary                           | The extra LA is converted effectively to AA (high FADS);                                                                                                                                                                |
| shark liver oil (alkylglycerols)               | Fatty acids                 | <b>BCO1:</b> slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable   | confirm retinol content                       | It can contain vitamin A preformed in variable quantity;                                                                                                                                                                |
| Grapeseed oil                                  | Fatty acids                 | <b>FADS1/FADS2:</b> high activity                                                 | Omega-6 unnecessary                           | The extra LA is converted effectively to AA (high FADS);                                                                                                                                                                |
| Blackcurrant oil (GLA)                         | Fatty acids                 | <b>FADS1/FADS2:</b> high activity                                                 | GLA can turn AA                               | With high activity FADS1, part of GLA→DGLA goes to arachidonic acid (pro-inflammatory), reducing the benefit.                                                                                                           |

| Supplement                          | Category                             | Genes (state)                                                                                                       | Predicted response              | What to do (summer)                                                                                                                                        |
|-------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primule oil (GLA)                   | Fatty acids                          | <b>FADS1/FADS2</b> : high activity                                                                                  | GLA can turn AA                 | With high activity FADS1, part of GLA→DGLA goes to arachidonic acid (pro-inflammatory), reducing the benefit.                                              |
| Omega-3-6-9                         | Fatty acids                          | <b>FADS1/FADS2</b> : high activity                                                                                  | Omega-6 unnecessary             | The extra LA is converted effectively to AA (high FADS);                                                                                                   |
| (–)-epicatechin                     | Polyphenols and other phytochemicals | <b>COMT</b> : Val/Val (high)                                                                                        | catechol                        | Epicatechin isolated at high dose is methylated by COMT (Val/Val);                                                                                         |
| allicin (stabilized garlic extract) | Polyphenols and other phytochemicals | <b>CYP3A5</b> :expressor (*1/*3)                                                                                    | P-gp/CYP3A4 inducer             | Allicin of fresh extracts induces P-gp/CYP3A4 (lowered saquinavir ~50%): it can lower sirolimus.                                                           |
| Saffron powder                      | Polyphenols and other phytochemicals | <b>UGT1A1</b> : IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>HLA-B*35:01</b> : probable negative | only cooking doses              | Less curcumin than extracts, but cases of liver injury include root powder.                                                                                |
| baicalein                           | Polyphenols and other phytochemicals | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                          | cases of liver injury           | Scutellaria baicalensis was associated with liver injury (LiverTox);                                                                                       |
| baicalin (Scutellaria baicalensis)  | Polyphenols and other phytochemicals | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                          | cases of liver injury           | Scutellaria baicalensis was associated with liver injury (LiverTox);                                                                                       |
| berberine                           | Polyphenols and other phytochemicals | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3)         | the genetic plan advises to add | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures;                   |
| dihydroberberine                    | Polyphenols and other phytochemicals | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3)         | the genetic plan advises to add | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures;                   |
| glabridine (liquorice)              | Polyphenols and other phytochemicals | <b>CYP3A5</b> :expressor (*1/*3)                                                                                    | inhibits CYP3A4                 | Glabridin inhibits CYP3A4 in vitro (liquorice).                                                                                                            |
| myricetin                           | Polyphenols and other phytochemicals | <b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                                                         | catechol: prefer low doses      | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). |
| naringin/naringenin                 | Polyphenols and other phytochemicals | <b>CYP3A5</b> :expressor (*1/*3)                                                                                    | weak inhibition of CYP3A4       | Naringenin contributes to the orange effect (weakly isolated).                                                                                             |
| piceatannol                         | Polyphenols and other phytochemicals | <b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                                                         | catechol: prefer low doses      | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). |
| piperine (black pepper extract)     | Polyphenols and other phytochemicals | <b>CYP3A5</b> :expressor (*1/*3)<br><b>UGT1A1</b> : IM (*1/*28)                                                     | increases drug exposure         | Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements.                                                           |
| polydatin                           | Polyphenols and other phytochemicals | <b>UGT1A1</b> : IM (*1/*28)                                                                                         | precursor of resveratrol        | Converted into resveratrol: the same logic (UGT1A1 *28, CYP3A4).                                                                                           |
| pterostilben                        | Polyphenols and other phytochemicals | <b>CETP/9p21/APOB</b> : genetic vascular risk<br><b>UGT1A1</b> : IM (*1/*28)                                        | can rise to LDL                 | In trial, pterostilbene rose to LDL (Riche 2014);                                                                                                          |
| resveratrol                         | Polyphenols and other phytochemicals | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3)                       | inhibits CYP3A4/2C9             | Glucuronidated by UGT1A1 (*28: a little higher exposure).                                                                                                  |

| Supplement                      | Category                             | Genes (state)                                                                                        | Predicted response                          | What to do (summer)                                                                                                                                        |
|---------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rutin                           | Polyphenols and other phytochemicals | <b>COMT:</b> Val/Val (high)<br><b>UGT1A1:</b> IM (*1/*28)                                            | catechol (via quercetin)                    | Rutin releases quercetin in the intestine: the same logic (COMT Val/Val, UGT1A1 *28) but lower exposure.                                                   |
| silibinin phytosome             | Polyphenols and other phytochemicals | <b>UGT1A1:</b> IM (*1/*28)<br><b>CYP2C9:</b> IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable | without proven gain; inhibits UGT1A1/CYP2C9 | Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3;     |
| silymarin (marian card)         | Polyphenols and other phytochemicals | <b>UGT1A1:</b> IM (*1/*28)<br><b>CYP2C9:</b> IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable | without proven gain; inhibits UGT1A1/CYP2C9 | Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3;     |
| taxifolin (dihydroquercetin)    | Polyphenols and other phytochemicals | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC                                            | catechol: prefer low doses                  | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). |
| caffeic acid                    | Polyphenols and other phytochemicals | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC                                            | catechol: prefer low doses                  | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). |
| chlorogenic acid (green coffee) | Polyphenols and other phytochemicals | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC                                            | can rise homocysteine                       | In trial, 2 g/day chlorogenic acid increased homocysteine (Olthof 2001);                                                                                   |
| gallic acid                     | Polyphenols and other phytochemicals | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC                                            | catechol: prefer low doses                  | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). |
| rosmarinic acid                 | Polyphenols and other phytochemicals | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC                                            | catechol: prefer low doses                  | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). |
| garlic/oil                      | Western medicinal plants             | <b>CYP3A5:</b> expressor (*1/*3)                                                                     | P-gp/CYP3A4 inducer                         | Allicin of fresh extracts induces P-gp/CYP3A4 (lowered saquinavir ~50%): it can lower sirolimus.                                                           |
| aloe vera (oral)                | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                            | monitoring liver                            | Cases of liver injury described (LiverTox);                                                                                                                |
| chinese angelic (quang quai)    | Western medicinal plants             | <b>CYP3A5:</b> expressor (*1/*3)<br><b>TERT/melanoma:</b> unfavourable                               | furanocoumarins                             | Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi).                                                                    |
| pocket                          | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                            | monitoring liver                            | Cases of liver injury described (LiverTox);                                                                                                                |
| willow bark (salicin)           | Western medicinal plants             | <b>UGT1A6:</b> *2 het<br><b>CYP2C9:</b> IM (*1/*3)                                                   | salicylate                                  | Salicylate is metabolized by UGT1A6 (*2 het) and CYP2C9 (*1/*3);                                                                                           |
| cimicifiga (black cohosh)       | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                            | monitoring liver                            | Cases of liver injury described (LiverTox);                                                                                                                |
| cascara-sagrada                 | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                            | only short term                             | Anthraquinones: chronic use associated with rare liver injury and melanotic colon;                                                                         |
| yerba (Melissa) + valerian      | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                            | monitoring liver (valerian)                 | Cases of liver injury described (LiverTox);                                                                                                                |
| American listener (skullcap)    | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                            | monitoring liver                            | Cases of liver injury described (LiverTox);                                                                                                                |
| Cerval thorn (frangula)         | Western medicinal plants             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                            | only short term                             | Anthraquinones: chronic use associated with rare liver injury and melanotic colon;                                                                         |

| Supplement                           | Category                                      | Genes (state)                                                                                               | Predicted response              | What to do (summer)                                                                                                                      |
|--------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| hydrate (goldenseal)                 | Western medicinal plants                      | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP3A5</b> :expressor (*1/*3)                               | strong inhibitor CYP3A4/2D6     | Hydrastin/berberine: clinical inhibition of CYP3A4 and CYP2D6 (Gurley 2008).                                                             |
| mahonia (Oregon grape)               | Western medicinal plants                      | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3) | the genetic plan advises to add | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; |
| pelargonium sidoides                 | Western medicinal plants                      | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                | Cases of liver injury described (LiverTox);                                                                                              |
| petasitis (butterbur, without PA)    | Western medicinal plants                      | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                | Even extracts without PA had cases of hepatitis (LiverTox);                                                                              |
| angelica root (Angelica arcangelica) | Western medicinal plants                      | <b>CYP3A5</b> :expressor (*1/*3)<br><b>TERT/melanoma</b> : unfavourable                                     | furanocoumarins                 | Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi).                                                  |
| rhubarb (root)                       | Western medicinal plants                      | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | only short term                 | Anthraquinones: chronic use associated with rare liver injury and melanotic colon;                                                       |
| ruta (ruda)                          | Western medicinal plants                      | <b>CYP3A5</b> :expressor (*1/*3)<br><b>TERT/melanoma</b> : unfavourable                                     | furanocoumarins                 | Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi).                                                  |
| sene                                 | Western medicinal plants                      | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | only short term                 | Anthraquinones: chronic use associated with rare liver injury and melanotic colon;                                                       |
| meadowsweet                          | Western medicinal plants                      | <b>UGT1A6</b> : *2 het<br><b>CYP2C9</b> : IM (*1/*3)                                                        | salicylate                      | Salicylate is metabolized by UGT1A6 (*2 het) and CYP2C9 (*1/*3);                                                                         |
| grape (brerberis)                    | Western medicinal plants                      | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3) | the genetic plan advises to add | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; |
| valerian                             | Western medicinal plants                      | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                | Cases of liver injury described (LiverTox);                                                                                              |
| gotu kola (centella)                 | Ayurvedic plants                              | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                | Cases of liver injury described (LiverTox);                                                                                              |
| guggul                               | Ayurvedic plants                              | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>CYP3A5</b> :expressor (*1/*3)                              | CYP3A4 + liver inducer          | Guggulsterone activates PXR (induces CYP3A4: it can lower sirolimus) and there are cases of hepatitis.                                   |
| mucuna pruriens (L-dopa)             | Ayurvedic plants                              | <b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                                                 | L-dopa: COMT/ homocysteine      | The L-dopa is O-methylated by COMT (Val/Val, high) spending SAM and up to homocysteine;                                                  |
| pippali (long pepper)                | Ayurvedic plants                              | <b>CYP3A5</b> :expressor (*1/*3)<br><b>UGT1A1</b> : IM (*1/*28)                                             | increases drug exposure         | Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements.                                         |
| trikatu (peppers + ginger)           | Ayurvedic plants                              | <b>CYP3A5</b> :expressor (*1/*3)<br><b>UGT1A1</b> : IM (*1/*28)                                             | increases drug exposure         | Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements.                                         |
| Sho-saiko-to / Xiao Chai Hu Tang     | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                | Cases of liver injury described (LiverTox);                                                                                              |

| Supplement                                | Category                                      | Genes (state)                                                                                               | Predicted response                     | What to do (summer)                                                                                                                      |
|-------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| angelica dahurica (bai zhi)               | Traditional Chinese medicine plants and Kampo | <b>CYP3A5</b> :expressor (*1/*3)<br><b>TERT/melanoma</b> : unfavourable                                     | furanocoumarins                        | Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi).                                                  |
| bupleur                                   | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                       | Cases of liver injury described (LiverTox);                                                                                              |
| coptis (huang lian)                       | Traditional Chinese medicine plants and Kampo | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3) | the genetic plan advises to add        | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; |
| corydalis (yan hu suo)                    | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                       | Cases of liver injury described (LiverTox);                                                                                              |
| Chinese mantle (Scutellaria baicalensis)  | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | cases of liver injury                  | Scutellaria baicalensis was associated with liver injury (LiverTox);                                                                     |
| schisandra (Schisandra chinensis)         | Traditional Chinese medicine plants and Kampo | <b>CYP3A5</b> :expressor (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                              | inhibits CYP3A4/P-gp                   | Schisandra rises the levels of tacrolimus (used for this in China);                                                                      |
| hu zhang (Polygonum cuspidatum, knotweed) | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | only short term                        | Anthraquinones: chronic use associated with rare liver injury and melanotic colon;                                                       |
| phellodendron                             | Traditional Chinese medicine plants and Kampo | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3) | the genetic plan advises to add        | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; |
| cassava seed (jue ming zi)                | Traditional Chinese medicine plants and Kampo | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | only short term                        | Anthraquinones: chronic use associated with rare liver injury and melanotic colon;                                                       |
| cinnamon-cassava (coumarin)               | Plants of the Americas and Africa; superfood  | <b>CYP2A6</b> : probable normal<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                               | hepatotoxic coumarin                   | Cassia cinnamon has coumarin (EFSA fixed a low tolerable daily intake);                                                                  |
| yohimbine HCl                             | Plants of the Americas and Africa; superfood  | <b>CYP2D6</b> : Probable NM (without CNV)                                                                   | CYP2D6 (confirm CNV)                   | Yohimbine depends on CYP2D6: slow metabolizers have often higher exposure.                                                               |
| noni                                      | Plants of the Americas and Africa; superfood  | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                       | Cases of liver injury described (LiverTox);                                                                                              |
| concentrated beet juice (nitrate)         | Plants of the Americas and Africa; superfood  | <b>NOS3</b> : -786CT + 298GT                                                                                | avoid concentrated nitrate (tadalafil) | NOS3 gives rational, but concentrated inorganic nitrate is added to the tadalafil (hypotension).                                         |
| yohimbe (shell)                           | Plants of the Americas and Africa; superfood  | <b>CYP2D6</b> : Probable NM (without CNV)                                                                   | CYP2D6 (confirm CNV)                   | Yohimbine depends on CYP2D6: slow metabolizers have often higher exposure.                                                               |
| reishi (Ganoderma lucidum)                | Medicinal mushrooms                           | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                       | Cases of liver injury described (LiverTox);                                                                                              |

| Supplement                          | Category              | Genes (state)                                                                                                                       | Predicted response                     | What to do (summer)                                                                                       |
|-------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| CBD (cannabidiol)                   | Nootropics            | <b>CYP2C19</b> : NM<br><b>CYP2C9</b> : IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>CYP3A5</b> :expressor (*1/*3) | inhibits CYP; transaminase             | CBD is metabolized by CYP2C19 (*1/*1, normal) and inhibits CYP2C19/2C9/3A4;                               |
| galantamine (winter bell)           | Nootropics            | <b>CYP2D6</b> : Probable NM (without CNV)                                                                                           | CYP2D6 (confirm CNV)                   | Galantamine is a substrate of CYP2D6/3A4;                                                                 |
| Hemp oil (total spectrum)           | Nootropics            | <b>CYP2C19</b> : NM<br><b>CYP2C9</b> : IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>CYP3A5</b> :expressor (*1/*3) | inhibits CYP; transaminase             | CBD is metabolized by CYP2C19 (*1/*1, normal) and inhibits CYP2C19/2C9/3A4;                               |
| BCAA + electrolytes                 | Sport and performance | <b>TCF7L2/FTO/GCKR</b> : unfavourable                                                                                               | the genetic plan advises to add        | High BCAA are associated with insulin resistance;                                                         |
| waxy corn starch                    | Sport and performance | <b>TCF7L2/FTO/GCKR</b> : unfavourable                                                                                               | prefer intact hydrates                 | TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training. |
| gainer (hydrates + protein)         | Sport and performance | <b>TCF7L2/FTO/GCKR</b> : unfavourable                                                                                               | prefer intact hydrates                 | TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training. |
| leucine + HMB (combined)            | Sport and performance | <b>TCF7L2/FTO/GCKR</b> : unfavourable                                                                                               | the genetic plan advises to add        | High BCAA are associated with insulin resistance;                                                         |
| maltodextrin/cyclic dextrin         | Sport and performance | <b>TCF7L2/FTO/GCKR</b> : unfavourable                                                                                               | prefer intact hydrates                 | TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training. |
| synephrine (Orange-yellow)          | Sport and performance | <b>CYP3A5</b> :expressor (*1/*3)<br><b>CETP/9p21/APOB</b> : genetic vascular risk                                                   | stimulant + CYP3A4                     | Yellow orange furanocoumarins inhibit CYP3A4 (grapefruit effect);                                         |
| deer horn velvet (IGF-1)            | Sport and performance | <b>TERT/melanoma</b> : unfavourable                                                                                                 | IGF-1 axis                             | Sold by IGF-1: the plan avoids the GH/IGF-1 axis (risk TERT alleles).                                     |
| Garcinia + green coffee (combined)  | Metabolism            | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                                          | monitoring liver                       | Cases of liver injury described (LiverTox);                                                               |
| sine detox tea                      | Metabolism            | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                                          | monitoring liver                       | Cases of liver injury described (LiverTox);                                                               |
| chromium + cinnamon (glucose)       | Metabolism            | <b>CYP2A6</b> : probable normal<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                       | hepatotoxic coumarin                   | Cassia cinnamon has coumarin (EFSA fixed a low tolerable daily intake);                                   |
| garcinia cambogia (HCA)             | Metabolism            | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                                          | monitoring liver                       | Cases of liver injury described (LiverTox);                                                               |
| sodium nitrate/potassium            | Metabolism            | <b>NOS3</b> : -786CT + 298GT                                                                                                        | avoid concentrated nitrate (tadalafil) | NOS3 gives rational, but concentrated inorganic nitrate is added to the tadalafil (hypotension).          |
| black pepper (extract)              | Metabolism            | <b>CYP3A5</b> :expressor (*1/*3)<br><b>UGT1A1</b> : IM (*1/*28)                                                                     | increases drug exposure                | Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements.          |
| thermogenic (caffeine + synephrine) | Metabolism            | <b>CYP3A5</b> :expressor (*1/*3)<br><b>CETP/9p21/APOB</b> : genetic vascular risk                                                   | stimulant + CYP3A4                     | Yellow orange furanocoumarins inhibit CYP3A4 (grapefruit effect);                                         |

| Supplement                                   | Category | Genes (state)                                                                                           | Predicted response                          | What to do (summer)                                                                                                                                        |
|----------------------------------------------|----------|---------------------------------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extract of fava (Vicia faba, L-dopa natural) | Others   | <b>G6PD</b> : normal<br><b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                     | Normal G6PD; L-dopa (COMT)                  | Normal G6PD (probable favism);                                                                                                                             |
| anti-stress formula (adaptogen)              | Others   | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                              | monitoring liver                            | Cases of liver injury described (LiverTox);                                                                                                                |
| hepatic support formula (detox)              | Others   | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                              | monitoring liver                            | Cases of liver injury described (LiverTox);                                                                                                                |
| hangover formula (DHM, dihydromyricetin)     | Others   | <b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                                             | catechol: prefer low doses                  | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). |
| hanging formula with NAC + cardo             | Others   | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable | without proven gain; inhibits UGT1A1/CYP2C9 | Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3;     |
| Sleep formula (mixture of herbs)             | Others   | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                              | monitoring liver (valerian)                 | Cases of liver injury described (LiverTox);                                                                                                                |
| nutritional yeast                            | Others   | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                                        | check fortification                         | Many brands are fortified with synthetic folic acid (DHFR CC): choosing not fortified or with 5-MTHF.                                                      |

## Undetermined (11)

| Supplement               | Category                                      | Genes (state)                  | Predicted response                  | What to do (summer)                                                                                                                                                                                                                                                                              |
|--------------------------|-----------------------------------------------|--------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ L-carnosine            | Amino acids                                   | <b>CNDP1</b> : indeterminate   | <b>indeterminate (carnosinase)</b>  | Current stack (2 g) — plan S21: KEEP. 1 g × 2 (non-evaluable carnosinase CNDP1). Serum carnosinase (CNDP1) is regulated by a repetition (CTG) <sub>n</sub> that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2). |
| anserine                 | Amino acids                                   | <b>CNDP1</b> : indeterminate   | indeterminate (carnosinase)         | Serum carnosinase (CNDP1) is regulated by a repetition (CTG) <sub>n</sub> that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).                                                                                  |
| licorice (root)          | Western medicinal plants                      | <b>HSD11B2</b> : indeterminate | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).                                                                |
| Hochuekkito (Kampo)      | Traditional Chinese medicine plants and Kampo | <b>HSD11B2</b> : indeterminate | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).                                                                |
| Juzentaiho (Kampo)       | Traditional Chinese medicine plants and Kampo | <b>HSD11B2</b> : indeterminate | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).                                                                |
| Kakkonto (Kampo)         | Traditional Chinese medicine plants and Kampo | <b>HSD11B2</b> : indeterminate | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).                                                                |
| Shakuyakukanzoto (Kampo) | Traditional Chinese medicine plants and Kampo | <b>HSD11B2</b> : indeterminate | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).                                                                |

| Supplement                   | Category                                      | Genes (state)                   | Predicted response                  | What to do (summer)                                                                                                                                                                                                               |
|------------------------------|-----------------------------------------------|---------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yokukansan (Kampo)           | Traditional Chinese medicine plants and Kampo | <b>HSD11B2</b> : indeterminate  | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks). |
| amylase                      | Enzymes                                       | <b>AMY1</b> : indeterminate     | non-valuable AMY1 copies            | The number of copies of AMY1 (salivary amylase) is not visible in the chip.                                                                                                                                                       |
| nicotine (chiplets/gums)     | Nootropics                                    | <b>CYP2A6</b> : probable normal | debugging not fully evaluated       | Nicotine is eliminated by CYP2A6 (*2/*9 absent; non-visible *4 deletions): non-determinable clearance rate. Dependence; no indication.                                                                                            |
| carnosine + ALCAR (combined) | Longevity                                     | <b>CNDP1</b> : indeterminate    | indeterminate (carnosinase)         | Serum carnosinase (CNDP1) is regulated by a repetition (CTG)n that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).                               |

### Nutrigenetic card (summary to take to the consultation)

| Axis                         | Genotype                                                                                                             | Indication (suggestion, not applied)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Liver × hepatotoxic plants   | PNPLA3 I148M C/G; MBOAT7 C/T; SERPINA1 Pi* <sup>MZ</sup> probable; HSD17B13 protector; HLA-B*35:probable negative 01 | Avoid green tea extract, kava, ashwagandha, red rice yeast, pyrrolizidine alkaloids (consolidation, tussilage), He Shou Wu, germander, celandine, chaparral, usnic acid, kratom, Tinospora, Psoralea and multi-ingredient fat burners. Plants with sporadic cases (black cohosh, valerian, garcinia, oral aloe, noni, gotu kola, reishi, CBD): only short term.                                                                                                                                                               |
| Gilbert × polyphenols        | UGT1A1 *1/*28                                                                                                        | Suggestion (not applied): pause curcumin, quercetin, luteolin and apigenin (test-pause), because they inhibit UGT1A1 (concentrate bilirubin and liver) and CYP3A4/P-gp (sirolimus); pause while ALT is high (current protocol); curcumin until ALT normalizes. Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. Resveratrol and silymarin: monitoring. |
| Methylation                  | MTHFR 677CT/1298AC DHFR CC; BHMT R239Q; GNMT TT; COMT Val/Val; CPS1 (low glycine)                                    | 5-MTHF instead of folic acid (multivitamins, complexes B, fortified yeasts). TMG ≤ 3 g, glycine 5 g, creatine. Avoid SAME, methionine and concentrated catechols (luteolin, quercetin, EGCG, chlorogenic acid, L-dopa). B6 ≤ 5 mg.                                                                                                                                                                                                                                                                                            |
| Iron                         | HFE H63D het (C282Y absent)                                                                                          | No iron or liver dried by routine; ferritin target below 100; vitamin C ≥ 2 h away from meals; calcium at meals.                                                                                                                                                                                                                                                                                                                                                                                                              |
| Caffeine and melatonin       | CYP1A2 *1F het + rs2472297 C/C; ADORA2A C/C; MTNR1B C/G                                                              | Total caffeine ≤ 200 mg/day to 12h (including guarana, mate, glue, green coffee, pre-training). Melatonin 0.3–1 mg, ≥ 3 h after the last meal. The suggested apigenin (pause) would prolong both (inibate CYP1A2).                                                                                                                                                                                                                                                                                                            |
| Telomerase × cancer          | 5 TERT risk alleles; p88 nevi                                                                                        | Avoid purified extracts of astragalus, cycloastragenol, astragaloside IV and astragalus; photoprotection (St John's wort, psoralens and angelica are photosensitizers; Oral polypodium can help).                                                                                                                                                                                                                                                                                                                             |
| Sirolimus × CYP3A4/P-gp      | CYP3A5 *1/*3 (expressor); CYP3A4 *1/*1                                                                               | Avoid St John's wort (strong inducer). Schistandra, berberine/hydraste, piperine, allicin, guggul, resveratrol, CBD, blackberry orange and furanocoumarin never in the rapamycin window.                                                                                                                                                                                                                                                                                                                                      |
| Omega 3                      | FADS1/2 of high activity; APOE ε3/ε3                                                                                 | EPA/DHA useful preformed; total 3.2–4 g/day (Coordination Note). GLA (primary, blurring) and extra Omega-6 without advantage.                                                                                                                                                                                                                                                                                                                                                                                                 |
| Zinc                         | SLC30A8 C/C + rs11558471 A/A                                                                                         | Total zinc 20–25 mg + copper 1.5–2 mg.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Phase II / NRF2              | Probable null GSTM1; GSTA1 *B/*B; NRF2 –617 C/C; NQO1 *1/*1                                                          | Sulforaphane and NAC coherent; CoQ10 without genetic indication.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Hemolysis / lactose / gluten | Normal G6PD; Persistent LCT; without HLA-DQ2/DQ8                                                                     | High-dose vitamin C, methylene blue, quinine, yellow melon and fava without genetic restriction of hemolysis; lactose and gluten tolerated.                                                                                                                                                                                                                                                                                                                                                                                   |

## Current x genetic stack

| Supplement (current dose)                                                                                      | Genes (state)                                                                                                                 | Interpretation                   | Plan                                | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ <b>solid lipid curcumin (SLCP)</b><br>1 g                                                                    | UGT1A1: IM (*1/*28)<br>PNPLA3/Pi*MZ (liver):<br>unfavourable<br>HLA-B*35:01: probable<br>negative<br>CYP3A5:expressor (*1/*3) | avoid/pause                      | <b>S26: PAUSE</b>                   | Pause (current protocol). UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                                                                            |
| ★ <b>luteolin</b><br>100 mg                                                                                    | COMT: Val/Val (high)<br>MTHFR: 677CT/1298AC<br>UGT1A1: IM (*1/*28)<br>CYP2C9: IM (*1/*3)                                      | avoid/pause                      | <b>S27: PAUSE</b>                   | Pause (current protocol). Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                        |
| ★ <b>apigenin</b><br>100 mg                                                                                    | UGT1A1: IM (*1/*28)<br>CYP3A5:expressor (*1/*3)<br>CYP1A2: normal to<br>intermediate<br>CYP2C9: IM (*1/*3)                    | avoid / pause (test-<br>pause)   | <b>S28: PAUSE (test-<br/>pause)</b> | Test-pause (current protocol). Test pause for the same reasons of luteolin: inhibits UGT1A1 (*1/*28: up to bilirubin and confuses liver reading) and CYP3A4/P-gp (sirolimus; CYP3A5 expressor). It is not catechol, but also inhibits CYP1A2 (already normal to intermediate: *1F het + rs2472297 C/C), which rises melatonin and caffeine, and CYP2C9 (*1/*3). Pause; pause while ALT is high (current protocol). Coordination note: in the current protocol, the pause lasts while ALT is high; if it returns, 25-50 mg at night, outside the rapamycin window (suggestion, not applied).                                                                                                                                  |
| ★ <b>quercetin</b><br>250 mg (3×/month)                                                                        | UGT1A1: IM (*1/*28)<br>COMT: Val/Val (high)<br>MTHFR: 677CT/1298AC<br>CYP3A5:expressor (*1/*3)                                | avoid/pause                      | <b>S29: PAUSE</b>                   | It is suggested to pause as an isolated supplement (genetic verdict); in the current protocol enters the monthly pulse with fisetin, only with normal ALT. UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. |
| ★ <b>fish oil (EPA+DHA, TG/rTG)</b><br>⊕<br>EPA 1125 + DHA 875 mg                                              | FADS1/FADS2: high activity<br>APOE: ε3/ε3                                                                                     | adjust dose (plan)               | <b>S1: ADJUST</b>                   | Up to 3.2-4 g/day of EPA+DHA (Coordination Note), form rTG, with the meal fatter and far from psyllium. FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2-4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                             |
| ★ <b>trace minerals (multineural)</b> ⊕<br>Zn 11, Cu 0,9, I 0,05, Se 0,055, Mn 2,3, Cr 0,035, Mo 0,045, B 1 mg | SLC30A8: C/C (usual zinc)<br>GPX1: Pro198Leu het<br>SLC39A8: normal<br>AOC1 (DAO): slightly ↓                                 | up zinc and copper               | <b>S12: ADJUST (Zn and Cu)</b>      | Total zinc 20-25 mg + total copper 1.5-2 mg; remaining equal. SLC30A8 C/C + rs11558471 A/A (genetically useful zinc); GPX1 Pro198Leu (selenium already suitable, without Brazil nuts); SLC39A8 C/C (manganese without adjustment); DAO is a copper enzyme (AOC1 het). Total zinc 20-25 mg + total copper 1.5-2 mg, ≥ 2 h away from NAC, Ca-AMG and fiber (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                          |
| ★ <b>vitamin C liposomal</b><br>500 mg                                                                         | HFE: H63D het<br>G6PD: normal<br>SLC23A1/2 (vit. C): typical                                                                  | drinking between<br>meals (hour) | <b>S16: ADJUST (hour)</b>           | Always ≥ 2 h away from meals. HFE H63D het: vitamin C with food increases the absorption of non-heme iron. 500 mg always ≥ 2 h away from meals. Normal G6PD; Typical SLC23A1/2 (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ★ <b>lycopene</b><br>35 mg                                                                                     | BCO1: slightly ↓ conversion                                                                                                   | lower dose (≤ 10 mg)             | <b>S23: ADJUST (descend)</b>        | 10 mg or stop. Genetic levels of lycopene tendentially high: no genetic indication for 35 mg. 10 mg or tomato cooked with olive oil (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Supplement (current dose)                                | Genes (state)                                                                            | Interpretation              | Plan                  | Comment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ fisetin<br>2.5 g (3×/month)                            | COMT: Val/Val (high)<br>UGT1A1: IM (*1/*28)<br>CYP3A5:expressor (*1/*3)                  | isolated monthly pulse      | S29: ADJUST           | Monthly pulse (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI; current protocol: with quercetin, only with normal ALT. Catechol (COMT Val/Val + MTHFR) and inhibitor of CYP3A4/P-gp and UGT1A1. 9p21 homozygous (CDKN2A/B region, senescence) gives senolytic rationale. Genetic reading: 1 pulse/month (2 days), ≥ 7 days away from rapamycin and FOXO4-DRI (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. |
| ★ melatonin high dose (≥ 10 mg)<br>up to 100 mg          | MTNR1B: C/G (carrier)<br>CYP1A2: normal to intermediate                                  | low (0.3–1 mg)              | S30: ADJUST (descent) | 0.3–1 mg, 30–60 min before bedtime. MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers; Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied). Doses ≥ 10 mg (up to 100 mg) have no genetic basis and prolong the effect on insulin.                                                                                                                                                                                |
| ★ PHGG (partially hydrolyzed guar gum) ⊕<br>6 g          | TCF7L2/FTO/GCKR: unfavourable<br>CETP/9p21/APOB: genetic vascular risk<br>FUT2: secretor | adjust dose (plan)          | S35: ADJUST (subir)   | 10 g/day. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ★ psyllium (ispaghula bark) ⊕<br>10 g                    | TCF7L2/FTO/GCKR: unfavourable<br>CETP/9p21/APOB: genetic vascular risk<br>FUT2: secretor | adjust dose (plan)          | S35: ADJUST (subir)   | 15–20 g/day in 2–3 takes with water; drugs ≥ 1 h before or 2–4 h after. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| ★ pyridoxal-5-phosphate (P5P)<br>15 mg                   | PDXK/ALPL/NBPF3 (B6): typical/high                                                       | lowering dose (≤ 5 mg/day)  | S6: ADJUST (descend)  | ≤ 5 mg/day or stop (above the upper limit of ingestion, without genetic indication). PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. ≤ 5 mg/day or pause (suggestion, not applied).                                                                                                                                                                                                                                                                             |
| ★ TMG (trimethylglycine, betaine anhydrous) ⊕<br>1500 mg | BHMT: R239Q het<br>GNMT: TT<br>MTHFR: 677CT/1298AC                                       | adjust dose (plan)          | S7: ADJUST (subir)    | Up to 3 g/day in 2 tomas; do not exceed 3 g (LDL). BHMT R239Q het, GNMT TT and MTHFR compound: the BHMT pathway compensates for folate. 3 g/day (≥ 4 g up to LDL) (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ★ glycine ⊕<br>3 g                                       | CPS1: low glycine<br>GSTM1: probable null<br>GNMT: TT                                    | adjust dose (plan)          | S8: ADJUST (subir)    | Up to 5 g/day (3 g at bedtime + 2 g in the morning). CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                |
| ★ L-carnosine<br>2 g                                     | CNDP1: indeterminate                                                                     | indeterminate (carnosinase) | S21: KEEP             | 1 g × 2 (non-valuable carnosinase CNDP1). Serum carnosinase (CNDP1) is regulated by a repetition (CTG) <sub>n</sub> that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).                                                                                                                                                                                                                                                                                                                                                            |
| ★ creatine monohydrate ⊕<br>5 g                          | AMPD1: normal<br>ACTN3: R/X<br>GNMT: TT                                                  | useful (muscle + save SAM)  | S10: KEEP             | The same. Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.                                                                                                                                                                                                                                                                                                                                                                                                 |
| ★ magnesium acetyl-aurate<br>750 mg                      | TRPM6: typical                                                                           | typical absorption          | S13: KEEP             | The same. TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Supplement (current dose)                                              | Genes (state)                                                 | Interpretation                             | Plan                    | Comment                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ magnesium glycinate<br>500 mg                                        | TRPM6: typical                                                | typical absorption                         | S13: KEEP               | The same. TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                                                              |
| ★ lithium orotate (low dose)<br>5 mg                                   | APOE: ε3/ε3                                                   | without a safety gene                      | S15: KEEP               | Similarly; pause in days of vomiting/diarrhea (telmisartan/dapagliflozin). No PGx for lithium in microdose; APOE ε3/ε3; Alzheimer's PGS of 39 variants p91 (genomic p52 to p85; genomic consensus p75): weak rational. With telmisartan (sodium lithium) and dapagliflozin (low-o): pause in days of vomiting/diarrhea.    |
| ★ NAC (N-acetylcysteine) ☺<br>1800 mg                                  | GSTM1: probable null<br>GSTA1: *B/*B<br>GPX1: Pro198Leu het   | useful support for glutathione             | S17: KEEP               | 1200–1800 mg, far from training. Probable null GSTM1, GSTA1 *B/*B, GCLM –588 C/T and GPX1 Pro198Leu: weak II → NAC phase is coherent. Far from the training; ≥ 2 h away from zinc/copper.                                                                                                                                  |
| ★ acetyl-L-carnitine (ALCAR)<br>500 mg                                 | SLC22A4/5 (OCTN1/2): typical<br>FMO3: typical                 | Typical transport                          | S19: KEEP               | 500 mg in the morning. OCTN2 (SLC22A5) typical; CPT2 V368I het; FMO3 typical (some TMAO). Keep 500 mg in the morning.                                                                                                                                                                                                      |
| ★ vitamin D3 (cholecalciferol)<br>10 000 IU                            | GC (DBP): favourable<br>VDR: typical<br>CYP2R1/DHCR7: typical | metabolism of vitamin D favourable/typical | S2: KEEP (coordination) | Maintain 10 000 IU (Coordination note: measurement-based decision). GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied). |
| ★ vitamin K2 MK-7<br>240 µg                                            | VKORC1: TT (sensitive)<br>CYP4F2: normal<br>GGCX: C/T         | normal (relevant only with coumarins)      | S2: KEEP                | The same; it only weighs if you take coumarins (VKORC1 TT). VKORC1 –1639 T/T + CYP4F2 C/C + GGCX C/T: no effect in the absence of anticoagulant. If you ever take warfarin/acenocoumarol, keep K intake stable and warn the doctor (VKORC1 TT makes INR very sensitive).                                                   |
| ★ ergothioneine<br>10 mg                                               | SLC22A4/5 (OCTN1/2): typical                                  | typical capture                            | S20: KEEP               | The same. OCTN1 (SLC22A4) L503 C/C: typical ergothioneine uptake. Maintain.                                                                                                                                                                                                                                                |
| ★ lutein<br>20 mg                                                      | CFH/ARMS2: CFH het<br>BCO1: slightly ↓ conversion             | useful (retina)                            | S24: KEEP               | The same, with fat. CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.                                                                                                                                                                                                                   |
| ★ zeaxanthin<br>4 mg                                                   | CFH/ARMS2: CFH het<br>BCO1: slightly ↓ conversion             | useful (retina)                            | S24: KEEP               | The same, with fat. CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.                                                                                                                                                                                                                   |
| ★ sulforaphane (glucoraphanine + myrosinase) ☺<br>glucoraphanin 100 mg | GSTM1: probable null<br>NFE2L2 (NRF2): normal                 | favourable response                        | S25: KEEP               | The same in the morning. Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1-NRF2 intact. Maintain.                                                                                                                                                           |
| ★ 5-MTHF (calcium L-methylfolate) ☺<br>800 µg                          | MTHFR: 677CT/1298AC<br>DHFR: slow reduction (CC)              | ideal form for the genotype                | S3: KEEP                | Keep 800 µg; no product with folic acid. 5-MTHF enters directly into the cycle without DHFR or MTHFR. Keep 800 µg (not climbing) (suggestion, not applied).                                                                                                                                                                |
| ★ pine bark extract (marine pine hull) ☺<br>100 mg                     | NOS3: –786CT + 298GT                                          | useful (NO)                                | S31: KEEP               | In the morning, outside the rapamycin window. NOS3 –786 C/T + Glu298Asp: pine bark extract improves endothelial function. Keep in the morning, outside the rapamycin window for prudence.                                                                                                                                  |
| ★ olive leaf extract (oleuropein) ☺<br>oleuropein 250 mg               | PON1: low expression                                          | useful (PON1)                              | S32: KEEP               | The same, with meal. Low expression PON1: olive oil/oliveira polyphenols increase the activity of PON1.                                                                                                                                                                                                                    |
| ★ hydroxytyrosol ☺<br>25 mg                                            | PON1: low expression<br>COMT: Val/Val (high)                  | useful in low dose                         | S32: KEEP               | The same, with meal. Low expression PON1: olive oil polyphenols rise to PON1. Low dose (25 mg) : small impact on COMT. Maintain.                                                                                                                                                                                           |
| ★ black garlic<br>10 g                                                 | NOS3: –786CT + 298GT<br>CYP3A5:expressor (*1/*3)              | low allicin                                | S33: optional           | Redundant with AGE. NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.                                                                                                                                                          |

| Supplement (current dose)                                                           | Genes (state)                                                                                              | Interpretation                          | Plan                        | Comment                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ <b>extract of aged garlic (AGE)</b><br>1.2 g                                      | <b>NOS3: –786CT + 298GT</b><br><b>CYP3A5:expressor (*1/*3)</b>                                             | <b>low allicin</b>                      | <b>S33: KEEP</b>            | The same. NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.                                                                                                                                                                               |
| ★ <b>multistrain (generic) multistrain</b>                                          | <b>FUT2: secretor</b>                                                                                      | <b>typical colonization (secretor)</b>  | <b>S36: KEEP</b>            | The same. FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                                                                                                               |
| ★ <b>methylcobalamine</b><br>1000 µg (not specified)                                | <b>TCN2: 776CG (light)</b><br><b>FUT2: secretor</b><br><b>MTR: normal (AA)</b><br><b>MTRR: normal (AA)</b> | <b>Transport and use of typical B12</b> | <b>S4: KEEP</b>             | 1000 µg (500 µg if acne appears); methyl- or adenosyl-cobalamine. MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears). |
| ★ <b>riboflavin-5-phosphate (R5P)</b><br>25 mg                                      | <b>MTHFR: 677CT/1298AC</b>                                                                                 | <b>modest benefit in 677CT</b>          | <b>S5: ADJUST (optimal)</b> | 5–10 mg arrives; maintaining 25 mg is at low risk. Riboflavin stabilizes thermolabile MTHFR; the benefit is clear in 677TT and only modest in heterozygotes. Low dose (5–10 mg) arrives; 25 mg is safe.                                                                                                                               |
| ★ <b>bovine hydrolyzed collagen</b><br>20 g                                         | <b>COL1A1/COL5A1: normal</b>                                                                               | <b>without risk alleles</b>             | <b>S9: KEEP</b>             | The same (glycine source). COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).                                                                                                                                                                                                                      |
| ★ <b>taurine</b><br>6 g                                                             | —                                                                                                          | <b>without known genetic relevance</b>  | <b>S11: KEEP</b>            | The same (no specific gene). No known gene-supplement interaction or relevant security gene.                                                                                                                                                                                                                                          |
| ★ <b>calcium alpha-ketoglutarate (Ca-AKG)</b><br>2 g                                | —                                                                                                          | <b>without known genetic relevance</b>  | <b>S14: KEEP</b>            | 1 g + 1 g at meals (calcium at meals reduces iron absorption); HFE H63D). No known gene-supplement interaction or relevant security gene.                                                                                                                                                                                             |
| ★ <b>R-ALA (R-lipoic acid)</b><br>240 mg                                            | —                                                                                                          | <b>without known genetic relevance</b>  | <b>S18: KEEP (optimal)</b>  | 240 mg fasting. No known gene-supplement interaction or relevant security gene.                                                                                                                                                                                                                                                       |
| ★ <b>astaxanthin</b><br>12 mg                                                       | —                                                                                                          | <b>without known genetic relevance</b>  | <b>S22: KEEP</b>            | 12 mg with fat (not pro-vitamin A). No known gene-supplement interaction or relevant security gene.                                                                                                                                                                                                                                   |
| ★ <b>ginger</b><br>400 mg                                                           | —                                                                                                          | <b>without known genetic relevance</b>  | <b>S34: KEEP</b>            | The same (useful for the nausea of the retatrutide). No known gene-supplement interaction or relevant security gene.                                                                                                                                                                                                                  |
| ★ <b>tributyrine</b><br>500 mg                                                      | —                                                                                                          | <b>without known genetic relevance</b>  | <b>S37: KEEP</b>            | The same. No known gene-supplement interaction or relevant security gene.                                                                                                                                                                                                                                                             |
| Not yet taken, with genetic indication: choline (citicoline or phosphatidylcholine) | <b>PEMT: V175M het</b><br><b>FMO3: typical</b>                                                             | preferred form                          | S38: ADD                    | 250–500 mg/day of choline; prefer citicolin/PC to bitartrate (TMAO) (suggestion, not applied).                                                                                                                                                                                                                                        |

## What this work adds to what was already known

- **AMPD1 Q12X absent** (rs17602729 G/G): no myoadenylate deaminase deficiency; D-ribose has no genetic indication. New.
- **CYP2E1 \*5 absent** (rs2031920 C/C, rs3813867 G/G): typical metabolism of solvents, ethanol and some essential oils constituents. New.
- **UGT1A6 \*2 heterozygous** (rs2070959 A/G, rs6759892 T/G): small effect on salicylate glucuronidation (sallow bark, almaria), added to CYP2C9 \*1/\*3. New.
- **EPHX1 Tyr113His T/T + His139Arg A/G**: intermediate activity of epoxide-hydrolase (informative). New.
- **Normal G6PD applied to supplements**: high-dose vitamin C, methylene blue, quinine (quine peel), yellow melon and fava extract without genetic risk of hemolysis. New application.
- **PAH without pathogenic variants tested**: phenylalanine, DLPA and spirulina without genetic restriction. New application.

- **Gene-nutrient interactions applied for the first time:** FADS1 high × GLA (primary/borage tend to give arachidonic acid); COMT/MTHFR × chlorogenic acid and L-dopa (homocysteine); priority LDL × pterostilbene and coconut oil FUT2 secretor × bifidobacteria and 2'-FL; SLC01B1 \*14 × red rice yeast (favourable, but doubles statin and has liver risk); CYP2D6 without CNV × yohimbine and galantamine; CYP2A6 × coumarin of cinnamon-cassava and nicotine. New.
- **Telomerase activators:** the precaution of epithelon extends to the purified extracts of astragalum, cycloaestrageenol, astagaloside IV and astragalum. Coherent with the plan.
- **Not determinable in the chip** and therefore in gray: CNDP1 (carnosinase → carnosine, anserin), HSD11B2 (sensitivity to liquor and formulas Kampo with liquor), AMY1 (amylase), CYP2A6 deletions (nicotin). Clarification.
- Confirmed without change and applied to each supplement: MTHFR compound, DHFR CC, COMT Val/Val, UGT1A1 \*1/\*28, HFE H63D, SLC30A8 C/C, FADS high, GSTM1 probable null, CYP1A2 \*1F, MTNR1B C/G, PNPLA3/MBOAT7/Pi\*MZ, HLA-B\*35:01 probable negative, TERT risk.

## 2. Phenotype per gene

| Gene                               | Genotype                                                                                                                                                       | Phenotype                                                        | Trust / origin                                                |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------|
| <b>PNPLA3/Pi*MZ (liver)</b>        | PNPLA3 I148M (rs738409) C/G; MBOAT7 rs641738 C/T; SERPINA1 Pi*Z (rs28929474) probable C/T; HSD17B13 rs72613567 T/TA (protector, imputation DR2 1.00)           | Hepatic genetic vulnerability (attenuated by HSD17B13)           | Moderate - chip + imputation (Pi*Z to clinically confirm)     |
| <b>TERT/melanoma</b>               | TERT rs2736100 C/C, rs10069690 T/T, rs7734992 C/C, rs2242652 G/G, rs2736098 C/C; nevus at the 88th percentile (melanoma has low polygenic risk: consensus p18) | TERT alleles of cancer risk + nevi p88                           | high (genotypes) · moderate (score)                           |
| <b>COMT</b>                        | Val158Met (rs4680) G/G (Val/Val); rs4633 C/C                                                                                                                   | high activity (methyl catechols fast; spend SAM)                 | high - direct chip                                            |
| <b>CYP1A2</b>                      | *1F heterozygous (rs762551 A/C); rs2472297 C/C; AHR rs4410790 C/C                                                                                              | clearance of normal to intermediate caffeine                     | high - direct chip                                            |
| <b>CYP2C9</b>                      | *1/*3 (rs1057910 A/C)                                                                                                                                          | intermediate metabolizer (AS 1.0)                                | high - direct chip                                            |
| <b>CYP3A5</b>                      | *1/*3 (rs776746 C/T)                                                                                                                                           | expresser (intermediate metabolizer)                             | high - direct chip                                            |
| <b>DHFR</b>                        | rs408626 C/C                                                                                                                                                   | slow reduction of synthetic folic acid (small effect)            | Moderate - direct chip (functional effect discussed)          |
| <b>HFE</b>                         | H63D (rs1799945) C/G heterozygous; C282Y (rs1800562) G/G (absent); S65C absent                                                                                 | H63D carrier (iron absorption slightly ↑)                        | high - direct chip                                            |
| <b>MTHFR</b>                       | 677 C/T (rs1801133 A/G) + 1298 A/C (rs1801131 T/G); probable compound heterozygous (undetermined phase)                                                        | activity ~50-60%                                                 | high - direct chip (undetermined phase)                       |
| <b>MTNR1B</b>                      | rs10830963 C/G                                                                                                                                                 | carrier G: high melatonin reduces insulin secretion              | high - direct chip                                            |
| <b>CETP/9p21/APOB</b>              | CETP rs708272 G/G, rs3764261 C/C; 9p21 rs1333049/rs10757278/rs4977574 homozygous risk; APOB rs693 G/A                                                          | genetic priority to low LDL/ApoB                                 | high - direct chip                                            |
| <b>CPS1</b>                        | rs715 T/T + rs1047891 C/C                                                                                                                                      | Genetically lower plasma glycine                                 | high - direct chip                                            |
| <b>CYP2D6</b>                      | Probable *1/*2 (rs16947 A/G, rs1080985 C/G; without *3/*4/*6/*9/*10/*41)                                                                                       | Probable normal metabolizer (AS 2)                               | moderate - chip + imputation; CNV (deletion *5, duplications) |
| <b>GNMT</b>                        | rs9296404 T/T                                                                                                                                                  | higher conversion of SAM to homocysteine (hypothesis)            | high - Imputed DR2 ≥ 0.97 + chip                              |
| <b>GSTM1</b>                       | null *0/*0 (inferred by rs366631)                                                                                                                              | without GSTM1 activity (probable)                                | moderate - inferred deletion                                  |
| <b>NOS3</b>                        | -786 T/C (rs2070744 C/T); Glu298Asp (rs1799983) G/T                                                                                                            | production of slightly reduced NO                                | high - chip + imputed                                         |
| <b>PON1</b>                        | Q192R (rs662) T/T (QQ); L55M M/M; -108 T/T (rs705379)                                                                                                          | low expression                                                   | high - direct chip                                            |
| <b>Sal (AGT/ADD1/GNB3/CYP11B2)</b> | AGT M235T (rs699) A/G; ADD1 G460W (rs4961) G/T; GNB3 825 C/T (rs5443); CYP11B2 -344 (rs1799998) A/A; ACE rs4343 G/A                                            | sensitivity to modest salt (polygenic)                           | high - direct chip (small effect)                             |
| <b>TCF7L2/FTO/GCKR</b>             | TCF7L2 rs7903146 C/T; FTO rs9939609 T/A; GCKR rs1260326 T/C                                                                                                    | Glucemic and hepatic response to less favourable sugars/fructose | high - direct chip                                            |
| <b>UGT1A1</b>                      | *1/*28 ((TA)7 heterozygous; rs887829 C/T)                                                                                                                      | intermediate metabolizer (Gilbert heterozygote)                  | high - DR2 1.00 + chip imputed                                |

| Gene                 | Genotype                                                                     | Phenotype                                                                            | Trust / origin                      |
|----------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------|
| <b>VKORC1</b>        | –1639 T/T (rs9923231); rs9934438 A/A                                         | high sensitivity to coumarins (only relevant if taking warfarin/acenocoumarol)       | high – DR2 0.99 + chip imputed      |
| <b>AMY1</b>          | number of non-visible copies in the chip                                     | indeterminate                                                                        | not determinable                    |
| <b>CNDP1</b>         | repeat (CTG)n D18S880 not testable                                           | indeterminate                                                                        | not determinable                    |
| <b>CYP2A6</b>        | without *2 or *9; deletions (*4) not visible                                 | probable normal                                                                      | low – deletions not evaluated       |
| <b>HSD11B2</b>       | Functional variants not tested in the chip                                   | indeterminate (sensitivity to liquor)                                                | not determinable                    |
| <b>ABCG5/ABCG8</b>   | ABCG8 D19H (rs11887534) G/G; rs4148217 C/C; ABCG5/8 rs6544713 C/C            | without hyperabsorption of plant sterols                                             | high - direct chip                  |
| <b>ACTN3</b>         | R577X (rs1815739) T/C (R/X)                                                  | mixed strength/resistance                                                            | high - direct chip                  |
| <b>ADORA2A</b>       | rs5751876 C/C                                                                | genotype with less anxiety sensitivity, but caffeine-sensitive sleep (small studies) | high - direct chip (weak effect)    |
| <b>AMPD1</b>         | Q12X (rs17602729) G/G                                                        | normal (without myoadenylate deaminase deficiency)                                   | high - direct chip                  |
| <b>AOC1 (DAO)</b>    | Thr16Met (rs10156191) C/T; His645Asp (rs1049793) C/G; rs1049742 C/C          | DAO activity slightly reduced                                                        | high - direct chip (small effect)   |
| <b>APOE</b>          | ε3/ε3 (rs429358 T/T, rs7412 C/C)                                             | common genotype                                                                      | high - chip + imputation DR2 ≥ 0.97 |
| <b>BCO1</b>          | R267S (rs12934922) A/T; A379V (rs7501331) C/C; rs6564851 G/T                 | conversion of slightly reduced β-carotene                                            | high - direct chip (small effect)   |
| <b>BHMT</b>          | rs3733890 G/A (R239Q het)                                                    | betaine pathway with common variant                                                  | high - direct chip (small effect)   |
| <b>CFH/ARMS2</b>     | CFH Y402H (rs1061170) C/T; ARMS2 A69S (rs10490924) G/G                       | Mild DMI risk (1 allele CFH)                                                         | high - direct chip                  |
| <b>COL1A1/COL5A1</b> | COL1A1 Sp1 (rs1800012) C/C; COL5A1 rs12722 T/T                               | without risk alleles                                                                 | high - direct chip                  |
| <b>CYP2C19</b>       | *1/*1 (without *2, *3, *4, *17)                                              | normal metabolizer                                                                   | high - direct chip                  |
| <b>CYP2R1/DHCR7</b>  | CYP2R1 rs10741657 G/A, rs10766197 G/G (imputed); DHCR7 rs12785878 G/G        | typical                                                                              | high – chip + imputation            |
| <b>CYP3A4</b>        | *1/*1 (without *22, rs35599367 G/G)                                          | normal                                                                               | high - direct chip                  |
| <b>CYP4F2</b>        | V433M (rs2108622) C/C                                                        | normal (metabolism of vitamins E and normal K)                                       | high - direct chip                  |
| <b>FADS1/FADS2</b>   | FADS1 rs174546 C/C; rs174537 G/G; FADS2 rs1535 A/A; ELOVL2 rs953413 G/G      | High activity desaturases (plus AA from LA; good conversion ALA→EPA)                 | high - direct chip                  |
| <b>FMO3</b>          | E158K (rs2266782) G/G; rs2266780 A/A                                         | FMO3 typical (TMAO form from choline/ carnitine)                                     | high - direct chip                  |
| <b>FUT2</b>          | rs601338 G/A (W143X heterozygous); rs602662 G/A                              | secretor (1 functional allele)                                                       | high - direct chip                  |
| <b>G6PD</b>          | B (normal), hemizygous; without c.202G>A (rs1050828) or c.563C>T (rs5030868) | normal                                                                               | high for the tested variants - chip |
| <b>GC (DBP)</b>      | rs2282679 T/T; rs4588 G/G; rs7041 C/C (without Gc2)                          | vitamin D binding protein                                                            | high - direct chip                  |
| <b>GGCX</b>          | rs699664 C/T                                                                 | common variant                                                                       | high - direct chip (small effect)   |
| <b>GPX1</b>          | Pro198Leu (rs1050450) A/G                                                    | slightly reduced activity                                                            | high - direct chip                  |

| Gene                        | Genotype                                                                              | Phenotype                                                          | Trust / origin                             |
|-----------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------|
| <b>GSTA1</b>                | *B/*B (rs3957357 A/A)                                                                 | reduced expression                                                 | high – chip + imputation                   |
| <b>HLA-B*35:01</b>          | non-probable carrier (imputed tags; VPN 0.97)                                         | without the HLA of liver injury by curcuma/<br>green tea/Polygonum | moderate - imputed tags                    |
| <b>HLA-DQ2/DQ8</b>          | rs2187668 C/C; rs7454108 T/T; rs7775228 T/T (without DQ2.5 or DQ8)                    | without HLA of celiac disease                                      | high - chip tags                           |
| <b>HNMT</b>                 | Thr105Ile (rs11558538) T/C                                                            | slightly reduced activity                                          | high - direct chip (small effect)          |
| <b>LCT</b>                  | –13910 C/T (rs4988235 A/G); rs182549 C/T                                              | lactase persistence                                                | high - direct chip                         |
| <b>MTR</b>                  | A2756G (rs1805087) A/A                                                                | normal                                                             | high - direct chip                         |
| <b>MTRR</b>                 | A66G (rs1801394) A/A; rs1532268 C/C                                                   | normal                                                             | high - direct chip                         |
| <b>NAT2</b>                 | Probable *5/*5 (341C, 481T, 803G homozygotes)                                         | slow acetylator                                                    | high - direct chip (7 SNPs)                |
| <b>NFE2L2 (NRF2)</b>        | –617 C/C (rs6721961)                                                                  | Normal promoter: intact NRF2 response                              | high - direct chip                         |
| <b>NQO1</b>                 | P187S (rs1800566) G/G (*1/*1)                                                         | normal (without null allele)                                       | high - direct chip                         |
| <b>OPRM1</b>                | 118 A/G (rs1799971)                                                                   | G allele carrier                                                   | high – chip (inconsistent clinical effect) |
| <b>PAH</b>                  | without pathogenic variants among the tested ones (chip ClinVar tracking)             | normal for the tested variants                                     | partial – chip (only known variants)       |
| <b>PDXK/ALPL/NBPF3 (B6)</b> | PDXK rs2010795 G/G; ALPL rs1256335 G/A; NBPF3 rs4654748 C/T                           | activation of typical to high B6                                   | high - direct chip                         |
| <b>PEMT</b>                 | rs7946 C/T (V175M het); rs12325817 C/C                                                | endogenous synthesis of reduced choline (mild)                     | high - direct chip                         |
| <b>SLC22A4/5 (OCTN1/2)</b>  | SLC22A4 L503F (rs1050152) C/C; SLC22A5 rs2631367 G/G                                  | Ergothioneine and carnitine transport                              | high - direct chip                         |
| <b>SLC23A1/2 (vit. C)</b>   | SLC23A1 rs33972313 C/C, rs10063949 C/C; SLC23A2 rs6053005 C/C                         | typical vitamin C transport                                        | high - direct chip                         |
| <b>SLC30A8</b>              | rs13266634 C/C (R325); rs11558471 A/A                                                 | Alleles that most benefit from zinc intake (glucose)               | high - direct chip                         |
| <b>SLC39A8</b>              | A391T (rs13107325) C/C; rs233804 C/C                                                  | Normal manganese transport                                         | high - direct chip                         |
| <b>SLCO1B1</b>              | *1/*14 (521 T/T; 388G and 463A heterozygotes)                                         | increased function                                                 | high - direct chip                         |
| <b>SOD2</b>                 | Val16Ala (rs4880) G/G (Ala/Ala)                                                       | Efficient mitochondrial import of MnSOD                            | high - direct chip                         |
| <b>TCN2</b>                 | 776 C/G (rs1801198 C/G); rs9606756 A/G                                                | slightly less efficient B12 transport (small effect)               | high - direct chip                         |
| <b>TMPRSS6/TF</b>           | TMPRSS6 rs855791 G/A; TF rs3811647 G/G, rs1799852 C/T                                 | small effects on iron                                              | high - direct chip                         |
| <b>TRPM6</b>                | rs3750425 C/C; rs2274924 T/T                                                          | typical magnesium absorption                                       | high - direct chip                         |
| <b>UGT1A6</b>               | *2 heterozygous (rs2070959 A/G, rs6759892 T/G)                                        | Glucuronidation of salicylate/phenols slightly different           | high - direct chip (small effect)          |
| <b>VDR</b>                  | FokI (rs2228570) G/A; TaqI (rs731236) G/A; BsmI (rs1544410) T/C; ApaI (rs7975232) A/A | typical (heterozygous in the main)                                 | high - direct chip (weak effect)           |

71 genes/axes used in this report. Colors of this table: red = vulnerability with consequence of avoiding; orange = phenotype that changes dose or shape; yellow = attention or small effect; gray = non-determinable; green = normal/favourable.

### 3. Supplements by category

In each category, the table shows the supplements with relevant gene, ordered by gravity. Below comes the compact list of those that do not have known genetic relevance, with the main pathway (when relevant) and, if it is a evaluated gene, its state in Sérgio F.

#### 3.1 Vitamins and forms — 70 supplements

prevention: 3 · dose adjustment: 24 · Monitoring: 1 · normal: 33 · without known genetic relevance: 9

| Supplement                                                   | Genes (state)                                                                        | Predicted response                              | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                       | Source · evidence                                                                              | Trust                                                                       |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| vitamin A (retinyl acetate)<br><b>Avoid</b>                  | <b>BCO1</b> : slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable    | avoid preformed retinol                         | The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied). | NIH ODS (vitamin A); genetic plan (b.3); isotretinoin label strong (interaction) · weak (BCO1) | BCO1: high<br>PNPLA3/Pi*MZ (liver): moderate                                |
| vitamin A (retinol, retinyl palmitate)<br><b>Avoid</b>       | <b>BCO1</b> : slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable    | avoid preformed retinol                         | The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied). | NIH ODS (vitamin A); genetic plan (b.3); isotretinoin label strong (interaction) · weak (BCO1) | BCO1: high<br>PNPLA3/Pi*MZ (liver): moderate                                |
| vitamin A + D (oil)<br><b>Avoid</b>                          | <b>BCO1</b> : slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable    | avoid preformed retinol                         | The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied). | NIH ODS (vitamin A); genetic plan (b.3); isotretinoin label strong (interaction) · weak (BCO1) | BCO1: high<br>PNPLA3/Pi*MZ (liver): moderate                                |
| acerola (natural vitamin C)<br><b>Adjust dose</b>            | <b>HFE</b> : H63D het<br><b>G6PD</b> : normal<br><b>SLC23A1/2 (vit. C)</b> : typical | drinking between meals (hour)                   | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                                                              | Genetic plan (S16); CPIC G6PD 2022 (analogy) moderate (iron interaction)                       | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high |
| alpha-GPC (glycerophosphocholine)<br><b>Adjust dose</b>      | <b>PEMT</b> : V175M het<br><b>FMO3</b> : typical                                     | prefer citicoline or phosphatidylcholine (form) | PEMT V175M het: a slightly smaller endogenous choline synthesis (rational to choline). Typical FMO3: bitartrate and alpha-GPC generate more TMAO than phosphatidylcholine/citicoline; with genetic vascular priority (9p21, CETP), preferring citicoline 250–500 mg or PC 1–2 g (suggestion, not applied).                                  | Genetic plan (S38); TMAO studies by choline weak-moderate                                      | PEMT: high<br>FMO3: high                                                    |
| calcium ascorbate (buffered vitamin C)<br><b>Adjust dose</b> | <b>HFE</b> : H63D het<br><b>G6PD</b> : normal<br><b>SLC23A1/2 (vit. C)</b> : typical | drinking between meals (hour)                   | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                                                              | Genetic plan (S16); CPIC G6PD 2022 (analogy) moderate (iron interaction)                       | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high |
| sodium ascorbate<br><b>Adjust dose</b>                       | <b>HFE</b> : H63D het<br><b>G6PD</b> : normal<br><b>SLC23A1/2 (vit. C)</b> : typical | drinking between meals (hour)                   | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                                                              | Genetic plan (S16); CPIC G6PD 2022 (analogy) moderate (iron interaction)                       | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high |

| Supplement                                        | Genes (state)                                                                                                                                                                  | Predicted response                              | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                          | Source · evidence                                         | Trust                                                                                  |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------|
| choline (bitartrate)<br><b>Adjust dose</b>        | <b>PEMT</b> : V175M het<br><b>FMO3</b> : typical                                                                                                                               | prefer citicoline or phosphatidylcholine (form) | PEMT V175M het: a slightly smaller endogenous choline synthesis (rational to choline). Typical FMO3: bitartrate and alpha-GPC generate more TMA/TMAO than phosphatidylcholine/citicoline; with genetic vascular priority (9p21, CETP), preferring citicoline 250–500 mg or PC 1–2 g (suggestion, not applied). | Genetic plan (S38); TMAO studies by choline weak-moderate | PEMT: high<br>FMO3: high                                                               |
| B complex<br><b>Adjust dose</b>                   | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                                                                 | choose formula (low B6, without folic acid)     | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                           | Genetic plan (S3, S6); EFSA UL B6 moderate                | PDXK/ALPL/NBPF3 (B6): high<br>MTHFR: high<br>DHFR: moderate                            |
| B complex in gums<br><b>Adjust dose</b>           | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                                                                 | choose formula (low B6, without folic acid)     | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                           | Genetic plan (S3, S6); EFSA UL B6 moderate                | PDXK/ALPL/NBPF3 (B6): high<br>MTHFR: high<br>DHFR: moderate                            |
| Methylated B complex<br><b>Adjust dose</b>        | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                                                                 | choose formula (low B6, without folic acid)     | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                           | Genetic plan (S3, S6); EFSA UL B6 moderate                | PDXK/ALPL/NBPF3 (B6): high<br>MTHFR: high<br>DHFR: moderate                            |
| iron + vitamin C (combined)<br><b>Adjust dose</b> | <b>HFE</b> : H63D het<br><b>TMPRSS6/TF</b> : small effect                                                                                                                      | only with confirmed deficit                     | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                            | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate   | HFE: high<br>TMPRSS6/TF: high                                                          |
| multivitamin 50+<br><b>Adjust dose</b>            | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)<br><b>HFE</b> : H63D het<br><b>BCO1</b> : slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high | choice formula                                  | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 $\leq 5$ mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                | Genetic plan (b.3) moderate                               | MTHFR: high<br>DHFR: moderate<br>HFE: high<br>BCO1: high<br>PDXK/ALPL/NBPF3 (B6): high |
| multivitamin in gums<br><b>Adjust dose</b>        | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)<br><b>HFE</b> : H63D het<br><b>BCO1</b> : slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high | choice formula                                  | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 $\leq 5$ mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                | Genetic plan (b.3) moderate                               | MTHFR: high<br>DHFR: moderate<br>HFE: high<br>BCO1: high<br>PDXK/ALPL/NBPF3 (B6): high |

| Supplement                                          | Genes (state)                                                                                                                                                                  | Predicted response            | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                       | Source · evidence                                                        | Trust                                                                                  |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| female multivitamin<br><b>Adjust dose</b>           | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)<br><b>HFE</b> : H63D het<br><b>BCO1</b> : slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  | Genetic plan (b.3) moderate                                              | MTHFR: high<br>DHFR: moderate<br>HFE: high<br>BCO1: high<br>PDXK/ALPL/NBPF3 (B6): high |
| General multivitamin<br><b>Adjust dose</b>          | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)<br><b>HFE</b> : H63D het<br><b>BCO1</b> : slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  | Genetic plan (b.3) moderate                                              | MTHFR: high<br>DHFR: moderate<br>HFE: high<br>BCO1: high<br>PDXK/ALPL/NBPF3 (B6): high |
| Multivitaminic male<br><b>Adjust dose</b>           | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)<br><b>HFE</b> : H63D het<br><b>BCO1</b> : slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  | Genetic plan (b.3) moderate                                              | MTHFR: high<br>DHFR: moderate<br>HFE: high<br>BCO1: high<br>PDXK/ALPL/NBPF3 (B6): high |
| multivitamin prenatal<br><b>Adjust dose</b>         | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)<br><b>HFE</b> : H63D het<br><b>BCO1</b> : slightly ↓ conversion<br><b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high | choice formula                | Choosing formula without folic acid (DHFR CC → 5-MTHF), without iron (HFE H63D; ferritin target below 100), without preformed retinol (isotretinoin); BCO1 without need) and with B6 ≤ 5 mg (PDXK). Most of the common multivitamins fail 2–3 of these criteria (suggestion, not applied).                                                                                                  | Genetic plan (b.3) moderate                                              | MTHFR: high<br>DHFR: moderate<br>HFE: high<br>BCO1: high<br>PDXK/ALPL/NBPF3 (B6): high |
| ascorbyl palmitate<br><b>Adjust dose</b>            | <b>HFE</b> : H63D het<br><b>G6PD</b> : normal<br><b>SLC23A1/2 (vit. C)</b> : typical                                                                                           | drinking between meals (hour) | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking ≥ 2 h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied).                                                                                                              | Genetic plan (S16); CPIC G6PD 2022 (analogy) moderate (iron interaction) | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high            |
| ★ pyridoxal-5-phosphate (P5P)<br><b>Adjust dose</b> | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high                                                                                                                                     | lowering dose (≤ 5 mg/day)    | Current stack (15 mg) — plan S6: ADJUST (descend). ≤ 5 mg/day or stop (above the upper limit of ingestion, without genetic indication). PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. ≤ 5 mg/day or pause (suggestion, not applied). | Genetic plan (S6); EFSA UL B6 (2023) moderate (safety limit)             | PDXK/ALPL/NBPF3 (B6): high                                                             |

| Supplement                                                               | Genes (state)                                                                     | Predicted response                                      | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                               | Source · evidence                                                           | Trust                                                                       |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| vitamin B6 (pyridoxine hydrochloride)<br><b>Adjust dose</b>              | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high                                         | lowering dose ( $\leq 5$ mg/day)                        | PDXK rs2010795 G/G (high activation) and ALPL/NBPF3 heterozygotes: no genetic need for extra B6. Chronic use of 15 mg passes the upper limit of ingestion (12 mg) and is associated with neuropathy. $\leq 5$ mg/day or pause (suggestion, not applied).                            | Genetic plan (S6); EFSA UL B6 (2023)<br>moderate (safety limit)             | PDXK/ALPL/NBPF3 (B6): high                                                  |
| vitamin C (ascorbic acid)<br><b>Adjust dose</b>                          | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical | drinking between meals (hour)                           | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied). | Genetic plan (S16); CPIC G6PD 2022 (analogy)<br>moderate (iron interaction) | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high |
| vitamin C + zinc<br><b>Adjust dose</b>                                   | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical | drinking between meals (hour)                           | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied). | Genetic plan (S16); CPIC G6PD 2022 (analogy)<br>moderate (iron interaction) | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high |
| vitamin C with metabolites (ascorbate + threonate)<br><b>Adjust dose</b> | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical | drinking between meals (hour)                           | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied). | Genetic plan (S16); CPIC G6PD 2022 (analogy)<br>moderate (iron interaction) | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high |
| ★ vitamin C liposomal<br><b>Adjust dose</b>                              | <b>HFE:</b> H63D het<br><b>G6PD:</b> normal<br><b>SLC23A1/2 (vit. C):</b> typical | <b>drinking between meals (hour)</b>                    | Current stack (500 mg) — plan S16: ADJUST (hour). Always $\geq 2$ h away from meals. HFE H63D het: vitamin C with food increases the absorption of non-heme iron. 500 mg always $\geq 2$ h away from meals. Normal G6PD; Typical SLC23A1/2 (suggestion, not applied).               | Genetic plan (S16)<br>moderate (iron interaction)                           | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high |
| folic acid (synthetic)<br><b>Adjust dose</b>                             | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)                    | prefer 5-MTHF (form)                                    | MTHFR 677CT + 1298AC (activity $\sim 50\text{--}60\%$ ) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).                           | Genetic plan (S3, b.3); MTHFR/DHFR literature<br>moderate                   | MTHFR: high<br>DHFR: moderate                                               |
| folic acid + B12 (combined)<br><b>Adjust dose</b>                        | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)                    | prefer 5-MTHF (form)                                    | MTHFR 677CT + 1298AC (activity $\sim 50\text{--}60\%$ ) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).                           | Genetic plan (S3, b.3); MTHFR/DHFR literature<br>moderate                   | MTHFR: high<br>DHFR: moderate                                               |
| formula AREDS2 (eyes)<br><b>Monitoring</b>                               | <b>CFH/ARMS2:</b> CFH het<br><b>SLC30A8:</b> C/C (usual zinc)                     | without indication; zinc x CFH controversy              | CFH Y402H het and ARMS2 G/G. One group (Awh et al.) suggested that, in this pattern, high dose zinc (80 mg) of AREDS worsens DMI; the reanalysis of AREDS (Chew 2014) did not confirm. Without DMI, the formula is not indicated; lutein/zeaxanthin alone arrives.                  | AREDS/AREDS2; Awh 2013/2015 vs Chew 2014<br>weak (controversy)              | CFH/ARMS2: high<br>SLC30A8: high                                            |
| ★ 5-MTHF (calcium L-methylfolate) ☺<br><b>Normal</b>                     | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)                    | <b>ideal form for the genotype</b>                      | Current stack (800 $\mu\text{g}$ ) — S3 plan: KEEP. Keep 800 $\mu\text{g}$ ; no product with folic acid. 5-MTHF enters directly into the cycle without DHFR or MTHFR. Keep 800 $\mu\text{g}$ (not climbing) (suggestion, not applied).                                              | Genetic plan (S3)<br>moderate                                               | MTHFR: high<br>DHFR: moderate                                               |
| 5-MTHF glucosamine salt ☺<br><b>Normal</b>                               | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)                    | ideal form for the genotype                             | The same logic of the calcium 5-MTHF does not depend on DHFR or MTHFR. Do not add to the 5-MTHF already taken (teto $\sim 800$ $\mu\text{g}$ ).                                                                                                                                     | Genetic plan (S3)<br>moderate                                               | MTHFR: high<br>DHFR: moderate                                               |
| PABA (paraaminobenzoic acid)<br><b>Normal</b>                            | <b>NAT2:</b> slow                                                                 | slow acetylation, with no relevance in nutritional dose | PABA is acetylated by NAT2 (slow $\ast 5/\ast 5$ ); with no known clinical consequence in supplement doses.                                                                                                                                                                         | Pharmacology of PABA<br>weak                                                | NAT2: high                                                                  |

| Supplement                                       | Genes (state)                                                                                                  | Predicted response                                       | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                        | Source · evidence                                                                                   | Trust                                                      |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| adenosylcobalamine<br><b>Normal</b>              | <b>TCN2</b> : 776CG (light)<br><b>FUT2</b> : secretor<br><b>MTR</b> : normal (AA)<br><b>MTRR</b> : normal (AA) | Transport and use of typical B12                         | MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).          | Genetic plan (S4); GWAS of B12 (Hazra 2008) weak                                                    | TCN2: high<br>FUT2: high<br>MTR: high<br>MTRR: high        |
| betacarotene<br><b>Normal</b>                    | <b>BCO1</b> : slightly ↓ conversion                                                                            | slightly reduced conversion, without practical relevance | BCO1 R267S het + rs6564851 G/T: conversion to retinol slightly smaller; A379V absent. It is the preferred form of vitamin A with isotretinoin (regulated conversion). Food or low dose; β-carotene at a high dose does not bring benefits (and is not advised in smokers).   | BCO1 studies (Leung 2009; Lietz 2012); NIH ODS moderate (genotype) · weak (practice)                | BCO1: high                                                 |
| biotin + zinc (hair) ⊕<br><b>Normal</b>          | <b>SLC30A8</b> : C/C (usual zinc)                                                                              | genetically useful zinc                                  | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied). | Kanoni 2011 (CHARGE); genetic plan (S12) moderate                                                   | SLC30A8: high                                              |
| calcifiediol (25-OH-D3)<br><b>Normal</b>         | <b>GC (DBP)</b> : favourable<br><b>VDR</b> : typical<br><b>CYP2R1/DHCR7</b> : typical                          | metabolism of vitamin D favourable/typical               | GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).                       | Genetic plan (S2 + coordination note); GWAS vitamin D (Wang 2010) moderate (genotype) · weak (dose) | GC (DBP): high<br>VDR: high<br>CYP2R1/DHCR7: high          |
| mixed carotenoids<br><b>Normal</b>               | <b>BCO1</b> : slightly ↓ conversion                                                                            | slightly reduced conversion, without practical relevance | BCO1 R267S het + rs6564851 G/T: conversion to retinol slightly smaller; A379V absent. It is the preferred form of vitamin A with isotretinoin (regulated conversion). Food or low dose; β-carotene at a high dose does not bring benefits (and is not advised in smokers).   | BCO1 studies (Leung 2009; Lietz 2012); NIH ODS moderate (genotype) · weak (practice)                | BCO1: high                                                 |
| citicoline (CDP-choline) ⊕<br><b>Normal</b>      | <b>PEMT</b> : V175M het<br><b>FMO3</b> : typical<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                 | Choline preferred form                                   | PEMT V175M het + genetic liver give rational to choline; this form generates little TMAO. 250–500 mg of choline/day in total (suggestion, not applied).                                                                                                                      | Genetic plan (S38) weak-moderate                                                                    | PEMT: high<br>FMO3: high<br>PNPLA3/Pi*MZ (liver): moderate |
| calcium + vitamin D3<br><b>Normal</b>            | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het                                  | typical absorption                                       | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                              | NIH ODS (calcium); genetic plan (S14) weak                                                          | VDR: high<br>GC (DBP): high<br>HFE: high                   |
| natural folate (food extract) ⊕<br><b>Normal</b> | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                                               | normal                                                   | Food folate (leaves, legumes) is always suitable for the genotype.                                                                                                                                                                                                           | Genetic plan (e.6) moderate                                                                         | MTHFR: high<br>DHFR: moderate                              |
| phosphatidylcholine ⊕<br><b>Normal</b>           | <b>PEMT</b> : V175M het<br><b>FMO3</b> : typical<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                 | Choline preferred form                                   | PEMT V175M het + genetic liver give rational to choline; this form generates little TMAO. 250–500 mg of choline/day in total (suggestion, not applied).                                                                                                                      | Genetic plan (S38) weak-moderate                                                                    | PEMT: high<br>FMO3: high<br>PNPLA3/Pi*MZ (liver): moderate |
| hydroxocobalamin<br><b>Normal</b>                | <b>TCN2</b> : 776CG (light)<br><b>FUT2</b> : secretor<br><b>MTR</b> : normal (AA)<br><b>MTRR</b> : normal (AA) | Transport and use of typical B12                         | MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).          | Genetic plan (S4); GWAS of B12 (Hazra 2008) weak                                                    | TCN2: high<br>FUT2: high<br>MTR: high<br>MTRR: high        |

| Supplement                                                                                                                       | Genes (state)                                                                                                  | Predicted response                                | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                             | Source · evidence                                                                                   | Trust                                                      |
|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| soybean lecithin/<br>gyrassol <br><b>Normal</b> | <b>PEMT</b> : V175M het<br><b>FMO3</b> : typical<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                 | Choline preferred form                            | PEMT V175M het + genetic liver give rational to choline; this form generates little TMAO. 250–500 mg of choline/day in total (suggestion, not applied).                                                                                                                                                                                                                                           | Genetic plan (S38)<br>weak–moderate                                                                 | PEMT: high<br>FMO3: high<br>PNPLA3/Pi*MZ (liver): moderate |
| ★ <b>methylcobalamine</b><br><b>Normal</b>                                                                                       | <b>TCN2</b> : 776CG (light)<br><b>FUT2</b> : secretor<br><b>MTR</b> : normal (AA)<br><b>MTRR</b> : normal (AA) | <b>Transport and use of typical B12</b>           | Current stack (1000 µg (unspecified form)) — S4 plan: KEEP. 1000 µg (500 µg if acne appears); methyl- or adenosyl-cobalamine. MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears). | Genetic plan (S4); GWAS of B12 (Hazra 2008)<br>weak                                                 | TCN2: high<br>FUT2: high<br>MTR: high<br>MTRR: high        |
| ★ <b>riboflavin-5-phosphate (R5P)</b><br><b>Normal</b>                                                                           | <b>MTHFR</b> : 677CT/<br>1298AC                                                                                | <b>modest benefit in 677CT</b>                    | Current stack (25 mg) — plan S5: ADJUST (optimal). 5–10 mg arrives; maintaining 25 mg is at low risk. Riboflavin stabilizes thermolabile MTHFR; the benefit is clear in 677TT and only modest in heterozygotes. Low dose (5–10 mg) arrives; 25 mg is safe.                                                                                                                                        | Riboflavin × MTHFR 677 (McNulty 2006) trials moderate (TT) · weak (CT)                              | MTHFR: high                                                |
| mixed tocopherols<br><b>Normal</b>                                                                                               | <b>CYP4F2</b> : normal                                                                                         | normal metabolism                                 | CYP4F2 V433M C/C (normal): vitamin E ω-hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.                                                                                                                                                                                                                                                                             | CYP4F2 × tocopherol (Bardowell 2012) studies weak                                                   | CYP4F2: high                                               |
| tocotrienols<br><b>Normal</b>                                                                                                    | <b>CYP4F2</b> : normal                                                                                         | normal metabolism                                 | CYP4F2 V433M C/C (normal): vitamin E ω-hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.                                                                                                                                                                                                                                                                             | CYP4F2 × tocopherol (Bardowell 2012) studies weak                                                   | CYP4F2: high                                               |
| vitamin B12 (cyanocobalamine)<br><b>Normal</b>                                                                                   | <b>TCN2</b> : 776CG (light)<br><b>FUT2</b> : secretor<br><b>MTR</b> : normal (AA)<br><b>MTRR</b> : normal (AA) | Transport and use of typical B12                  | MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).                                                                                                                               | Genetic plan (S4); GWAS of B12 (Hazra 2008)<br>weak                                                 | TCN2: high<br>FUT2: high<br>MTR: high<br>MTRR: high        |
| vitamin B12 sublingual/<br>spray<br><b>Normal</b>                                                                                | <b>TCN2</b> : 776CG (light)<br><b>FUT2</b> : secretor<br><b>MTR</b> : normal (AA)<br><b>MTRR</b> : normal (AA) | Transport and use of typical B12                  | MTR A/A and MTRR A/A (common); TCN2 776 C/G and TCN1 G/A with small effect FUT2 secretor. No genetic requirement of high dose or special; methyl/adenosyl are options, cyanocobalamin also works (relevant cyanide wear). 1000 µg arrives (500 µg if acne appears).                                                                                                                               | Genetic plan (S4); GWAS of B12 (Hazra 2008)<br>weak                                                 | TCN2: high<br>FUT2: high<br>MTR: high<br>MTRR: high        |
| vitamin B2 (riboflavin)<br><b>Normal</b>                                                                                         | <b>MTHFR</b> : 677CT/<br>1298AC                                                                                | modest benefit in 677CT                           | Riboflavin stabilizes thermolabile MTHFR; the benefit is clear in 677TT and only modest in heterozygotes. Low dose (5–10 mg) arrives; 25 mg is safe.                                                                                                                                                                                                                                              | Riboflavin × MTHFR 677 (McNulty 2006) trials moderate (TT) · weak (CT)                              | MTHFR: high                                                |
| vitamin D2 (ergocalciferol)<br><b>Normal</b>                                                                                     | <b>GC (DBP)</b> : favourable<br><b>CYP2R1/DHCR7</b> : typical                                                  | typical (D3 preferable)                           | Favourable genetics; D2 binds less to DBP and rises less to 25(OH)D than D3 (form question, not genes).                                                                                                                                                                                                                                                                                           | NIH ODS (vitamin D) moderate                                                                        | GC (DBP): high<br>CYP2R1/DHCR7: high                       |
| ★ <b>vitamin D3 (cholecalciferol)</b><br><b>Normal</b>                                                                           | <b>GC (DBP)</b> : favourable<br><b>VDR</b> : typical<br><b>CYP2R1/DHCR7</b> : typical                          | <b>metabolism of vitamin D favourable/typical</b> | Current stack (10 000 IU) — plan S2: KEEP (coordination). Maintain 10 000 IU (Coordination note: measurement-based decision). GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).              | Genetic plan (S2 + coordination note); GWAS vitamin D (Wang 2010) moderate (genotype) · weak (dose) | GC (DBP): high<br>VDR: high<br>CYP2R1/DHCR7: high          |

| Supplement                                                 | Genes (state)                                                                      | Predicted response                            | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                            | Source · evidence                                                                                         | Trust                                             |
|------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| vitamin D3 + K2<br><b>Normal</b>                           | <b>GC (DBP):</b> favourable<br><b>VDR:</b> typical<br><b>CYP2R1/DHCR7:</b> typical | metabolism of vitamin D<br>favourable/typical | GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).                                                           | Genetic plan (S2 + coordination note);<br>GWAS vitamin D (Wang 2010)<br>moderate (genotype) · weak (dose) | GC (DBP): high<br>VDR: high<br>CYP2R1/DHCR7: high |
| vitamin D3 weekly high dose (50 000 IU)<br><b>Normal</b>   | <b>GC (DBP):</b> favourable<br><b>VDR:</b> typical<br><b>CYP2R1/DHCR7:</b> typical | metabolism of vitamin D<br>favourable/typical | GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).                                                           | Genetic plan (S2 + coordination note);<br>GWAS vitamin D (Wang 2010)<br>moderate (genotype) · weak (dose) | GC (DBP): high<br>VDR: high<br>CYP2R1/DHCR7: high |
| vitamin D3 vegan (lichen)<br><b>Normal</b>                 | <b>GC (DBP):</b> favourable<br><b>VDR:</b> typical<br><b>CYP2R1/DHCR7:</b> typical | metabolism of vitamin D<br>favourable/typical | GC without Gc2 (DBP favourable), typical CYP2R1/DHCR7, typical VDR: genetics does not predict increased need. The dose is defined by the analysis; by coordination, it remains 10 000 IU of D3 (measurement-based decision) (suggestion, not applied).                                                           | Genetic plan (S2 + coordination note);<br>GWAS vitamin D (Wang 2010)<br>moderate (genotype) · weak (dose) | GC (DBP): high<br>VDR: high<br>CYP2R1/DHCR7: high |
| vitamin E (tocopheryl acetate)<br><b>Normal</b>            | <b>CYP4F2:</b> normal                                                              | normal metabolism                             | CYP4F2 V433M C/C (normal): vitamin E $\omega$ -hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.                                                                                                                                                                                    | CYP4F2 × tocopherol (Bardowell 2012) studies weak                                                         | CYP4F2: high                                      |
| vitamin E (d-alpha-tocopherol)<br><b>Normal</b>            | <b>CYP4F2:</b> normal                                                              | normal metabolism                             | CYP4F2 V433M C/C (normal): vitamin E $\omega$ -hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.                                                                                                                                                                                    | CYP4F2 × tocopherol (Bardowell 2012) studies weak                                                         | CYP4F2: high                                      |
| synthetic vitamin E (dl-alpha-tocopherol)<br><b>Normal</b> | <b>CYP4F2:</b> normal                                                              | normal metabolism                             | CYP4F2 V433M C/C (normal): vitamin E $\omega$ -hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.                                                                                                                                                                                    | CYP4F2 × tocopherol (Bardowell 2012) studies weak                                                         | CYP4F2: high                                      |
| vitamin K1 (phyloquinone)<br><b>Normal</b>                 | <b>VKORC1:</b> TT (sensitive)<br><b>CYP4F2:</b> normal<br><b>GGCX:</b> C/T         | normal (relevant only with coumarins)         | VKORC1 –1639 T/T + CYP4F2 C/C + GGCX C/T: no effect in the absence of anticoagulant. If you ever take warfarin/acenocoumarol, keep K intake stable and warn the doctor (VKORC1 TT makes INR very sensitive).                                                                                                     | CPIC warfarin 2017 (context); genetic plan (S2)<br>strong (coumarinic context)                            | VKORC1: high<br>CYP4F2: high<br>GGCX: high        |
| vitamin K2 MK-4<br><b>Normal</b>                           | <b>VKORC1:</b> TT (sensitive)<br><b>CYP4F2:</b> normal<br><b>GGCX:</b> C/T         | normal (relevant only with coumarins)         | VKORC1 –1639 T/T + CYP4F2 C/C + GGCX C/T: no effect in the absence of anticoagulant. If you ever take warfarin/acenocoumarol, keep K intake stable and warn the doctor (VKORC1 TT makes INR very sensitive).                                                                                                     | CPIC warfarin 2017 (context); genetic plan (S2)<br>strong (coumarinic context)                            | VKORC1: high<br>CYP4F2: high<br>GGCX: high        |
| ★ vitamin K2 MK-7<br><b>Normal</b>                         | <b>VKORC1:</b> TT (sensitive)<br><b>CYP4F2:</b> normal<br><b>GGCX:</b> C/T         | <b>normal (relevant only with coumarins)</b>  | Current stack (240 µg) — S2 plan: KEEP. The same; it only weighs if you take coumarins (VKORC1 TT). VKORC1 –1639 T/T + CYP4F2 C/C + GGCX C/T: no effect in the absence of anticoagulant. If you ever take warfarin/acenocoumarol, keep K intake stable and warn the doctor (VKORC1 TT makes INR very sensitive). | CPIC warfarin 2017 (context); genetic plan (S2)<br>strong (coumarinic context)                            | VKORC1: high<br>CYP4F2: high<br>GGCX: high        |
| folinic acid (calcium folinate)<br><b>Normal</b>           | <b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC)                     | avoids DHFR                                   | The folinic jumps to DHFR, but still needs the MTHFR to give 5-MTHF; with the 5-MTHF already in the stack is redundant.                                                                                                                                                                                          | Biochemistry of folate cycle weak                                                                         | MTHFR: high<br>DHFR: moderate                     |

No known genetic relevance (9):

benfotiamine (SLC19A2/3)  
biotin (B7) (renal excretion)  
D-qui-ro-inositol

inositol (mio-inositol)  
pantetin (CoA)  
sulbutiamine (liposoluble)

TTFD (fursultiamine) (liposoluble)  
vitamin B1 (tiamine HCl) (SLC19A2/3)  
pantotenic acid (B5) (CoA)

## 3.2 Minerals and trace elements — 74 supplements

dose adjustment: 10 · Monitoring: 6 · normal: 41 · without known genetic relevance: 17

| Supplement                                             | Genes (state)                                                                                                                 | Predicted response                          | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Source · evidence                                       | Trust                                                            |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------|
| ZMA (zinc, magnesium, B6)<br><b>Adjust dose</b>        | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                                                                                                                                                                                                                         | Genetic plan (S3, S6); EFSA UL B6 moderate              | PDXK/ALPL/NBPF3 (B6): high<br>MTHFR: high<br>DHFR: moderate      |
| iron bisglycinate<br><b>Adjust dose</b>                | <b>HFE</b> : H63D het<br><b>TMPRSS6/TF</b> : small effect                                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                          | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate | HFE: high<br>TMPRSS6/TF: high                                    |
| iron-polysaccharide complex<br><b>Adjust dose</b>      | <b>HFE</b> : H63D het<br><b>TMPRSS6/TF</b> : small effect                                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                          | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate | HFE: high<br>TMPRSS6/TF: high                                    |
| iron carbonyl<br><b>Adjust dose</b>                    | <b>HFE</b> : H63D het<br><b>TMPRSS6/TF</b> : small effect                                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                          | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate | HFE: high<br>TMPRSS6/TF: high                                    |
| iron heme (polypeptide)<br><b>Adjust dose</b>          | <b>HFE</b> : H63D het<br><b>TMPRSS6/TF</b> : small effect                                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                          | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate | HFE: high<br>TMPRSS6/TF: high                                    |
| liposomal / sucrosomal iron<br><b>Adjust dose</b>      | <b>HFE</b> : H63D het<br><b>TMPRSS6/TF</b> : small effect                                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                          | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate | HFE: high<br>TMPRSS6/TF: high                                    |
| ferrous fumarate<br><b>Adjust dose</b>                 | <b>HFE</b> : H63D het<br><b>TMPRSS6/TF</b> : small effect                                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                          | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate | HFE: high<br>TMPRSS6/TF: high                                    |
| ferrous gluconate<br><b>Adjust dose</b>                | <b>HFE</b> : H63D het<br><b>TMPRSS6/TF</b> : small effect                                                                     | only with confirmed deficit                 | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                                                                                                                                                                                                                                          | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate | HFE: high<br>TMPRSS6/TF: high                                    |
| ★ trace minerals (multineural) ⓘ<br><b>Adjust dose</b> | <b>SLC30A8</b> : C/C (usual zinc)<br><b>GPX1</b> : Pro198Leu het<br><b>SLC39A8</b> : normal<br><b>AOC1 (DAO)</b> : slightly ↓ | <b>up zinc and copper</b>                   | Current stack (Zn 11, Cu 0.9, I 0.05, Se 0.055, Mn 2.3, Cr 0.035, Mo 0.045, B 1 mg) — plan S12: ADJUST (Zn and Cu). Total zinc 20–25 mg + total copper 1.5–2 mg; remaining equal. SLC30A8 C/C + rs11558471 A/A (genetically useful zinc); GPX1 Pro198Leu (selenium already suitable, without Brazil nuts); SLC39A8 C/C (manganese without adjustment); DAO is a copper enzyme (AOC1 het). Total zinc 20–25 mg + total copper 1.5–2 mg, $\geq 2$ h away from NAC, Ca-AMG and fiber (suggestion, not applied). | Genetic plan (S12) moderate                             | SLC30A8: high<br>GPX1: high<br>SLC39A8: high<br>AOC1 (DAO): high |

| Supplement                                                          | Genes (state)                                           | Predicted response          | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                               | Source · evidence                                       | Trust                              |
|---------------------------------------------------------------------|---------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|------------------------------------|
| ferrous sulfate<br><b>Adjust dose</b>                               | <b>HFE:</b> H63D het<br><b>TMPRSS6/TF:</b> small effect | only with confirmed deficit | HFE H63D het (C282Y and S65C absent): low risk of overload, but routine iron has no genetic indication and the target is not to accumulate iron (target ferritin below 100). Use only if a doctor confirms a deficit, for a limited time (suggestion, not applied).                 | Genetic plan (b.3); HFE guidelines (EASL 2022) moderate | HFE: high<br>TMPRSS6/TF: high      |
| potassium bicarbonate<br><b>Monitoring</b>                          | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)  | potassium only for food     | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied). | Genetic plan (b.3, f.12) moderate (interaction)         | Salt (AGT/ADD1/GNB3/CYP11B2): high |
| potassium citrate<br><b>Monitoring</b>                              | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)  | potassium only for food     | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied). | Genetic plan (b.3, f.12) moderate (interaction)         | Salt (AGT/ADD1/GNB3/CYP11B2): high |
| potassium chloride<br><b>Monitoring</b>                             | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)  | potassium only for food     | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied). | Genetic plan (b.3, f.12) moderate (interaction)         | Salt (AGT/ADD1/GNB3/CYP11B2): high |
| salt tablets (sodium chloride)<br><b>Monitoring</b>                 | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)  | avoid routine use           | Genetic sensitivity to salt: salt tablets only for intense and prolonged sweat.                                                                                                                                                                                                     | Genetic plan (e.3) moderate                             | Salt (AGT/ADD1/GNB3/CYP11B2): high |
| potassium gluconate<br><b>Monitoring</b>                            | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)  | potassium only for food     | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied). | Genetic plan (b.3, f.12) moderate (interaction)         | Salt (AGT/ADD1/GNB3/CYP11B2): high |
| salt with potassium chloride (salt substitute)<br><b>Monitoring</b> | <b>Sal (AGT/ADD1/GNB3/CYP11B2):</b> sensitive (modest)  | potassium only for food     | Genetic sensitivity to salt (AGT M235T, ADD1 G460W, GNB3 825T, CYP11B2 –344) is rational for more potassium, but the telmisartan of the stack retains potassium: supplement or salt with KCl only with medical control. To prefer vegetables and legumes (suggestion, not applied). | Genetic plan (b.3, f.12) moderate (interaction)         | Salt (AGT/ADD1/GNB3/CYP11B2): high |
| Magnesium aspartate<br><b>Normal</b>                                | <b>TRPM6:</b> typical                                   | typical absorption          | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                                 | TRPM6 (Song 2009) studies; genetic plan (S13) weak      | TRPM6: high                        |
| Magnesium l-threonate<br><b>Normal</b>                              | <b>TRPM6:</b> typical                                   | typical absorption          | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                                 | TRPM6 (Song 2009) studies; genetic plan (S13) weak      | TRPM6: high                        |
| zinc acetate (chipelets) ☯<br><b>Normal</b>                         | <b>SLC30A8:</b> C/C (usual zinc)                        | genetically useful zinc     | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).        | Kanoni 2011 (CHARGE); genetic plan (S12) moderate       | SLC30A8: high                      |
| ★ magnesium acetyl-taurate<br><b>Normal</b>                         | <b>TRPM6:</b> typical                                   | <b>typical absorption</b>   | Current stack (750 mg) — plan S13: KEEP. The same. TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                              | TRPM6 (Song 2009) studies; genetic plan (S13) weak      | TRPM6: high                        |

| Supplement                                     | Genes (state)                                                                 | Predicted response      | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                        | Source · evidence                                        | Trust                                    |
|------------------------------------------------|-------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------|
| lithium aspartate<br><b>Normal</b>             | <b>APOE</b> : ε3/ε3                                                           | without a safety gene   | No PGx for lithium in microdose; APOE ε3/ε3; Alzheimer's PGS of 39 variants p91 (genomic p52 to p85; genomic consensus p75): weak rational. With telmisartan (sodium lithium) and dapagliflozin (low-o): pause in days of vomiting/diarrhea.                                 | Genetic plan (S15, f.19)<br>weak                         | APOE: high                               |
| copper bisglycinate<br><b>Normal</b>           | <b>SLC30A8</b> : C/C (usual zinc)<br><b>AOC1 (DAO)</b> : slightly ↓           | useful if zinc rises    | Necessary to maintain Zn:Cu ≈ 10–15:1 if zinc rises (SLC30A8); DAO is a copper enzyme (AOC1 het). Total 1.5–2 mg/day (suggestion, not applied).                                                                                                                              | Genetic plan (S12)<br>moderate                           | SLC30A8: high<br>AOC1 (DAO): high        |
| zinc bisglycinate ☉<br><b>Normal</b>           | <b>SLC30A8</b> : C/C (usual zinc)                                             | genetically useful zinc | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied). | Kanoni 2011 (CHARGE);<br>genetic plan (S12)<br>moderate  | SLC30A8: high                            |
| calcium carbonate<br><b>Normal</b>             | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het | typical absorption      | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                              | NIH ODS (calcium);<br>genetic plan (S14)<br>weak         | VDR: high<br>GC (DBP): high<br>HFE: high |
| calcium citrate<br><b>Normal</b>               | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het | typical absorption      | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                              | NIH ODS (calcium);<br>genetic plan (S14)<br>weak         | VDR: high<br>GC (DBP): high<br>HFE: high |
| magnesium citrate<br><b>Normal</b>             | <b>TRPM6</b> : typical                                                        | typical absorption      | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                          | TRPM6 (Song 2009)<br>studies; genetic plan (S13)<br>weak | TRPM6: high                              |
| zinc citrate ☉<br><b>Normal</b>                | <b>SLC30A8</b> : C/C (usual zinc)                                             | genetically useful zinc | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied). | Kanoni 2011 (CHARGE);<br>genetic plan (S12)<br>moderate  | SLC30A8: high                            |
| magnesium chloride<br><b>Normal</b>            | <b>TRPM6</b> : typical                                                        | typical absorption      | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                          | TRPM6 (Song 2009)<br>studies; genetic plan (S13)<br>weak | TRPM6: high                              |
| calcium + magnesium<br><b>Normal</b>           | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het | typical absorption      | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                              | NIH ODS (calcium);<br>genetic plan (S14)<br>weak         | VDR: high<br>GC (DBP): high<br>HFE: high |
| Algal calcium (Lithothamnion)<br><b>Normal</b> | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het | typical absorption      | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                              | NIH ODS (calcium);<br>genetic plan (S14)<br>weak         | VDR: high<br>GC (DBP): high<br>HFE: high |
| choral calcium<br><b>Normal</b>                | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het | typical absorption      | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                              | NIH ODS (calcium);<br>genetic plan (S14)<br>weak         | VDR: high<br>GC (DBP): high<br>HFE: high |
| calcium-magnesium-zinc<br><b>Normal</b>        | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het | typical absorption      | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                              | NIH ODS (calcium);<br>genetic plan (S14)<br>weak         | VDR: high<br>GC (DBP): high<br>HFE: high |

| Supplement                                           | Genes (state)                                                              | Predicted response           | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                           | Source · evidence                                  | Trust                                    |
|------------------------------------------------------|----------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------|
| ★ <b>magnesium glycinate</b><br><b>Normal</b>        | <b>TRPM6: typical</b>                                                      | <b>typical absorption</b>    | Current stack (500 mg) — plan S13: KEEP. The same. TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                                                          | TRPM6 (Song 2009) studies; genetic plan (S13) weak | TRPM6: high                              |
| copper gluconate<br><b>Normal</b>                    | <b>SLC30A8:</b> C/C (usual zinc)<br><b>AOC1 (DAO):</b> slightly ↓          | useful if zinc rises         | Necessary to maintain Zn:Cu ≈ 10–15:1 if zinc rises (SLC30A8); DAO is a copper enzyme (AOC1 het). Total 1.5–2 mg/day (suggestion, not applied).                                                                                                                                                                                                                 | Genetic plan (S12) moderate                        | SLC30A8: high<br>AOC1 (DAO): high        |
| zinc gluconate ☺<br><b>Normal</b>                    | <b>SLC30A8:</b> C/C (usual zinc)                                           | genetically useful zinc      | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).                                                                                    | Kanoni 2011 (CHARGE); genetic plan (S12) moderate  | SLC30A8: high                            |
| hydroxyapatite (MCHC)<br><b>Normal</b>               | <b>VDR:</b> typical<br><b>GC (DBP):</b> favourable<br><b>HFE:</b> H63D het | typical absorption           | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                                                                                                                 | NIH ODS (calcium); genetic plan (S14) weak         | VDR: high<br>GC (DBP): high<br>HFE: high |
| iodine + selenium (thyroid)<br><b>Normal</b>         | <b>GPX1:</b> Pro198Leu het                                                 | Nutritional dose             | GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.                                                                                                                                                                   | SELECT; genetic plan (S12) weak                    | GPX1: high                               |
| calcium lactate/ gluconate<br><b>Normal</b>          | <b>VDR:</b> typical<br><b>GC (DBP):</b> favourable<br><b>HFE:</b> H63D het | typical absorption           | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                                                                                                                 | NIH ODS (calcium); genetic plan (S14) weak         | VDR: high<br>GC (DBP): high<br>HFE: high |
| selenium yeast<br><b>Normal</b>                      | <b>GPX1:</b> Pro198Leu het                                                 | Nutritional dose             | GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.                                                                                                                                                                   | SELECT; genetic plan (S12) weak                    | GPX1: high                               |
| magnesium + zinc<br><b>Normal</b>                    | <b>TRPM6:</b> typical                                                      | typical absorption           | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                                                                                                             | TRPM6 (Song 2009) studies; genetic plan (S13) weak | TRPM6: high                              |
| magnesium malate<br><b>Normal</b>                    | <b>TRPM6:</b> typical                                                      | typical absorption           | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                                                                                                             | TRPM6 (Song 2009) studies; genetic plan (S13) weak | TRPM6: high                              |
| manganese (bisglycine/ gluconate)<br><b>Normal</b>   | <b>SLC39A8:</b> normal                                                     | Normal transport             | SLC39A8 A391T absent (C/C): manganese without adjustment; not exceeding 11 mg/day.                                                                                                                                                                                                                                                                              | SLC39A8 studies (Park 2015) moderate               | SLC39A8: high                            |
| zinc monomethionine ☺<br><b>Normal</b>               | <b>SLC30A8:</b> C/C (usual zinc)                                           | genetically useful zinc      | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied).                                                                                    | Kanoni 2011 (CHARGE); genetic plan (S12) moderate  | SLC30A8: high                            |
| ★ <b>lithium orotate (low dose)</b><br><b>Normal</b> | <b>APOE:</b> ε3/ε3                                                         | <b>without a safety gene</b> | Current stack (5 mg) — plan S15: KEEP. Similarly; pause in days of vomiting/ diarrhea (telmisartan/dapagliflozin). No PGx for lithium in microdose; APOE ε3/ε3; Alzheimer's PGS of 39 variants p91 (genomic p52 to p85; genomic consensus p75): weak rational. With telmisartan (sodium lithium) and dapagliflozin (low-o): pause in days of vomiting/diarrhea. | Genetic plan (S15, f.19) weak                      | APOE: high                               |

| Supplement                                      | Genes (state)                                                       | Predicted response      | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                        | Source · evidence                                  | Trust                                    |
|-------------------------------------------------|---------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------|
| magnesium orotato<br><b>Normal</b>              | <b>TRPM6</b> : typical                                              | typical absorption      | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                          | TRPM6 (Song 2009) studies; genetic plan (S13) weak | TRPM6: high                              |
| zinc picolinate ☺<br><b>Normal</b>              | <b>SLC30A8</b> : C/C (usual zinc)                                   | genetically useful zinc | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied). | Kanoni 2011 (CHARGE); genetic plan (S12) moderate  | SLC30A8: high                            |
| sodium selenate<br><b>Normal</b>                | <b>GPX1</b> : Pro198Leu het                                         | Nutritional dose        | GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.                                                                                | SELECT; genetic plan (S12) weak                    | GPX1: high                               |
| sodium selenite<br><b>Normal</b>                | <b>GPX1</b> : Pro198Leu het                                         | Nutritional dose        | GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.                                                                                | SELECT; genetic plan (S12) weak                    | GPX1: high                               |
| selenomethionine<br><b>Normal</b>               | <b>GPX1</b> : Pro198Leu het                                         | Nutritional dose        | GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.                                                                                | SELECT; genetic plan (S12) weak                    | GPX1: high                               |
| copper sulfate<br><b>Normal</b>                 | <b>SLC30A8</b> : C/C (usual zinc)<br><b>AOC1 (DAO)</b> : slightly ↓ | useful if zinc rises    | Necessary to maintain Zn:Cu ≈ 10–15:1 if zinc rises (SLC30A8); DAO is a copper enzyme (AOC1 het). Total 1.5–2 mg/day (suggestion, not applied).                                                                                                                              | Genetic plan (S12) moderate                        | SLC30A8: high<br>AOC1 (DAO): high        |
| magnesium sulfate (epsom salt)<br><b>Normal</b> | <b>TRPM6</b> : typical                                              | typical absorption      | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                          | TRPM6 (Song 2009) studies; genetic plan (S13) weak | TRPM6: high                              |
| zinc sulfate ☺<br><b>Normal</b>                 | <b>SLC30A8</b> : C/C (usual zinc)                                   | genetically useful zinc | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied). | Kanoni 2011 (CHARGE); genetic plan (S12) moderate  | SLC30A8: high                            |
| magnesium taurate<br><b>Normal</b>              | <b>TRPM6</b> : typical                                              | typical absorption      | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                          | TRPM6 (Song 2009) studies; genetic plan (S13) weak | TRPM6: high                              |
| zinc + copper (combined) ☺<br><b>Normal</b>     | <b>SLC30A8</b> : C/C (usual zinc)                                   | genetically useful zinc | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied). | Kanoni 2011 (CHARGE); genetic plan (S12) moderate  | SLC30A8: high                            |
| zinc carnosine (polaprezinc) ☺<br><b>Normal</b> | <b>SLC30A8</b> : C/C (usual zinc)<br><b>CNDP1</b> : indeterminate   | useful zinc             | Zinc (SLC30A8 C/C) with local carnosine in the stomach; carnosinase (CNDP1) is not evaluated, but here the effect is topic.                                                                                                                                                  | Kanoni 2011; polaprezinc trials weak               | SLC30A8: high<br>CNDP1: non-determinable |
| magnesium oxide<br><b>Normal</b>                | <b>TRPM6</b> : typical                                              | typical absorption      | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                          | TRPM6 (Song 2009) studies; genetic plan (S13) weak | TRPM6: high                              |

| Supplement             | Genes (state)             | Predicted response      | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                        | Source · evidence                                 | Trust         |
|------------------------|---------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------|
| zinc oxide ⓘ<br>Normal | SLC30A8: C/C (usual zinc) | genetically useful zinc | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20–25 mg/day with 1.5–2 mg of copper (Zn:Cu ≈ 10–15:1; DAO depends on copper, AOC1 het) (suggestion, not applied). | Kanoni 2011 (CHARGE); genetic plan (S12) moderate | SLC30A8: high |

No known genetic relevance (17):

|                                |                                  |                                  |
|--------------------------------|----------------------------------|----------------------------------|
| boron (citrate/glycinate)      | ionic mineral droplets (various) | colloidal silver                 |
| strontium citrate (bone)       | potassium iodide (thyroid)       | Lugol solution (thyroid)         |
| chromium GTF (light)           | algal iodine (kelp) (thyroid)    | silica / orthosilic acid (renal) |
| organic sulfur (MSM)           | iodine (thyroid)                 | sodium/electrolytes (renal)      |
| Phosphorus (phosphate) (renal) | molybdenum (glycinate)           | vanadium (vanadyl sulfate)       |
| organic germanium (Ge-132)     | chromium picolinate              |                                  |

### 3.3 Amino acids, proteins and derivatives — 80 supplements

dose adjustment: 2 · Monitoring: 6 · indeterminate: 2 · normal: 30 · without known genetic relevance: 40

| Supplement                                                                    | Genes (state)                                                                | Predicted response                 | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                            | Source · evidence                                | Trust                                       |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------|
| ★ <b>TMG</b><br>(trimethylglycine, betaine anhydrous) ⓘ<br><b>Adjust dose</b> | <b>BHMT: R239Q het</b><br><b>GNMT: TT</b><br><b>MTHFR: 677CT/1298AC</b>      | <b>adjust dose (plan)</b>          | Current stack (1500 mg) — plan S7: ADJUST (subir). Up to 3 g/day in 2 tomas; do not exceed 3 g (LDL). BHMT R239Q het, GNMT TT and MTHFR compound: the BHMT pathway compensates for folate. 3 g/day (≥ 4 g up to LDL) (suggestion, not applied).                                                  | Genetic plan (S7) moderate                       | BHMT: high<br>GNMT: high<br>MTHFR: high     |
| ★ <b>glycine</b> ⓘ<br><b>Adjust dose</b>                                      | <b>CPS1: low glycine</b><br><b>GSTM1: probable null</b><br><b>GNMT: TT</b>   | <b>adjust dose (plan)</b>          | Current stack (3 g) — plan S8: ADJUST (subir). Up to 5 g/day (3 g at bedtime + 2 g in the morning). CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).             | GWAS glycine (CPS1); genetic plan (S8) moderate  | CPS1: high<br>GSTM1: moderate<br>GNMT: high |
| BCAA (leucine, isoleucine, valine)<br><b>Monitoring</b>                       | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                         | the genetic plan advises to add    | High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).                                                                                                                | Genetic plan (b.3) weak                          | TCF7L2/FTO/GCKR: high                       |
| L-isoleucine<br><b>Monitoring</b>                                             | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                         | the genetic plan advises to add    | High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).                                                                                                                | Genetic plan (b.3) weak                          | TCF7L2/FTO/GCKR: high                       |
| L-leucine<br><b>Monitoring</b>                                                | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                         | the genetic plan advises to add    | High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).                                                                                                                | Genetic plan (b.3) weak                          | TCF7L2/FTO/GCKR: high                       |
| L-methionine<br><b>Monitoring</b>                                             | <b>COMT: Val/Val (high)</b><br><b>GNMT: TT</b><br><b>MTHFR: 677CT/1298AC</b> | avoiding isolated supplement       | Extra methionine turns SAM and then homocysteine; with compound MTHFR, GNMT TT and COMT Val/Val (uses up SAM) the genetic score of homocysteine is at the 92nd percentile. No indication.                                                                                                        | Genetic plan (f.2) weak                          | COMT: high<br>GNMT: high<br>MTHFR: high     |
| L-valine<br><b>Monitoring</b>                                                 | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                         | the genetic plan advises to add    | High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).                                                                                                                | Genetic plan (b.3) weak                          | TCF7L2/FTO/GCKR: high                       |
| SAMe (S-adenosylmethionine)<br><b>Monitoring</b>                              | <b>COMT: Val/Val (high)</b><br><b>GNMT: TT</b><br><b>MTHFR: 677CT/1298AC</b> | the genetic plan advises to add    | COMT Val/Val + GNMT TT: plus SAM tends to be transformed into homocysteine (genetic score p92). For methylation, prefer TMG + 5-MTHF + creatine (suggestion, not applied).                                                                                                                       | Genetic plan (b.3) weak                          | COMT: high<br>GNMT: high<br>MTHFR: high     |
| ★ <b>L-carnosine</b><br><b>Undetermined</b>                                   | <b>CNDP1: indeterminate</b>                                                  | <b>indeterminate (carnosinase)</b> | Current stack (2 g) — plan S21: KEEP. 1 g × 2 (non-evaluable carnosinase CNDP1). Serum carnosinase (CNDP1) is regulated by a repetition (CTG) <sub>n</sub> that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2). | CNDP1 Literature (Janssen 2005) not determinable | CNDP1: non-determinable                     |
| anserine<br><b>Undetermined</b>                                               | <b>CNDP1: indeterminate</b>                                                  | indeterminate (carnosinase)        | Serum carnosinase (CNDP1) is regulated by a repetition (CTG) <sub>n</sub> that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).                                                                                  | CNDP1 Literature (Janssen 2005) not determinable | CNDP1: non-determinable                     |
| AAKG (alpha-ketoglutarate arginine)<br><b>Normal</b>                          | <b>NOS3: –786CT + 298GT</b>                                                  | rational NO (small effect)         | NOS3 –786 T/C + Glu298Asp G/T: a slightly smaller production of NO. Oral arginine has low bioavailability; with tadalafil the vasodilator effect is added (mild).                                                                                                                                | NOS3 studies; NIH ODS (arginine) weak            | NOS3: high                                  |
| DL-phenylalanine (DLPA)<br><b>Normal</b>                                      | <b>PAH: no variants</b>                                                      | normal for the tested variants     | No pathogenic variants of PAH among those tested (improbable phenylketonuria).                                                                                                                                                                                                                   | ClinVar (chip) moderate                          | PAH: partial                                |

| Supplement                                           | Genes (state)                                                                       | Predicted response                      | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                               | Source · evidence                               | Trust                                        |
|------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------|
| GlyNAC (glycine + NAC)<br>⊕<br><b>Normal</b>         | <b>CPS1</b> : low glycine<br><b>GSTM1</b> : probable null<br><b>GNMT</b> : TT       | genetically useful glycine              | CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).                                                    | GWAS glycine (CPS1); genetic plan (S8) moderate | CPS1: high<br>GSTM1: moderate<br>GNMT: high  |
| L-arginine<br><b>Normal</b>                          | <b>NOS3</b> : –786CT + 298GT                                                        | rational NO (small effect)              | NOS3 –786 T/C + Glu298Asp G/T: a slightly smaller production of NO. Oral arginine has low bioavailability; with tadalafil the vasodilator effect is added (mild).                                                                   | NOS3 studies; NIH ODS (arginine) weak           | NOS3: high                                   |
| L-carnitine (tartrate)<br><b>Normal</b>              | <b>SLC22A4/5 (OCTN1/2)</b> : typical<br><b>FMO3</b> : typical                       | Typical transport; TMAO form            | Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.                                                                         | Genetic plan (S19); Koeth 2013 weak             | SLC22A4/5 (OCTN1/2): high<br>FMO3: high      |
| L-carnitine base<br><b>Normal</b>                    | <b>SLC22A4/5 (OCTN1/2)</b> : typical<br><b>FMO3</b> : typical                       | Typical transport; TMAO form            | Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.                                                                         | Genetic plan (S19); Koeth 2013 weak             | SLC22A4/5 (OCTN1/2): high<br>FMO3: high      |
| L-citrulline ⊕<br><b>Normal</b>                      | <b>NOS3</b> : –786CT + 298GT                                                        | rational NO                             | NOS3 –786 T/C + Glu298Asp: citrulline rises arginine better than arginine itself. With tadalafil, slight vasodilator sum.                                                                                                           | NOS3 studies; citrulline trials weak            | NOS3: high                                   |
| L-phenylalanine<br><b>Normal</b>                     | <b>PAH</b> : no variants                                                            | normal for the tested variants          | No pathogenic variants of PAH among those tested (improbable phenylketonuria).                                                                                                                                                      | ClinVar (chip) moderate                         | PAH: partial                                 |
| L-tyrosine<br><b>Normal</b>                          | <b>COMT</b> : Val/Val (high)                                                        | without robust interaction              | COMT Val/Val: rapid dopamine degradation; tyrosine is safe, without robust evidence of genotype-dependent effect.                                                                                                                   | COMT × tyrosine studies (small) weak            | COMT: high                                   |
| N-acetyl-L-tyrosine (NALT)<br><b>Normal</b>          | <b>COMT</b> : Val/Val (high)                                                        | without robust interaction              | COMT Val/Val: rapid dopamine degradation; tyrosine is safe, without robust evidence of genotype-dependent effect.                                                                                                                   | COMT × tyrosine studies (small) weak            | COMT: high                                   |
| ★ <b>NAC (N-acetylcysteine)</b> ⊕<br><b>Normal</b>   | <b>GSTM1</b> : probable null<br><b>GSTA1</b> : *B/*B<br><b>GPX1</b> : Pro198Leu het | <b>useful support for glutathione</b>   | Current stack (1800 mg) — plan S17: KEEP. 1200–1800 mg, far from training. Probable null GSTM1, GSTA1 *B/*B, GCLM –588 C/T and GPX1 Pro198Leu: weak II → NAC phase is coherent. Far from the training; ≥ 2 h away from zinc/copper. | Genetic plan (S17) moderate                     | GSTM1: moderate<br>GSTA1: high<br>GPX1: high |
| S-acetyl-glutathione<br><b>Normal</b>                | <b>GSTM1</b> : probable null<br><b>GSTA1</b> : *B/*B                                | rational (redundant with NAC + glycine) | Probable null GSTM1/GSTA1 *B/*B gives rationale; with NAC + glycine is redundant.                                                                                                                                                   | Genetic plan (d) weak                           | GSTM1: moderate<br>GSTA1: high               |
| ★ <b>acetyl-L-carnitine (ALCAR)</b><br><b>Normal</b> | <b>SLC22A4/5 (OCTN1/2)</b> : typical<br><b>FMO3</b> : typical                       | <b>Typical transport</b>                | Current stack (500 mg) — plan S19: KEEP. 500 mg in the morning. OCTN2 (SLC22A5) typical; CPT2 V368I het; FMO3 typical (some TMAO). Keep 500 mg in the morning.                                                                      | Genetic plan (S19) weak                         | SLC22A4/5 (OCTN1/2): high<br>FMO3: high      |
| betaine HCl (with pepsin) ⊕<br><b>Normal</b>         | <b>BHMT</b> : R239Q het<br><b>GNMT</b> : TT<br><b>MTHFR</b> : 677CT/1298AC          | genetically useful betaine pathway      | BHMT R239Q het, GNMT TT and MTHFR compound: the BHMT pathway compensates for folate. 3 g/day (≥ 4 g up to LDL) (suggestion, not applied).                                                                                           | Genetic plan (S7) moderate                      | BHMT: high<br>GNMT: high<br>MTHFR: high      |
| citrulline malate ⊕<br><b>Normal</b>                 | <b>NOS3</b> : –786CT + 298GT                                                        | rational NO                             | NOS3 –786 T/C + Glu298Asp: citrulline rises arginine better than arginine itself. With tadalafil, slight vasodilator sum.                                                                                                           | NOS3 studies; citrulline trials weak            | NOS3: high                                   |
| ★ <b>bovine hydrolyzed collagen</b><br><b>Normal</b> | <b>COL1A1/COL5A1</b> : normal                                                       | <b>without risk alleles</b>             | Current stack (20 g) — plan S9: KEEP. The same (glycine source). COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).                                                                              | Genetic plan (S9) weak                          | COL1A1/COL5A1: high                          |

| Supplement                                            | Genes (state)                                                | Predicted response                      | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                      | Source · evidence                                          | Trust                                    |
|-------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------|
| marine collagen<br><b>Normal</b>                      | <b>COL1A1/COL5A1:</b> normal                                 | without risk alleles                    | COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).                                                                                                                                                                      | Genetic plan (S9) weak                                     | COL1A1/COL5A1: high                      |
| creatine HCl ☺<br><b>Normal</b>                       | <b>AMPD1:</b> normal<br><b>ACTN3:</b> R/X<br><b>GNMT:</b> TT | useful (muscle + save SAM)              | Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.                                                 | Genetic plan (S10); ISSN (muscle) · moderate (methylation) | AMPD1: high<br>ACTN3: high<br>GNMT: high |
| creatine ethyl ester ☺<br><b>Normal</b>               | <b>AMPD1:</b> normal<br><b>ACTN3:</b> R/X<br><b>GNMT:</b> TT | useful (muscle + save SAM)              | Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.                                                 | Genetic plan (S10); ISSN (muscle) · moderate (methylation) | AMPD1: high<br>ACTN3: high<br>GNMT: high |
| ★ creatine monohydrate ☺<br><b>Normal</b>             | <b>AMPD1: normal</b><br><b>ACTN3: R/X</b><br><b>GNMT: TT</b> | <b>useful (muscle + save SAM)</b>       | Current stack (5 g) — plan S10: KEEP. The same. Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence. | Genetic plan (S10); ISSN (muscle) · moderate (methylation) | AMPD1: high<br>ACTN3: high<br>GNMT: high |
| buffered creatine ☺<br><b>Normal</b>                  | <b>AMPD1:</b> normal<br><b>ACTN3:</b> R/X<br><b>GNMT:</b> TT | useful (muscle + save SAM)              | Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.                                                 | Genetic plan (S10); ISSN (muscle) · moderate (methylation) | AMPD1: high<br>ACTN3: high<br>GNMT: high |
| glycine-propionyl-L-carnitine (GPLC)<br><b>Normal</b> | <b>SLC22A4/5 (OCTN1/2):</b> typical<br><b>FMO3:</b> typical  | Typical transport; TMAO form            | Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.                                                                                                | Genetic plan (S19); Koeth 2013 weak                        | SLC22A4/5 (OCTN1/2): high<br>FMO3: high  |
| glutathione liposomal<br><b>Normal</b>                | <b>GSTM1:</b> probable null<br><b>GSTA1:</b> *B/*B           | rational (redundant with NAC + glycine) | Probable null GSTM1/GSTA1 *B/*B gives rationale; with NAC + glycine is redundant.                                                                                                                                                                          | Genetic plan (d) weak                                      | GSTM1: moderate<br>GSTA1: high           |
| reduced oral glutathione<br><b>Normal</b>             | <b>GSTM1:</b> probable null<br><b>GSTA1:</b> *B/*B           | rational (redundant with NAC + glycine) | Probable null GSTM1/GSTA1 *B/*B gives rationale; with NAC + glycine is redundant.                                                                                                                                                                          | Genetic plan (d) weak                                      | GSTM1: moderate<br>GSTA1: high           |
| creatine nitrate ☺<br><b>Normal</b>                   | <b>AMPD1:</b> normal<br><b>ACTN3:</b> R/X<br><b>GNMT:</b> TT | useful (muscle + save SAM)              | Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.                                                 | Genetic plan (S10); ISSN (muscle) · moderate (methylation) | AMPD1: high<br>ACTN3: high<br>GNMT: high |
| propionyl-L-carnitine<br><b>Normal</b>                | <b>SLC22A4/5 (OCTN1/2):</b> typical<br><b>FMO3:</b> typical  | Typical transport; TMAO form            | Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.                                                                                                | Genetic plan (S19); Koeth 2013 weak                        | SLC22A4/5 (OCTN1/2): high<br>FMO3: high  |
| whey protein<br><b>Normal</b>                         | <b>LCT:</b> persistent                                       | Lactose tolerated                       | LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.                                                                                                                                                                               | Genetic plan (e.6) strong                                  | LCT: high                                |
| specific bioactive collagen peptides<br><b>Normal</b> | <b>COL1A1/COL5A1:</b> normal                                 | without risk alleles                    | COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).                                                                                                                                                                      | Genetic plan (S9) weak                                     | COL1A1/COL5A1: high                      |
| whey hydrolyzed<br><b>Normal</b>                      | <b>LCT:</b> persistent                                       | Lactose tolerated                       | LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.                                                                                                                                                                               | Genetic plan (e.6) strong                                  | LCT: high                                |
| whey isolated<br><b>Normal</b>                        | <b>LCT:</b> persistent                                       | Lactose tolerated                       | LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.                                                                                                                                                                               | Genetic plan (e.6) strong                                  | LCT: high                                |

#### No known genetic relevance (40):

5-HTP (AADC)  
agmatin

beta-alanine (CNDP1) — CNDP1: undetermined  
Bone juice powder

casein micellar  
collagen type II undenatured

DMG (dimethylglycine)  
EAA (essential amino acids) (various)  
GABA  
gelatin  
glutamine peptide  
HICA (alpha-hydroxyisocaproic acid)  
hydroxyproline  
HMB (beta-hydroxy-betamethylbutyrate)  
L-alanine  
L-aspartate  
L-cysteine (CBS/GSH)

L-glutamine  
L-histidine (HDC)  
L-lysine  
L-ornithine (OTC)  
L-proline  
L-serine (PHGDH)  
L-theanine  
L-threonine  
L-tryptophan (TPH/IDO)  
OKG (ornithine alpha-ketoglutarate)  
ornithine-aspartate (LOLA) (urea cycle)

rice protein  
meat protein (beef insulator)  
hemp protein  
pea protein  
cricket protein (insects)  
egg protein (albumin)  
soy protein isolated  
sarcosin  
★ **taurine**  
D-aspartic acid  
fermented gamma-aminobutyric acid  
glutamic acid

### 3.4 Fatty acids, oils and phospholipids — 45 supplements

prevention: 1 · dose adjustment: 1 · Monitoring: 9 · normal: 19 · without known genetic relevance: 15

| Supplement                                            | Genes (state)                                                                       | Predicted response                            | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                                                                     | Source · evidence                                                                              | Trust                                                   |
|-------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| cod liver oil<br><b>Avoid</b>                         | <b>BCO1</b> : slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable   | avoid preformed retinol                       | The conversion of β-carotene is only slightly reduced (BCO1 R267S het; without A379V), so there is no genetic need for retinol. With the isotretinoin of the stack, retinol supplemental sum retinoids (toxicity, transaminases); the genetic liver (PNPLA3/Pi*MZ) reinforces. Obtain vitamin A from carotenoids (suggestion, not applied).                                                                                               | NIH ODS (vitamin A); genetic plan (b.3); isotretinoin label strong (interaction) · weak (BCO1) | BCO1: high<br>PNPLA3/Pi*MZ (liver): moderate            |
| ★ fish oil (EPA+DHA, TG/rTG) ⊕<br><b>Adjust dose</b>  | <b>FADS1/FADS2</b> : high activity<br><b>APOE</b> : ε3/ε3                           | <b>adjust dose (plan)</b>                     | Current stack (EPA 1125 + DHA 875 mg) — plan S1: ADJUST. Up to 3.2–4 g/day of EPA+DHA (Coordination Note), form rTG, with the meal fatter and far from psyllium. FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate                                   | FADS1/FADS2: high<br>APOE: high                         |
| CLA (conjugated linoleic acid)<br><b>Monitoring</b>   | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>TCF7L2/FTO/GCKR</b> : unfavourable | without benefit; possible metabolic worsening | CLA t10,c12 increased insulin resistance and hepatic fat in trials; with PNPLA3/TCF7L2 at risk, avoid.                                                                                                                                                                                                                                                                                                                                    | CLA (Risérus 2002) trials weak                                                                 | PNPLA3/Pi*MZ (liver): moderate<br>TCF7L2/FTO/GCKR: high |
| Sewer oil (GLA)<br><b>Monitoring</b>                  | <b>FADS1/FADS2</b> : high activity<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable    | GLA→AA + alkaloids                            | High FADS1 (GLA→AA) and possible presence of hepatotoxic pyrrolizidine alkaloids (genetic liver): only certified products without PA.                                                                                                                                                                                                                                                                                                     | FADS1 studies; EMA (PA) weak                                                                   | FADS1/FADS2: high<br>PNPLA3/Pi*MZ (liver): moderate     |
| coconut oil<br><b>Monitoring</b>                      | <b>CETP/9p21/APOB</b> : genetic vascular risk                                       | saturated fat (LDL)                           | Genetic priority to low LDL/ApoB (high CETP, 9p21, APOB): coconut oil rises to LDL; not use as a supplement.                                                                                                                                                                                                                                                                                                                              | Genetic plan (e.3) moderate                                                                    | CETP/9p21/APOB: high                                    |
| safflower oil<br><b>Monitoring</b>                    | <b>FADS1/FADS2</b> : high activity                                                  | Omega-6 unnecessary                           | The extra LA is converted effectively into AA (high FADS); the diet already has enough omega-6. Prefer omega-3 isolated.                                                                                                                                                                                                                                                                                                                  | FADS × LA studies weak                                                                         | FADS1/FADS2: high                                       |
| shark liver oil (alkylglycerols)<br><b>Monitoring</b> | <b>BCO1</b> : slightly ↓ conversion<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable   | confirm retinol content                       | It can contain vitamin A preformed in variable quantity; with isotretinoin and genetic liver, only one product without retinol (suggestion, not applied).                                                                                                                                                                                                                                                                                 | NIH ODS (vitamin A) weak                                                                       | BCO1: high<br>PNPLA3/Pi*MZ (liver): moderate            |
| Grapeseed oil<br><b>Monitoring</b>                    | <b>FADS1/FADS2</b> : high activity                                                  | Omega-6 unnecessary                           | The extra LA is converted effectively into AA (high FADS); the diet already has enough omega-6. Prefer omega-3 isolated.                                                                                                                                                                                                                                                                                                                  | FADS × LA studies weak                                                                         | FADS1/FADS2: high                                       |
| Blackcurrant oil (GLA)<br><b>Monitoring</b>           | <b>FADS1/FADS2</b> : high activity                                                  | GLA can turn AA                               | With high activity FADS1, part of GLA→DGLA goes to arachidonic acid (pro-inflammatory), reducing the benefit. No genetic indication.                                                                                                                                                                                                                                                                                                      | FADS1 × GLA (Gillingham 2013) studies weak                                                     | FADS1/FADS2: high                                       |
| Primule oil (GLA)<br><b>Monitoring</b>                | <b>FADS1/FADS2</b> : high activity                                                  | GLA can turn AA                               | With high activity FADS1, part of GLA→DGLA goes to arachidonic acid (pro-inflammatory), reducing the benefit. No genetic indication.                                                                                                                                                                                                                                                                                                      | FADS1 × GLA (Gillingham 2013) studies weak                                                     | FADS1/FADS2: high                                       |
| Omega-3-6-9<br><b>Monitoring</b>                      | <b>FADS1/FADS2</b> : high activity                                                  | Omega-6 unnecessary                           | The extra LA is converted effectively into AA (high FADS); the diet already has enough omega-6. Prefer omega-3 isolated.                                                                                                                                                                                                                                                                                                                  | FADS × LA studies weak                                                                         | FADS1/FADS2: high                                       |

| Supplement                                            | Genes (state)                                                              | Predicted response                       | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                    | Source · evidence                                            | Trust                                     |
|-------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------|
| DHA concentrate ☺<br><b>Normal</b>                    | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3                    | EPA/DHA useful preformed                 | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high           |
| EPA concentrate ☺<br><b>Normal</b>                    | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3                    | EPA/DHA useful preformed                 | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high           |
| Plant stanols ☺<br><b>Normal</b>                      | <b>ABCG5/ABCG8:</b> normal<br><b>CETP/9p21/APOB:</b> genetic vascular risk | useful for LDL (without hyperabsorption) | ABCG5/8 without hyperabsorption variants (safe); low genetic priority to LDL (CETP/9p21/APOB). 2 g/day lower LDL ~8–10%; compatible with ezetimibe (suggestion, not applied).                                                                                            | Genetic plan (1 annex); consensus EAS phytosterols moderate  | ABCG5/ABCG8: high<br>CETP/9p21/APOB: high |
| phytosterol (beta-sitosterol) ☺<br><b>Normal</b>      | <b>ABCG5/ABCG8:</b> normal<br><b>CETP/9p21/APOB:</b> genetic vascular risk | useful for LDL (without hyperabsorption) | ABCG5/8 without hyperabsorption variants (safe); low genetic priority to LDL (CETP/9p21/APOB). 2 g/day lower LDL ~8–10%; compatible with ezetimibe (suggestion, not applied).                                                                                            | Genetic plan (1 annex); consensus EAS phytosterols moderate  | ABCG5/ABCG8: high<br>CETP/9p21/APOB: high |
| krill phospholipids ☺<br><b>Normal</b>                | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3                    | EPA/DHA useful preformed                 | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high           |
| caprylic acid (C8)<br><b>Normal</b>                   | <b>APOE:</b> ε3/ε3                                                         | typical response                         | APOE ε3/ε3: typical ketone metabolism (the ketogenic advantage described in ε4 does not apply).                                                                                                                                                                          | APOE × ketones studies weak                                  | APOE: high                                |
| oleic acid / olive oil in capsules ☺<br><b>Normal</b> | <b>PON1:</b> low expression                                                | useful (PON1)                            | Low expression PON1: olive oil polyphenols increase the activity of PON1; extra virgin olive oil in food is better than capsules.                                                                                                                                        | Genetic plan (S32, e.2) weak                                 | PON1: high                                |
| MCT oil<br><b>Normal</b>                              | <b>APOE:</b> ε3/ε3                                                         | typical response                         | APOE ε3/ε3: typical ketone metabolism (the ketogenic advantage described in ε4 does not apply).                                                                                                                                                                          | APOE × ketones studies weak                                  | APOE: high                                |
| Algal oil (DHA vegan) ☺<br><b>Normal</b>              | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3                    | EPA/DHA useful preformed                 | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high           |
| chia oil<br><b>Normal</b>                             | <b>FADS1/FADS2:</b> high activity                                          | good conversion ALA→EPA                  | With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.                                                                                                                                            | FADS × ALA studies moderate                                  | FADS1/FADS2: high                         |
| Equium oil (SDA)<br><b>Normal</b>                     | <b>FADS1/FADS2:</b> high activity                                          | good conversion ALA→EPA                  | With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.                                                                                                                                            | FADS × ALA studies moderate                                  | FADS1/FADS2: high                         |
| Focus oil ☺<br><b>Normal</b>                          | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3                    | EPA/DHA useful preformed                 | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high           |
| krill oil ☺<br><b>Normal</b>                          | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3                    | EPA/DHA useful preformed                 | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high           |

| Supplement                                              | Genes (state)                                           | Predicted response       | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                    | Source · evidence                                            | Trust                           |
|---------------------------------------------------------|---------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------|
| linseed oil (ALA)<br><b>Normal</b>                      | <b>FADS1/FADS2:</b> high activity                       | good conversion ALA→EPA  | With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.                                                                                                                                            | FADS × ALA studies moderate                                  | FADS1/FADS2: high               |
| Squid oil ⊕<br><b>Normal</b>                            | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3 | EPA/DHA useful preformed | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high |
| Green lips mussel oil ⊕<br><b>Normal</b>                | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3 | EPA/DHA useful preformed | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high |
| Concentrated fish oil (ethyl esters) ⊕<br><b>Normal</b> | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3 | EPA/DHA useful preformed | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high |
| perila oil<br><b>Normal</b>                             | <b>FADS1/FADS2:</b> high activity                       | good conversion ALA→EPA  | With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.                                                                                                                                            | FADS × ALA studies moderate                                  | FADS1/FADS2: high               |
| Salmon oil ⊕<br><b>Normal</b>                           | <b>FADS1/FADS2:</b> high activity<br><b>APOE:</b> ε3/ε3 | EPA/DHA useful preformed | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied). | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high |

**No known genetic relevance (15):**

|                                  |                                         |                                                      |
|----------------------------------|-----------------------------------------|------------------------------------------------------|
| plant ceramides (phytoceramides) | PEA (palmitoylethanolamide) (FAAH/NAAA) | Nigela oil (black cumin)                             |
| squalene                         | plasmalogens                            | pine nut oil (pinenic acid)                          |
| phosphatidylethanolamine         | polycosanol                             | pumpkin seed oil                                     |
| phosphatidylinositol             | phosphatidic acid                       | Hemp seed oil (FADS1/2) — FADS1/FADS2: high activity |
| phosphatidylserine               | avocado oil (capsule)                   | Omega-7 (palmitoleic acid, seabream)                 |

3.5 Polyphenols and other phytochemicals — 91 supplements

prevention: 14 · dose adjustment: 2 · Monitoring: 23 · normal: 17 · without known genetic relevance: 35

| Supplement                                           | Genes (state)                                                                                                                                       | Predicted response         | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Source · evidence                                                    | Trust                                                                                   |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| EGCG / green tea extract<br><b>Avoid</b>             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>COMT:</b> Val/Val (high)<br><b>HLA-B*35:01:</b> probable negative<br><b>UGT1A1:</b> IM (*1/*28)     | avoid (liver)              | Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                   | LiverTox; EFSA 2018 (catechins); genetic plan (b.3)<br>moderate      | PNPLA3/Pi*MZ (liver): moderate<br>COMT: high<br>HLA-B*35:01: moderate<br>UGT1A1: high   |
| ★ apigenin<br><b>Avoid</b>                           | <b>UGT1A1:</b> IM (*1/*28)<br><b>CYP3A5:expressor (*1/*3)</b><br><b>CYP1A2: normal to intermediate</b><br><b>CYP2C9:</b> IM (*1/*3)                 | avoid / pause (test-pause) | Current stack (100 mg) — plan S28: PAUSE (test-pause). Test-pause (current protocol). Test pause for the same reasons of luteolin: inhibits UGT1A1 (*1/*28: up to bilirubin and confuses liver reading) and CYP3A4/P-gp (sirolimus; CYP3A5 expressor). It is not catechol, but also inhibits CYP1A2 (already normal to intermediate: *1F het + rs2472297 C/C), which rises melatonin and caffeine, and CYP2C9 (*1/*3). Pause; pause while ALT is high (current protocol). Coordination note: in the current protocol, the pause lasts while ALT is high; if it returns, 25–50 mg at night, outside the rapamycin window (suggestion, not applied). | Genetic plan (S28 + coordination note)<br>weak                       | UGT1A1: high<br>CYP3A5: high<br>CYP1A2: high<br>CYP2C9: high                            |
| curcumin (95% extract)<br><b>Avoid</b>               | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative<br><b>CYP3A5:expressor (*1/*3)</b> | avoid/pause                | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                            | Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01)<br>weak-moderate | UGT1A1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate<br>CYP3A5: high |
| curcumin + piperine<br><b>Avoid</b>                  | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative<br><b>CYP3A5:expressor (*1/*3)</b> | avoid/pause                | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                            | Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01)<br>weak-moderate | UGT1A1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate<br>CYP3A5: high |
| colloidal curcumin (nanoparticles)<br><b>Avoid</b>   | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative<br><b>CYP3A5:expressor (*1/*3)</b> | avoid/pause                | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                            | Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01)<br>weak-moderate | UGT1A1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate<br>CYP3A5: high |
| curcumin with curcuman essential oil<br><b>Avoid</b> | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative<br><b>CYP3A5:expressor (*1/*3)</b> | avoid/pause                | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                            | Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01)<br>weak-moderate | UGT1A1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate<br>CYP3A5: high |

| Supplement                                           | Genes (state)                                                                                                                                        | Predicted response | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Source · evidence                                                       | Trust                                                                                   |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| curcumin phytosome<br><b>Avoid</b>                   | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative<br><b>CYP3A5:</b> expressor (*1/*3) | avoid/pause        | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                     | Genetic plan (S26);<br>LiverTox (curcuma, HLA-B*35:01)<br>weak-moderate | UGT1A1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate<br>CYP3A5: high |
| ★ solid lipid curcumin (SLCP)<br><b>Avoid</b>        | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative<br><b>CYP3A5:</b> expressor (*1/*3) | avoid/pause        | Current stack (1 g) — plan S26: PAUSE. Pause (current protocol). UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied).                                                                                                                                                                                                                                                                    | Genetic plan (S26);<br>LiverTox (curcuma, HLA-B*35:01)<br>weak-moderate | UGT1A1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate<br>CYP3A5: high |
| isoquercytrin enzymatically modified<br><b>Avoid</b> | <b>UGT1A1:</b> IM (*1/*28)<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>CYP3A5:</b> expressor (*1/*3)                          | avoid/pause        | UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.                                                                                                                                                                                                                                           | Genetic plan (S29 + coordination note)<br>weak                          | UGT1A1: high<br>COMT: high<br>MTHFR: high<br>CYP3A5: high                               |
| ★ luteolin<br><b>Avoid</b>                           | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>UGT1A1:</b> IM (*1/*28)<br><b>CYP2C9:</b> IM (*1/*3)                                 | avoid/pause        | Current stack (100 mg) — plan S27: PAUSE. Pause (current protocol). Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Genetic plan (S27)<br>weak                                              | COMT: high<br>MTHFR: high<br>UGT1A1: high<br>CYP2C9: high                               |
| luteolin + PEA<br><b>Avoid</b>                       | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>UGT1A1:</b> IM (*1/*28)<br><b>CYP2C9:</b> IM (*1/*3)                                 | avoid/pause        | Catechol: methylated by COMT (Val/Val), spent SAM and produces homocysteine (MTHFR compound, p92 score). It inhibits UGT1A1 (*28) and CYP2C9 (*1/*3 IM) in vitro, and CYP3A4/P-gp (sirolimus). Pause; pause while ALT is high (current protocol) (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Genetic plan (S27)<br>weak                                              | COMT: high<br>MTHFR: high<br>UGT1A1: high<br>CYP2C9: high                               |
| ★ quercetin<br><b>Avoid</b>                          | <b>UGT1A1:</b> IM (*1/*28)<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>CYP3A5:</b> expressor (*1/*3)                          | avoid/pause        | Current stack (250 mg (3x/month)) — plan S29: PAUSE (suggestion, not applied). It is suggested to pause as an isolated supplement (genetic verdict); in the current protocol enters the monthly pulse with fisetin, only with normal ALT. UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. | Genetic plan (S29 + coordination note)<br>weak                          | UGT1A1: high<br>COMT: high<br>MTHFR: high<br>CYP3A5: high                               |

| Supplement                                               | Genes (state)                                                                                                               | Predicted response     | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Source · evidence                              | Trust                                                                   |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------|
| quercetin phytosome<br><b>Avoid</b>                      | <b>UGT1A1:</b> IM (*1/*28)<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>CYP3A5:</b> expressor (*1/*3) | avoid/pause            | UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.                                                                                                                   | Genetic plan (S29 + coordination note)<br>weak | UGT1A1: high<br>COMT: high<br>MTHFR: high<br>CYP3A5: high               |
| resveratrol + quercetin (combined)<br><b>Avoid</b>       | <b>UGT1A1:</b> IM (*1/*28)<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC<br><b>CYP3A5:</b> expressor (*1/*3) | avoid/pause            | UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed.                                                                                                                   | Genetic plan (S29 + coordination note)<br>weak | UGT1A1: high<br>COMT: high<br>MTHFR: high<br>CYP3A5: high               |
| tea catechins (drink/matcha)<br><b>Adjust dose</b>       | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C<br><b>COMT:</b> Val/Val (high)                                 | count caffeine         | Tea/matcha in drink does not have the liver risk of the extract; counts for the caffeine ceiling ( $\leq 200$ mg/day, only in the morning; Normal to intermediate CYP1A2).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Genetic plan (e.4)<br>moderate                 | CYP1A2: high<br>ADORA2A: high<br>COMT: high                             |
| ★ fisetin<br><b>Adjust dose</b>                          | <b>COMT:</b> Val/Val (high)<br><b>UGT1A1:</b> IM (*1/*28)<br><b>CYP3A5:</b> expressor (*1/*3)                               | isolated monthly pulse | Current stack (2.5 g (3×/month)) — plan S29: ADJUST. Monthly pulse (2 days), $\geq 7$ days away from rapamycin and FOXO4-DRI; current protocol: with quercetin, only with normal ALT. Catechol (COMT Val/Val + MTHFR) and inhibitor of CYP3A4/P-gp and UGT1A1. 9p21 homozygous (CDKN2A/B region, senescence) gives senolytic rationale. Genetic reading: 1 pulse/month (2 days), $\geq 7$ days away from rapamycin and FOXO4-DRI (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. | Genetic plan (S29)<br>weak                     | COMT: high<br>UGT1A1: high<br>CYP3A5: high                              |
| (–)-epicatechin<br><b>Monitoring</b>                     | <b>COMT:</b> Val/Val (high)                                                                                                 | catechol               | Epicatechin isolated at high dose is methylated by COMT (Val/Val); prefer cocoa.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Genetic plan (f.2)<br>weak                     | COMT: high                                                              |
| allicin (stabilized garlic extract)<br><b>Monitoring</b> | <b>CYP3A5:</b> expressor (*1/*3)                                                                                            | P-gp/CYP3A4 inducer    | Allicin of fresh extracts induces P-gp/CYP3A4 (lowered saquinavir ~50%): it can lower sirolimus. Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Piscitelli 2002<br>moderate (interaction)      | CYP3A5: high                                                            |
| Saffron powder<br><b>Monitoring</b>                      | <b>UGT1A1:</b> IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative            | only cooking doses     | Less curcumin than extracts, but cases of liver injury include root powder. Cooking doses without problem; capsules such as extracts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | LiverTox (curcuma)<br>weak                     | UGT1A1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate |
| baicalein<br><b>Monitoring</b>                           | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | cases of liver injury  | Scutellaria baicalensis was associated with liver injury (LiverTox); with genetic liver, avoiding prolonged use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | LiverTox (hearing)<br>weak                     | PNPLA3/Pi*MZ (liver): moderate                                          |
| baicalin (Scutellaria baicalensis)<br><b>Monitoring</b>  | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                   | cases of liver injury  | Scutellaria baicalensis was associated with liver injury (LiverTox); with genetic liver, avoiding prolonged use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | LiverTox (hearing)<br>weak                     | PNPLA3/Pi*MZ (liver): moderate                                          |

| Supplement                                           | Genes (state)                                                                                               | Predicted response                           | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                   | Source · evidence                                       | Trust                                                          |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------|
| berberine<br><b>Monitoring</b>                       | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3) | the genetic plan advises to add              | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose).     | Guo 2012; genetic plan (b.3)<br>moderate (interaction)  | CYP2D6: moderate<br>CYP2C9: high<br>CYP3A5: high               |
| dihydroberberine<br><b>Monitoring</b>                | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3) | the genetic plan advises to add              | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose).     | Guo 2012; genetic plan (b.3)<br>moderate (interaction)  | CYP2D6: moderate<br>CYP2C9: high<br>CYP3A5: high               |
| glabridine (liquorice)<br><b>Monitoring</b>          | <b>CYP3A5</b> :expressor (*1/*3)                                                                            | inhibits CYP3A4                              | Glabridin inhibits CYP3A4 in vitro (liquorice). Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                                                                                   | In vitro weak                                           | CYP3A5: high                                                   |
| myricetin<br><b>Monitoring</b>                       | <b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                                                 | catechol: prefer low doses                   | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.                                                                                                | Genetic plan (f.2)<br>weak                              | COMT: high<br>MTHFR: high                                      |
| naringin/naringenin<br><b>Monitoring</b>             | <b>CYP3A5</b> :expressor (*1/*3)                                                                            | weak inhibition of CYP3A4                    | Naringenin contributes to the orange effect (weakly isolated). Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                                                                    | Grapevine pharmacology weak                             | CYP3A5: high                                                   |
| piceatannol<br><b>Monitoring</b>                     | <b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                                                 | catechol: prefer low doses                   | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.                                                                                                | Genetic plan (f.2)<br>weak                              | COMT: high<br>MTHFR: high                                      |
| piperine (black pepper extract)<br><b>Monitoring</b> | <b>CYP3A5</b> :expressor (*1/*3)<br><b>UGT1A1</b> : IM (*1/*28)                                             | increases drug exposure                      | Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements. Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                                  | Bhardwaj 2002<br>moderate (interaction)                 | CYP3A5: high<br>UGT1A1: high                                   |
| polydatin<br><b>Monitoring</b>                       | <b>UGT1A1</b> : IM (*1/*28)                                                                                 | precursor of resveratrol                     | Converted into resveratrol: the same logic (UGT1A1 *28, CYP3A4).                                                                                                                                                                                                                        | Pharmacokinetics weak                                   | UGT1A1: high                                                   |
| pterostilben<br><b>Monitoring</b>                    | <b>CETP/9p21/APOB</b> : genetic vascular risk<br><b>UGT1A1</b> : IM (*1/*28)                                | can rise to LDL                              | In the trial, pterostilbene rose to LDL (Riche 2014); with low genetic priority to LDL/ApoB, avoid or monitor.                                                                                                                                                                          | 2014 Rich<br>weak-moderate                              | CETP/9p21/APOB: high<br>UGT1A1: high                           |
| resveratrol<br><b>Monitoring</b>                     | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3)               | inhibits CYP3A4/2C9                          | Glucuronidated by UGT1A1 (*28: a little higher exposure). With 1 g/day inhibits CYP3A4, CYP2C9 and CYP2D6 in humans (Chow 2010): added to CYP2C9 *1/*3, increases exposure to NSAIDs; and moves on the sirolimus. Never in the rapamycin window (3 days before 7 days after each dose). | Chow 2010; genetic plan (f.1)<br>moderate (interaction) | UGT1A1: high<br>CYP2C9: high<br>CYP3A5: high                   |
| rutin<br><b>Monitoring</b>                           | <b>COMT</b> : Val/Val (high)<br><b>UGT1A1</b> : IM (*1/*28)                                                 | catechol (via quercetin)                     | Rutin releases quercetin in the intestine: the same logic (COMT Val/Val, UGT1A1 *28) but lower exposure. Food doses OK.                                                                                                                                                                 | Pharmacokinetics of rutin weak                          | COMT: high<br>UGT1A1: high                                     |
| silibinin phytosome<br><b>Monitoring</b>             | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable     | without proven gain; inhibits UGT1A1/ CYP2C9 | Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3; Han 2009). It does not protect the liver in a proven way in PNPLA3/Pi*MZ.                                                        | Han 2009; silymarin NASH trials weak                    | UGT1A1: high<br>CYP2C9: high<br>PNPLA3/Pi*MZ (liver): moderate |
| silymarin (marian card)<br><b>Monitoring</b>         | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable     | without proven gain; inhibits UGT1A1/ CYP2C9 | Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3; Han 2009). It does not protect the liver in a proven way in PNPLA3/Pi*MZ.                                                        | Han 2009; silymarin NASH trials weak                    | UGT1A1: high<br>CYP2C9: high<br>PNPLA3/Pi*MZ (liver): moderate |

| Supplement                                           | Genes (state)                                                        | Predicted response         | Recommendation (summary) and implication for Sérgio F                                                                                                                                                        | Source · evidence            | Trust                           |
|------------------------------------------------------|----------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------|
| taxifolin<br>(dihydroquercetin)<br><b>Monitoring</b> | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/<br>1298AC        | catechol: prefer low doses | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.                     | Genetic plan (f.2)<br>weak   | COMT: high<br>MTHFR: high       |
| caffeic acid<br><b>Monitoring</b>                    | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/<br>1298AC        | catechol: prefer low doses | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.                     | Genetic plan (f.2)<br>weak   | COMT: high<br>MTHFR: high       |
| chlorogenic acid (green coffee)<br><b>Monitoring</b> | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/<br>1298AC        | can rise homocysteine      | In the trial, 2 g/day of chlorogenic acid rose to homocysteine (Olthof 2001); with COMT Val/Val + MTHFR compound, avoiding concentrated extracts.                                                            | Olthof 2001<br>weak-moderate | COMT: high<br>MTHFR: high       |
| gallic acid<br><b>Monitoring</b>                     | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/<br>1298AC        | catechol: prefer low doses | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.                     | Genetic plan (f.2)<br>weak   | COMT: high<br>MTHFR: high       |
| rosmarinic acid<br><b>Monitoring</b>                 | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/<br>1298AC        | catechol: prefer low doses | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.                     | Genetic plan (f.2)<br>weak   | COMT: high<br>MTHFR: high       |
| DIM (di-indolmethane)<br><b>Normal</b>               | <b>CYP1A2:</b> normal to intermediate<br><b>GSTM1:</b> probable null | induces CYP1A2             | CYP1A2 inducers: accelerate caffeine and melatonin clearance (which is normal to intermediate). Null GSTM1: no known negative effect.                                                                        | Pharmacology DIM/I3C<br>weak | CYP1A2: high<br>GSTM1: moderate |
| I3C (indol-3-carbinol)<br><b>Normal</b>              | <b>CYP1A2:</b> normal to intermediate<br><b>GSTM1:</b> probable null | induces CYP1A2             | CYP1A2 inducers: accelerate caffeine and melatonin clearance (which is normal to intermediate). Null GSTM1: no known negative effect.                                                                        | Pharmacology DIM/I3C<br>weak | CYP1A2: high<br>GSTM1: moderate |
| garlic powder (capsules)<br><b>Normal</b>            | <b>NOS3:</b> –786CT + 298GT<br><b>CYP3A5:</b> expressor (*1/*3)      | low allicin                | NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.                                                                | Genetic plan (S33)<br>weak   | NOS3: high<br>CYP3A5: high      |
| ★ black garlic<br><b>Normal</b>                      | <b>NOS3:</b> –786CT + 298GT<br><b>CYP3A5:</b> expressor (*1/*3)      | low allicin                | Current stack (10 g) — S33 plan: optional. Redundant with AGE. NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant. | Genetic plan (S33)<br>weak   | NOS3: high<br>CYP3A5: high      |
| chrysin<br><b>Normal</b>                             | <b>UGT1A1:</b> IM (*1/*28)                                           | no clinical effect         | Inducer of UGT1A1 in vitro, but without effect on humans (very low bioavailability).                                                                                                                         | Walle 2001<br>weak           | UGT1A1: high                    |
| ★ extract of aged garlic (AGE)<br><b>Normal</b>      | <b>NOS3:</b> –786CT + 298GT<br><b>CYP3A5:</b> expressor (*1/*3)      | low allicin                | Current stack (1.2 g) — plan S33: KEEP. The same. NOS3 gives vascular rationale; AGE and black garlic have little allicin → minor effect on sirolimus. Maintain AGE; black garlic is redundant.              | Genetic plan (S33)<br>weak   | NOS3: high<br>CYP3A5: high      |
| pine bark extract (generic) ⊕<br><b>Normal</b>       | <b>NOS3:</b> –786CT + 298GT                                          | useful (NO)                | NOS3 –786 C/T + Glu298Asp: pine bark extract improves endothelial function. Keep in the morning, outside the rapamycin window for prudence.                                                                  | Genetic plan (S31)<br>weak   | NOS3: high                      |
| ★ olive leaf extract (oleuropein) ⊕<br><b>Normal</b> | <b>PON1:</b> low expression                                          | useful (PON1)              | Current stack (oleuropein 250 mg) — plan S32: KEEP. The same, with meal. Low expression PON1: olive oil/oliveira polyphenols increase the activity of PON1.                                                  | Genetic plan (S32)<br>weak   | PON1: high                      |

| Supplement                                                      | Genes (state)                                               | Predicted response         | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                              | Source · evidence                        | Trust                                  |
|-----------------------------------------------------------------|-------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------|
| pomegranate extract (punicalagines) ⊕<br><b>Normal</b>          | <b>PON1:</b> low expression                                 | useful (PON1)              | Low expression PON1: pomegranate increases the activity of PON1 (Aviram 2004). No relevant effect on CYP3A4 in humans.                                                                                                             | 2004 weak                                | PON1: high                             |
| flavanols from cocoa<br><b>Normal</b>                           | <b>COMT:</b> Val/Val (high)                                 | low catechol load          | Flavanols of cocoa in standard dose (500–900 mg): low catechol load; NOS3 gives vascular rationale.                                                                                                                                | COSMOS weak                              | COMT: high                             |
| glucoraphanin (without myrosinase) ⊕<br><b>Normal</b>           | <b>GSTM1:</b> probable null<br><b>NFE2L2 (NRF2):</b> normal | favourable response        | Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.                                                                                            | Gasper 2005; genetic plan (S25) moderate | GSTM1: moderate<br>NFE2L2 (NRF2): high |
| ★ hydroxytyrosol ⊕<br><b>Normal</b>                             | <b>PON1: low expression</b><br><b>COMT: Val/Val (high)</b>  | <b>useful in low dose</b>  | Current stack (25 mg) — plan S32: KEEP. The same, with meal. Low expression PON1: olive oil polyphenols rise to PON1. Low dose (25 mg) : small impact on COMT. Maintain.                                                           | Genetic plan (S32) weak                  | PON1: high<br>COMT: high               |
| moringin/ isothiocyanates of moringa ⊕<br><b>Normal</b>         | <b>GSTM1:</b> probable null<br><b>NFE2L2 (NRF2):</b> normal | favourable response        | Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.                                                                                            | Gasper 2005; genetic plan (S25) moderate | GSTM1: moderate<br>NFE2L2 (NRF2): high |
| ★ pine bark extract (marine pine hull) ⊕<br><b>Normal</b>       | <b>NOS3: –786CT + 298GT</b>                                 | <b>useful (NO)</b>         | Current stack (100 mg) — plan S31: KEEP. In the morning, outside the rapamycin window. NOS3 –786 C/T + Glu298Asp: pine bark extract improves endothelial function. Keep in the morning, outside the rapamycin window for prudence. | Genetic plan (S31) weak                  | NOS3: high                             |
| polyphenols of olive oil ⊕<br><b>Normal</b>                     | <b>PON1:</b> low expression                                 | useful (PON1)              | Low expression PON1: olive oil/oliveira polyphenols increase the activity of PON1.                                                                                                                                                 | Genetic plan (S32) weak                  | PON1: high                             |
| Broccoli shoot powder ⊕<br><b>Normal</b>                        | <b>GSTM1:</b> probable null<br><b>NFE2L2 (NRF2):</b> normal | favourable response        | Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.                                                                                            | Gasper 2005; genetic plan (S25) moderate | GSTM1: moderate<br>NFE2L2 (NRF2): high |
| ★ sulforaphane (glucoraphanine + myrosinase) ⊕<br><b>Normal</b> | <b>GSTM1: probable null</b><br><b>NFE2L2 (NRF2): normal</b> | <b>favourable response</b> | Current stack (glucorphanin 100 mg) — plan S25: KEEP. The same in the morning. Probable null GSTM1: isothiocyanates are more in circulation (associated with more effect); NRF2 –617 C/C: KEAP1–NRF2 intact. Maintain.             | Gasper 2005; genetic plan (S25) moderate | GSTM1: moderate<br>NFE2L2 (NRF2): high |

### No known genetic relevance (35):

anthocyanins  
capsaicin (CYP2E1/1A2) — CYP2E1: normal  
diosmin  
equal (S-equil)  
ericytrin (lime extract)  
spermidine (wheat germ)  
aronia extract  
cranberry extract (PAC)  
extract of maqui  
blueberry extract  
grape seed extract (OPC)

genistein (UGT)  
gingerols  
hesperetin  
hesperidin  
holokiol / magnol  
soybean isoflavones (UGT; microbiota (equil))  
isoflavones of clover (UGT)  
kaempferol (UGT)  
linseed lignans (SDG) (microbiota)  
methylhesperidin chalcone  
neohesperidin dihydrochalcone

oleanolic  
oleocanthal  
oligonol (lychee)  
salidroside  
sesamine (CYP)  
tangerine/nobiletin (PMF citrus fruits)  
theaflavins (black tea)  
uroletin A (microbiota)  
betulinic acid  
ellagic acid (microbiota)  
ferulic acid

ursolic acid

garlic oil (CYP2E1 (inhib)) CYP2E1: normal

3.6 Carotenoids and pigments — 13 supplements

dose adjustment: 1 · normal: 5 · without known genetic relevance: 7

| Supplement                   | Genes (state)                                     | Predicted response                                       | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                      | Source · evidence                                                                    | Trust                         |
|------------------------------|---------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------|
| ★ lycopene<br>Adjust dose    | BCO1: slightly ↓ conversion                       | lower dose (≤ 10 mg)                                     | Current stack (35 mg) — plan S23: ADJUST (descend). 10 mg or stop. Genetic levels of lycopene tendentially high: no genetic indication for 35 mg. 10 mg or tomato cooked with olive oil (suggestion, not applied).                                                         | Genetic plan (S23)<br>weak                                                           | BCO1: high                    |
| alpha-carotene<br>Normal     | BCO1: slightly ↓ conversion                       | slightly reduced conversion, without practical relevance | BCO1 R267S het + rs6564851 G/T: conversion to retinol slightly smaller; A379V absent. It is the preferred form of vitamin A with isotretinoin (regulated conversion). Food or low dose; β-carotene at a high dose does not bring benefits (and is not advised in smokers). | BCO1 studies (Leung 2009; Lietz 2012); NIH ODS moderate (genotype) · weak (practice) | BCO1: high                    |
| beta-cryptoxanthin<br>Normal | BCO1: slightly ↓ conversion                       | slightly reduced conversion, without practical relevance | BCO1 R267S het + rs6564851 G/T: conversion to retinol slightly smaller; A379V absent. It is the preferred form of vitamin A with isotretinoin (regulated conversion). Food or low dose; β-carotene at a high dose does not bring benefits (and is not advised in smokers). | BCO1 studies (Leung 2009; Lietz 2012); NIH ODS moderate (genotype) · weak (practice) | BCO1: high                    |
| ★ lutein<br>Normal           | CFH/ARMS2: CFH het<br>BCO1: slightly ↓ conversion | useful (retina)                                          | Current stack (20 mg) — plan S24: KEEP. The same, with fat. CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.                                                                                                                           | Genetic plan (S24); AREDS2<br>weak                                                   | CFH/ARMS2: high<br>BCO1: high |
| meso-zeaxanthin<br>Normal    | CFH/ARMS2: CFH het<br>BCO1: slightly ↓ conversion | useful (retina)                                          | CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.                                                                                                                                                                                       | Genetic plan (S24); AREDS2<br>weak                                                   | CFH/ARMS2: high<br>BCO1: high |
| ★ zeaxanthin<br>Normal       | CFH/ARMS2: CFH het<br>BCO1: slightly ↓ conversion | useful (retina)                                          | Current stack (4 mg) — plan S24: KEEP. The same, with fat. CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.                                                                                                                            | Genetic plan (S24); AREDS2<br>weak                                                   | CFH/ARMS2: high<br>BCO1: high |

No known genetic relevance (7):

★ astaxanthin  
betalains (beet)  
cantaxanthin

chlorophylline  
crocin/crocetin (saffron)  
phycocyanin (blue spirulin)

fucoxanthin

### 3.7 Western medicinal plants — 140 supplements

prevention: 6 · Monitoring: 21 · indeterminate: 1 · normal: 7 · without known genetic relevance: 105

| Supplement                                        | Genes (state)                                                                                                         | Predicted response              | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                     | Source · evidence                                            | Trust                                                                |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------|
| greater celandine<br><b>Avoid</b>                 | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | avoid (liver)                   | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                             | LiverTox moderate                                            | PNPLA3/Pi*MZ (liver): moderate                                       |
| comfrey (conferred)<br><b>Avoid</b>               | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | avoid (pyrrolizidine alkaloids) | Pyrrolizidine alkaloids cause veno-occlusive liver disease; oral use is prohibited in several countries. Genetic liver reinforces.                                                                                                                                        | EMA/HMPC; LiverTox strong                                    | PNPLA3/Pi*MZ (liver): moderate                                       |
| germander (Teucrium)<br><b>Avoid</b>              | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | avoid (liver)                   | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                             | LiverTox moderate                                            | PNPLA3/Pi*MZ (liver): moderate                                       |
| St. John's wort<br><b>Avoid</b>                   | <b>CYP3A4:</b> normal<br><b>CYP3A5:</b> expressor (*1/*3)<br><b>CYP2C19:</b> NM<br><b>TERT/melanoma:</b> unfavourable | avoid (strong inducer)          | Strong inducer of CYP3A4, P-gp and CYP2C19: low cyclosporine/tacrolimus 30–60% → predictable descent of sirolimus, which has already reduced exposure to CYP3A5 expressor; also low tadalafil. Photosensitizing hypericin (p88 + isotretinoin) (suggestion, not applied). | EMA/FDA labels (interactions); Stockley strong (interaction) | CYP3A4: high<br>CYP3A5: high<br>CYP2C19: high<br>TERT/melanoma: high |
| pore (oil)<br><b>Avoid</b>                        | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | avoid (liver)                   | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                             | LiverTox moderate                                            | PNPLA3/Pi*MZ (liver): moderate                                       |
| tussiling<br><b>Avoid</b>                         | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | avoid (pyrrolizidine alkaloids) | Pyrrolizidine alkaloids cause veno-occlusive liver disease; oral use is prohibited in several countries. Genetic liver reinforces.                                                                                                                                        | EMA/HMPC; LiverTox strong                                    | PNPLA3/Pi*MZ (liver): moderate                                       |
| garlic/oil<br><b>Monitoring</b>                   | <b>CYP3A5:</b> expressor (*1/*3)                                                                                      | P-gp/CYP3A4 inducer             | Allicin of fresh extracts induces P-gp/CYP3A4 (lowered saquinavir ~50%): it can lower sirolimus. Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                    | Piscitelli 2002 moderate (interaction)                       | CYP3A5: high                                                         |
| aloe vera (oral)<br><b>Monitoring</b>             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | monitoring liver                | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                   | LiverTox weak                                                | PNPLA3/Pi*MZ (liver): moderate                                       |
| chinese angelic (quang quai)<br><b>Monitoring</b> | <b>CYP3A5:</b> expressor (*1/*3)<br><b>TERT/melanoma:</b> unfavourable                                                | furanocoumarins                 | Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi). Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                             | Pharmacology of furanocoumarins weak                         | CYP3A5: high<br>TERT/melanoma: high                                  |
| pocket<br><b>Monitoring</b>                       | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | monitoring liver                | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                   | LiverTox weak                                                | PNPLA3/Pi*MZ (liver): moderate                                       |
| willow bark (salicin)<br><b>Monitoring</b>        | <b>UGT1A6:</b> *2 het<br><b>CYP2C9:</b> IM (*1/*3)                                                                    | salicylate                      | Salicylate is metabolized by UGT1A6 (*2 het) and CYP2C9 (*1/*3); it adds antiplatelet effect to fish oil, garlic, ginger and pine bark extract.                                                                                                                           | Pharmacology of salicylate weak                              | UGT1A6: high<br>CYP2C9: high                                         |
| cimicifiga (black cohosh)<br><b>Monitoring</b>    | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | monitoring liver                | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                   | LiverTox weak                                                | PNPLA3/Pi*MZ (liver): moderate                                       |
| cascara-sagrada<br><b>Monitoring</b>              | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                             | only short term                 | Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.                                                                                                                                                 | LiverTox; EMA/HMPC weak                                      | PNPLA3/Pi*MZ (liver): moderate                                       |

| Supplement                                                | Genes (state)                                                                                             | Predicted response              | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                               | Source · evidence                                   | Trust                                            |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| yerba (Melissa) + valerian<br><b>Monitoring</b>           | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                 | monitoring liver (valerian)     | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used. (valerian/scullery in mixtures).                                                                                                                            | LiverTox weak                                       | PNPLA3/Pi*MZ (liver): moderate                   |
| American listener (skullcap)<br><b>Monitoring</b>         | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                 | monitoring liver                | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                             | LiverTox weak                                       | PNPLA3/Pi*MZ (liver): moderate                   |
| Cervul thorn (frangula)<br><b>Monitoring</b>              | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                 | only short term                 | Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.                                                                                                                                                           | LiverTox; EMA/HMPC weak                             | PNPLA3/Pi*MZ (liver): moderate                   |
| hydrate (goldenseal)<br><b>Monitoring</b>                 | <b>CYP2D6:</b> Probable NM (without CNV)<br><b>CYP3A5:</b> expressor (*1/*3)                              | strong inhibitor CYP3A4/2D6     | Hydrastin/berberine: clinical inhibition of CYP3A4 and CYP2D6 (Gurley 2008). Interaction with sirolimus; the probable normal CYP2D6 (without CNV) can become phenoconverted. Never in the rapamycin window (3 days before 7 days after each dose).                                  | Gurley 2008 moderate (interaction)                  | CYP2D6: moderate<br>CYP3A5: high                 |
| mahonia (Oregon grape)<br><b>Monitoring</b>               | <b>CYP2D6:</b> Probable NM (without CNV)<br><b>CYP2C9:</b> IM (*1/*3)<br><b>CYP3A5:</b> expressor (*1/*3) | the genetic plan advises to add | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012); with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose). | Guo 2012; genetic plan (b.3) moderate (interaction) | CYP2D6: moderate<br>CYP2C9: high<br>CYP3A5: high |
| pelargonium sidoides<br><b>Monitoring</b>                 | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                 | monitoring liver                | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                             | LiverTox weak                                       | PNPLA3/Pi*MZ (liver): moderate                   |
| petasitis (butterbur, without PA)<br><b>Monitoring</b>    | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                 | monitoring liver                | Even extracts without PA had cases of hepatitis (LiverTox); with genetic liver, short use.                                                                                                                                                                                          | LiverTox (butterbur) weak                           | PNPLA3/Pi*MZ (liver): moderate                   |
| angelica root (Angelica arcangelica)<br><b>Monitoring</b> | <b>CYP3A5:</b> expressor (*1/*3)<br><b>TERT/melanoma:</b> unfavourable                                    | furanocoumarins                 | Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi). Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                                       | Pharmacology of furanocoumarins weak                | CYP3A5: high<br>TERT/melanoma: high              |
| rhubarb (root)<br><b>Monitoring</b>                       | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                 | only short term                 | Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.                                                                                                                                                           | LiverTox; EMA/HMPC weak                             | PNPLA3/Pi*MZ (liver): moderate                   |
| ruta (ruda)<br><b>Monitoring</b>                          | <b>CYP3A5:</b> expressor (*1/*3)<br><b>TERT/melanoma:</b> unfavourable                                    | furanocoumarins                 | Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi). Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                                       | Pharmacology of furanocoumarins weak                | CYP3A5: high<br>TERT/melanoma: high              |
| sene<br><b>Monitoring</b>                                 | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                 | only short term                 | Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.                                                                                                                                                           | LiverTox; EMA/HMPC weak                             | PNPLA3/Pi*MZ (liver): moderate                   |
| meadowsweet<br><b>Monitoring</b>                          | <b>UGT1A6:</b> *2 het<br><b>CYP2C9:</b> IM (*1/*3)                                                        | salicylate                      | Salicylate is metabolized by UGT1A6 (*2 het) and CYP2C9 (*1/*3); it adds antiplatelet effect to fish oil, garlic, ginger and pine bark extract.                                                                                                                                     | Pharmacology of salicylate weak                     | UGT1A6: high<br>CYP2C9: high                     |
| grape (brerberis)<br><b>Monitoring</b>                    | <b>CYP2D6:</b> Probable NM (without CNV)<br><b>CYP2C9:</b> IM (*1/*3)<br><b>CYP3A5:</b> expressor (*1/*3) | the genetic plan advises to add | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012); with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose). | Guo 2012; genetic plan (b.3) moderate (interaction) | CYP2D6: moderate<br>CYP2C9: high<br>CYP3A5: high |

| Supplement                                   | Genes (state)                                | Predicted response                  | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                             | Source · evidence                              | Trust                                              |
|----------------------------------------------|----------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------------------------------------|
| valerian<br><b>Monitoring</b>                | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable    | monitoring liver                    | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                           | LiverTox weak                                  | PNPLA3/Pi*MZ (liver): moderate                     |
| licorice (root)<br><b>Undetermined</b>       | <b>HSD11B2:</b> indeterminate                | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks). | EMA/HMPC; EFSA (glycyrrhizin) not determinable | HSD11B2: non-determinable                          |
| deglycerizin (DGL)<br><b>Normal</b>          | <b>HSD11B2:</b> indeterminate                | without glicirrizine                | DGL has no glicirrizine: no mineralocorticoid effect.                                                                                                                                                                             | EMA/HMPC weak                                  | HSD11B2: non-determinable                          |
| alfalfa<br><b>Normal</b>                     | <b>VKORC1:</b> TT (sensitive)                | only relevant with coumarins        | Vitamin K/coumarins: irrelevant without anticoagulant (VKORC1 TT).                                                                                                                                                                | CPIC warfarin context weak                     | VKORC1: high                                       |
| green oat (Avena sativa)<br><b>Normal</b>    | <b>HLA-DQ2/DQ8:</b> negative                 | without celiac HLA                  | Without DQ2.5 or DQ8: very low genetic risk of celiac disease; gluten/cereals tolerated.                                                                                                                                          | Genetic plan (e.6) strong                      | HLA-DQ2/DQ8: high                                  |
| cardo-marian (integer seed)<br><b>Normal</b> | <b>UGT1A1:</b> IM (*1/*28)                   | food dose                           | Whole/milled seed: low dose of silymarin.                                                                                                                                                                                         | Pharmacokinetics weak                          | UGT1A1: high                                       |
| ginkgo biloba<br><b>Normal</b>               | <b>CYP2C19:</b> NM                           | Normal CYP2C19                      | Ginkgo induces CYP2C19 (omeprazole –40%); with CYP2C19 *1/*1 and without drug CYP2C19 in the stack, without practical effect. Light antiplatelet effect (sum of fish oil).                                                        | Yin 2004 weak                                  | CYP2C19: high                                      |
| mellilote (church clover)<br><b>Normal</b>   | <b>VKORC1:</b> TT (sensitive)                | only relevant with coumarins        | Vitamin K/coumarins: irrelevant without anticoagulant (VKORC1 TT).                                                                                                                                                                | CPIC warfarin context weak                     | VKORC1: high                                       |
| cinchona<br><b>Normal</b>                    | <b>G6PD:</b> normal<br><b>CYP3A4:</b> normal | Normal G6PD                         | Quinine: hemolysis in G6PD deficit (normal here); metabolized by CYP3A4 (normal). Arrhythmias/thrombocytopenia do not depend on these genes.                                                                                      | FDA quinine label moderate (G6PD)              | G6PD: high for the tested variants<br>CYP3A4: high |

### No known genetic relevance (105):

absinthe (artemisia)  
 agnocasto (chasteberry)  
 agrimony  
 agripalma (motherwort)  
 AKBA (boswelina enriched in AKBA)  
 artichoke (leaf)  
 rosemary (carnosic acid)  
 leachate  
 alquemyl  
 love-perfect (heartsease)  
 Anise-star  
 artemisia (mugwort)  
 burdock  
 birch (leave/seed)  
 calendula (oral)  
 chamomile  
 cardo-santo

oak bark  
 sweet orange peel  
 horse chestnut (aescin)  
 horse (thioaminase)  
 centaurea-menor  
 Java tea (Orthosiphon)  
 cimbalaria / hera (leaf)  
 cranberry (concentrated juice)  
 D-mannose  
 dandelion (root)  
 echinacea (CYP3A (module, weak)  
 pale echinacea/angustifolia (CYP3A)  
 eucalyptus (oral oil) (CYP)  
 eufasia  
 elderberry flower  
 raspberry leaf  
 walnut leaf

leaf of nettle  
 red grape leaf  
 fumitory  
 fennel  
 Devil's claw (CYP2C9 (inhib, in vitro)) CYP2C9: IM (\*1/\*3)  
 gerenciane  
 ★ **ginger**  
 ginger + equinacea (immunity) (CYP3A)  
 Butcher's broom  
 grass  
 grindelia  
 blackberries (leaves)  
 hamamelis (oral)  
 hibiscus  
 hyssop  
 peppermint (enteric oil) (CYP3A4) CYP3A4: normal  
 wild yam

oral lavender (standard oil)  
Levistic  
lobelia  
lima  
hops  
brown-white  
black maroon  
matrix (ferrafew)  
melissa (lemon balm)  
millefolium  
myrrh  
blueberry (leaf)  
island moss  
black walnut (fruit bark)  
olmo-americano (slippery elm)  
olibane (Boswellia serrata)  
ononis (restarrow)  
Calyformia poppy

passiflora (passion fruit)  
pyloriteiro (pinheiro-alvar)  
primule (root)  
propolis (extract)  
pygeum  
rose (marshmallow)  
root of nettle  
rhodiola rosea (CYP3A4/2C9 (inhibits, in vitro)) CYP3A4: normal  
rooibs  
rosemary (hyp rose)  
elderberry (berry)  
parsley (leave/root) (furanocoumarins)  
salsaparril  
saw palmetto (Serenoa)  
pumpkin seed (extract)  
always-novel (knotgrass)  
sawdust (wildworm)  
sage

tanching  
Menor tanching (ribwort)  
tea tree (oral no; topic)  
Thyme  
tormentil  
red clover  
tilapia (flower)  
white nettle  
Ursin grape  
Gold rod  
fundaments  
verbena  
viburn (crack)  
violet  
visco (Viscum album)  
juniper  
Innula (elecampane)  
Oregano oil

### 3.8 Ayurvedic plants — 47 supplements

prevention: 6 · Monitoring: 5 · normal: 2 · without known genetic relevance: 34

| Supplement                                               | Genes (state)                                                                   | Predicted response                 | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                             | Source · evidence                                   | Trust                                                 |
|----------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|
| ashwagandha (full spectrum root extract)<br><b>Avoid</b> | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                       | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025). | LiverTox (ashwagandha); genetic plan (b.3) moderate | PNPLA3/Pi*MZ (liver): moderate                        |
| ashwagandha (root and leaf extract)<br><b>Avoid</b>      | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                       | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025). | LiverTox (ashwagandha); genetic plan (b.3) moderate | PNPLA3/Pi*MZ (liver): moderate                        |
| ashwagandha (root)<br><b>Avoid</b>                       | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                       | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025). | LiverTox (ashwagandha); genetic plan (b.3) moderate | PNPLA3/Pi*MZ (liver): moderate                        |
| babchi (Psoralea corylyfolia)<br><b>Avoid</b>            | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>TERT/melanoma:</b> unfavourable | avoid (liver + photosensitization) | Babchi: photosensitizing psoralens (p88 + isotretinoin) and hepatitis (LiverTox) cases.                                                                                                                                                                                           | LiverTox (Psoralea) moderate                        | PNPLA3/Pi*MZ (liver): moderate<br>TERT/melanoma: high |
| brahmi + ashwagandha (combined)<br><b>Avoid</b>          | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                       | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Ashwagandha: growing cholestatic cases (2017–2025). | LiverTox (ashwagandha); genetic plan (b.3) moderate | PNPLA3/Pi*MZ (liver): moderate                        |
| giloy / guduchi (Tinospora cordifolia)<br><b>Avoid</b>   | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                       | avoid (liver)                      | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                     | LiverTox moderate                                   | PNPLA3/Pi*MZ (liver): moderate                        |
| gotu kola (centella)<br><b>Monitoring</b>                | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                       | monitoring liver                   | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                           | LiverTox weak                                       | PNPLA3/Pi*MZ (liver): moderate                        |
| guggul<br><b>Monitoring</b>                              | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>CYP3A5:</b> expressor (*1/*3)   | CYP3A4 + liver inducer             | Guggulsterone activates PXR (induces CYP3A4: it can lower sirolimus) and there are cases of hepatitis. Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                      | LiverTox; Brobst 2004 weak                          | PNPLA3/Pi*MZ (liver): moderate<br>CYP3A5: high        |
| mucuna pruriens (L-dopa)<br><b>Monitoring</b>            | <b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC                       | L-dopa: COMT/ homocysteine         | The L-dopa is O-methylated by COMT (Val/Val, high) spending SAM and up to homocysteine; Compound MTHFR and p92 score reinforce.                                                                                                                                                   | L-dopa × homocysteine literature moderate           | COMT: high<br>MTHFR: high                             |
| pippali (long pepper)<br><b>Monitoring</b>               | <b>CYP3A5:</b> expressor (*1/*3)<br><b>UGT1A1:</b> IM (*1/*28)                  | increases drug exposure            | Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements. Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                            | Bhardwaj 2002 moderate (interaction)                | CYP3A5: high<br>UGT1A1: high                          |
| trikatu (peppers + ginger)<br><b>Monitoring</b>          | <b>CYP3A5:</b> expressor (*1/*3)<br><b>UGT1A1:</b> IM (*1/*28)                  | increases drug exposure            | Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements. Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                            | Bhardwaj 2002 moderate (interaction)                | CYP3A5: high<br>UGT1A1: high                          |

| Supplement                                             | Genes (state)                                                            | Predicted response | Recommendation (summary) and implication for Sérgio F                                                                                            | Source · evidence                              | Trust                                 |
|--------------------------------------------------------|--------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------|
| curcuma longa (paste/<br>golden milk)<br><b>Normal</b> | <b>UGT1A1</b> : IM (*1/*28)<br><b>HLA-B*35:01</b> :<br>probable negative | OK cooking dose    | Culinary quantities of curcuma: not relevant for UGT1A1 *28 or liver.                                                                            | LiverTox (curcuma)<br>weak                     | UGT1A1: high<br>HLA-B*35:01: moderate |
| yellow melon (karela)<br><b>Normal</b>                 | <b>G6PD</b> : normal                                                     | Normal G6PD        | The melon-yellow seeds have vicin (G6PD deficit haemolysis); Normal G6PD<br>→ without genetic risk. It is added to the drugs that lower glucose. | CPIC G6PD 2022<br>(context)<br>moderate (G6PD) | G6PD: high for the<br>tested variants |

No known genetic relevance (34):

|                              |                                |                      |
|------------------------------|--------------------------------|----------------------|
| ajwain                       | coleus forskohlii (forskoline) | nagarmotha           |
| amla (brown-indian)          | fenugreek                      | neem                 |
| andrographis (kalmegh) (CYP) | garcinia indica (kokum)        | punarnava            |
| arjuna (shell)               | gymnema sylvestre              | safe musli           |
| ashoka (Saraca)              | haritaki                       | Shankhpushpi         |
| assa-fetida (hing)           | jatamansi                      | shatavari            |
| bacopa (brahmi)              | kalonji (nigel seed)           | shilajit             |
| bhringraj                    | kutki (Picrorhiza kurroa)      | tribulus terrestris  |
| bibhitaki                    | lodhra                         | tryphal              |
| cardamom (extract)           | manjistha                      | tulsi (sacred basil) |
| chyawanprash                 | moringa (leaf)                 | vivari kand          |
|                              | musli-negro                    |                      |

### 3.9 Traditional Chinese medicine plants and Kampo — 76 supplements

prevention: 8 · Monitoring: 10 · indeterminate: 5 · without known genetic relevance: 53

| Supplement                                                            | Genes (state)                                                                      | Predicted response           | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                     | Source · evidence                             | Trust                                                   |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------|
| He Shou Wu (Polygonum multiflorum, fo-ti)<br><b>Avoid</b>             | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>HLA-B*35:01:</b> probable negative | avoid (liver)                | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Polygonum multiflorum: HLA-B*35:01-linked lesion (probable negative), but intrinsic hepatotoxicity remains. | LiverTox; Li 2019 (HLA-B*35:01) moderate      | PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate |
| Jin Bu Huan (tetra-hydropalmatin)<br><b>Avoid</b>                     | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                          | avoid (liver)                | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                                                             | LiverTox moderate                             | PNPLA3/Pi*MZ (liver): moderate                          |
| astragalosid IV<br><b>Avoid</b>                                       | <b>TERT/melanoma:</b> unfavourable                                                 | avoid (telomerase activator) | Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma has low polygenic risk: consensus p18), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).              | Genetic plan (d); GWAS TERT weak (precaution) | TERT/melanoma: high                                     |
| astralis (root)<br><b>Avoid</b>                                       | <b>TERT/melanoma:</b> unfavourable                                                 | avoid (telomerase activator) | Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma has low polygenic risk: consensus p18), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).              | Genetic plan (d); GWAS TERT weak (precaution) | TERT/melanoma: high                                     |
| cycloastragenol<br><b>Avoid</b>                                       | <b>TERT/melanoma:</b> unfavourable                                                 | avoid (telomerase activator) | Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma has low polygenic risk: consensus p18), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).              | Genetic plan (d); GWAS TERT weak (precaution) | TERT/melanoma: high                                     |
| purified extract of astragalum (telomerase activator)<br><b>Avoid</b> | <b>TERT/melanoma:</b> unfavourable                                                 | avoid (telomerase activator) | Used as telomerase activators. With 5 alleles TERT of oncologic risk and nevi at the 88th percentile (melanoma has low polygenic risk: consensus p18), precaution is asymmetric (hypothetical benefit vs theoretical risk). Avoid purified extracts of astragalus, cycloastragenol and astragalus (suggestion, not applied).              | Genetic plan (d); GWAS TERT weak (precaution) | TERT/melanoma: high                                     |
| ma huang (ephedra)<br><b>Avoid</b>                                    | <b>CETP/9p21/APOB:</b> genetic vascular risk<br><b>NOS3:</b> −786CT + 298GT        | avoid                        | Prohibited sympathomimetics (UE/EUA): Stroke/farce; 9p21 risk homozygous and NOS3 reinforce.                                                                                                                                                                                                                                              | FDA 2004; NCCIH strong                        | CETP/9p21/APOB: high<br>NOS3: high                      |
| tripterygium (thunder Godvine)<br><b>Avoid</b>                        | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                          | avoid                        | Tripterygium: hepatic, renal and reproductive toxicity; immunosuppressant.                                                                                                                                                                                                                                                                | NCCIH; LiverTox moderate                      | PNPLA3/Pi*MZ (liver): moderate                          |
| Sho-saiko-to / Xiao Chai Hu Tang<br><b>Monitoring</b>                 | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                          | monitoring liver             | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/ Pi*MZ), use only for short periods, if used.                                                                                                                                                                                                                  | LiverTox weak                                 | PNPLA3/Pi*MZ (liver): moderate                          |

| Supplement                                                     | Genes (state)                                                                                               | Predicted response                  | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                               | Source · evidence                                   | Trust                                            |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| angelica dahurica (bai zhi)<br><b>Monitoring</b>               | <b>CYP3A5</b> :expressor (*1/*3)<br><b>TERT/melanoma</b> : unfavourable                                     | furanocoumarins                     | Furanocoumarins: inhibition of CYP3A4 and photosensitization (p88 + isotretinoin nevi). Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                                       | Pharmacology of furanocoumarins weak                | CYP3A5: high<br>TERT/melanoma: high              |
| bupleur<br><b>Monitoring</b>                                   | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                    | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                             | LiverTox weak                                       | PNPLA3/Pi*MZ (liver): moderate                   |
| coptis (huang lian)<br><b>Monitoring</b>                       | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3) | the genetic plan advises to add     | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose). | Guo 2012; genetic plan (b.3) moderate (interaction) | CYP2D6: moderate<br>CYP2C9: high<br>CYP3A5: high |
| corydalis (yan hu suo)<br><b>Monitoring</b>                    | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | monitoring liver                    | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                             | LiverTox weak                                       | PNPLA3/Pi*MZ (liver): moderate                   |
| Chinese mantle (Scutellaria baicalensis)<br><b>Monitoring</b>  | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | cases of liver injury               | Scutellaria baicalensis was associated with liver injury (LiverTox); with genetic liver, avoiding prolonged use.                                                                                                                                                                    | LiverTox (hearing) weak                             | PNPLA3/Pi*MZ (liver): moderate                   |
| schisandra (Schisandra chinensis)<br><b>Monitoring</b>         | <b>CYP3A5</b> :expressor (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                              | inhibits CYP3A4/P-gp                | Schisandra rises the levels of tacrolimus (used for this in China); by analogy, it rises the sirolimus. Never in the rapamycin window (3 days before 7 days after each dose).                                                                                                       | Wuzhi x tacrolimus studies moderate (interaction)   | CYP3A5: high<br>PNPLA3/Pi*MZ (liver): moderate   |
| hu zhang (Polygonum cuspidatum, knotweed)<br><b>Monitoring</b> | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | only short term                     | Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.                                                                                                                                                           | LiverTox; EMA/HMPC weak                             | PNPLA3/Pi*MZ (liver): moderate                   |
| phellodendron<br><b>Monitoring</b>                             | <b>CYP2D6</b> : Probable NM (without CNV)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>CYP3A5</b> :expressor (*1/*3) | the genetic plan advises to add     | It inhibits CYP2D6, CYP2C9 and CYP3A4 in humans (Guo 2012): with CYP2C9 *1/*3 and sirolimus (CYP3A5 expressor), unpredictable exposures; it is added to 4 drugs that lower glucose (suggestion, not applied). Never in the rapamycin window (3 days before 7 days after each dose). | Guo 2012; genetic plan (b.3) moderate (interaction) | CYP2D6: moderate<br>CYP2C9: high<br>CYP3A5: high |
| cassava seed (jue ming zi)<br><b>Monitoring</b>                | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                  | only short term                     | Anthraquinones: chronic use associated with rare liver injury and melanotic colon; with genetic liver, only punctual use.                                                                                                                                                           | LiverTox; EMA/HMPC weak                             | PNPLA3/Pi*MZ (liver): moderate                   |
| Hochuekkito (Kampo)<br><b>Undetermined</b>                     | <b>HSD11B2</b> : indeterminate                                                                              | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).                                                   | EMA/HMPC; EFSA (glycyrrhizin) not determinable      | HSD11B2: non-determinable                        |
| Juzentaiho (Kampo)<br><b>Undetermined</b>                      | <b>HSD11B2</b> : indeterminate                                                                              | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).                                                   | EMA/HMPC; EFSA (glycyrrhizin) not determinable      | HSD11B2: non-determinable                        |
| Kakkonto (Kampo)<br><b>Undetermined</b>                        | <b>HSD11B2</b> : indeterminate                                                                              | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks).                                                   | EMA/HMPC; EFSA (glycyrrhizin) not determinable      | HSD11B2: non-determinable                        |

| Supplement                                      | Genes (state)                    | Predicted response                  | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                             | Source · evidence                                 | Trust                     |
|-------------------------------------------------|----------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------|
| Shakuyakukanzoto (Kampo)<br><b>Undetermined</b> | <b>HSD11B2:</b><br>indeterminate | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks). | EMA/HMPC; EFSA (glycyrrhizin)<br>not determinable | HSD11B2: non-determinable |
| Yokukansan (Kampo)<br><b>Undetermined</b>       | <b>HSD11B2:</b><br>indeterminate | non-valuable individual sensitivity | Glycyrrhizin inhibits 11β-HSD2 → sodium retention and hypokalemia; sensitivity varies between people (HSD11B2), but the chip does not read it. With telmisartan and genetic sensitivity to salt, avoid prolonged use (> 2 weeks). | EMA/HMPC; EFSA (glycyrrhizin)<br>not determinable | HSD11B2: non-determinable |

No known genetic relevance (53):

|                                                              |                                                                      |                                    |
|--------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------|
| achyranthes                                                  | evodia (evodiamine) (CYP1A2/3A4) — CYP1A2: normal to intermediate    | ligusticum (chuanxiong)            |
| blackberry (fruit, sang shen)                                | lotus leaf                                                           | ligustrum                          |
| ashitaba (chalcones)                                         | forsythia                                                            | madresilva (lonicera)              |
| atractylodes                                                 | gardenia (zhi zi) (genipina)                                         | morinda officinalis                |
| magnolia peel                                                | gastrodia (tian ma)                                                  | notoginseng (Panax notoginseng)    |
| tangerine peel (chen pi)                                     | american ginseng                                                     | ophiopogon (male dong)             |
| chrysanthemum tea                                            | ginseng (Panax ginseng) (CYP3A4 (module); ginsenoids) CYP3A4: normal | white peony (bai shao)             |
| cystanche                                                    | black ginseng                                                        | Chinese pestiler (shan zha)        |
| codonopsis (dang shen)                                       | siberian ginseng (eleuter)                                           | pinellia                           |
| coix (yi yi ren)                                             | Korean red ginseng (CYP3A4) CYP3A4: normal                           | polygonatum                        |
| cuscuta                                                      | goji (Lycium barbarum) (CYP2C9 (inhib, cases)) CYP2C9: IM (*1/*3)    | polygala (yuan zhi)                |
| Daikenchuto (Kampo)                                          | houttuynia                                                           | Pueraria mirifica (phytoestrogens) |
| danshen (Salvia miltiorrhiza) (CYP; antiplatelet)            | huperzine A (CYP1A2) — CYP1A2: normal to intermediate                | rehmannia                          |
| dendrobium                                                   | isatis (ban lan gen)                                                 | scrophilia (xuan shen)             |
| dioscorea (shan yao)                                         | jiaogulan (Gynostemma)                                               | lotus seed                         |
| epimedium (horny drop weed, icariin) (PDE5 (inhibits, weak)) | jujube (fruit)                                                       | black sesame                       |
| Seabream (sea buckthorn)                                     | kudzu (Pueraria lobata)                                              | yin chen (Artemisia capillaris)    |
| eucommia                                                     |                                                                      | ziphus (suan zao ren)              |

### 3.10 Plants of the Americas and Africa; superfoods — 61 supplements

prevention: 3 · dose adjustment: 6 · Monitoring: 5 · normal: 9 · without known genetic relevance: 38

| Supplement                                   | Genes (state)                                                                                                             | Predicted response          | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                              | Source · evidence                          | Trust                                                             |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|
| pharmacaria (Larrea)<br><b>Avoid</b>         | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                 | avoid (liver)               | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                                                                                                                      | LiverTox moderate                          | PNPLA3/Pi*MZ (liver): moderate                                    |
| kava (Piper methylicum)<br><b>Avoid</b>      | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>CYP2D6:</b> Probable NM (without CNV)                                     | avoid (liver)               | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied). Kava: severe hepatitis and transplants; removal in several countries.                                                                                                | LiverTox (kava) moderate                   | PNPLA3/Pi*MZ (liver): moderate<br>CYP2D6: moderate                |
| kratom (Mitragyna speciosa)<br><b>Avoid</b>  | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>CYP2D6:</b> Probable NM (without CNV)<br><b>OPRM1:</b> A/G                | avoid                       | Atypical opioid (dependence), cholestatic liver injury (LiverTox); CYP3A4/2D6 (CNV not evaluated) and OPRM1 A/G.                                                                                                                                                                                                                                                                                   | LiverTox (kratom); FDA moderate            | PNPLA3/Pi*MZ (liver): moderate<br>CYP2D6: moderate<br>OPRM1: high |
| green coffee (extract)<br><b>Adjust dose</b> | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C<br><b>COMT:</b> Val/Val (high)<br><b>MTHFR:</b> 677CT/1298AC | caffeine + chlorogenic acid | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). The concentrated chlorogenic acid can rise to homocysteine (COMT/MTHFR). | Cornelis 2006; Olthof 2001 moderate        | CYP1A2: high<br>ADORA2A: high<br>COMT: high<br>MTHFR: high        |
| yerba mate<br><b>Adjust dose</b>             | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                              | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                          | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high                                     |
| guarana<br><b>Adjust dose</b>                | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                              | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                          | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high                                     |
| guayusa<br><b>Adjust dose</b>                | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                              | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                          | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high                                     |
| walnut<br><b>Adjust dose</b>                 | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                              | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                          | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high                                     |
| yaupon<br><b>Adjust dose</b>                 | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                              | limit dose and time         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                          | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high                                     |

| Supplement                                             | Genes (state)                                                                 | Predicted response                       | Recommendation (summary) and implication for Sérgio F                                                                                                                        | Source · evidence                  | Trust                                         |
|--------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------|
| cinnamon-cassava (coumarin)<br><b>Monitoring</b>       | <b>CYP2A6</b> : probable normal<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable | hepatotoxic coumarin                     | Cassia cinnamon has coumarin (EFSA fixed a low tolerable daily intake); 7-hydroxylation depends on CYP2A6 (*2/*9 absent; non-visible deletions). To prefer cinnamon-do-ceil. | EFSA 2008 (coumarin) weak-moderate | CYP2A6: low<br>PNPLA3/Pi*MZ (liver): moderate |
| yohimbine HCl<br><b>Monitoring</b>                     | <b>CYP2D6</b> : Probable NM (without CNV)                                     | CYP2D6 (confirm CNV)                     | Yohimbine depends on CYP2D6: slow metabolizers have often higher exposure. Probable *1/*2 (normal), but not evaluated CNV; is also cardiovascular stimulant.                 | Le Corre 2004 moderate             | CYP2D6: moderate                              |
| noni<br><b>Monitoring</b>                              | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                    | monitoring liver                         | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                      | LiverTox weak                      | PNPLA3/Pi*MZ (liver): moderate                |
| concentrated beet juice (nitrate)<br><b>Monitoring</b> | <b>NOS3</b> : –786CT + 298GT                                                  | avoid concentrated nitrate (tadalafil)   | NOS3 gives rational, but concentrated inorganic nitrate is added to the tadalafil (hypotension). Only vegetal nitrates (suggestion, not applied).                            | Genetic plan (b.3, f.13) weak      | NOS3: high                                    |
| yohimbe (shell)<br><b>Monitoring</b>                   | <b>CYP2D6</b> : Probable NM (without CNV)                                     | CYP2D6 (confirm CNV)                     | Yohimbine depends on CYP2D6: slow metabolizers have often higher exposure. Probable *1/*2 (normal), but not evaluated CNV; is also cardiovascular stimulant.                 | Le Corre 2004 moderate             | CYP2D6: moderate                              |
| beet powder<br><b>Normal</b>                           | <b>NOS3</b> : –786CT + 298GT                                                  | OK food dose                             | Nitrate quantity of a food powder is moderate; with tadalafil, avoid "shot" doses.                                                                                           | Genetic plan (f.13) weak           | NOS3: high                                    |
| cocoa (theobromine extract)<br><b>Normal</b>           | <b>CYP1A2</b> : normal to intermediate                                        | theobromine (CYP1A2)                     | Theobromine is metabolized by CYP1A2 (normal to intermediate) but it is a weak stimulant; counting in large amounts.                                                         | Pharmacology weak                  | CYP1A2: high                                  |
| raw cocoa powder<br><b>Normal</b>                      | <b>CYP1A2</b> : normal to intermediate                                        | theobromine (CYP1A2)                     | Theobromine is metabolized by CYP1A2 (normal to intermediate) but it is a weak stimulant; counting in large amounts.                                                         | Pharmacology weak                  | CYP1A2: high                                  |
| Cinnamon<br><b>Normal</b>                              | <b>CYP2A6</b> : probable normal                                               | low coumarin                             | Ceylon cinnamon almost without coumarin.                                                                                                                                     | EFSA 2008 weak                     | CYP2A6: low                                   |
| chlorella<br><b>Normal</b>                             | <b>VKORC1</b> : TT (sensitive)                                                | vitamin K (only relevant with coumarins) | Rich in vitamin K; it only weighs if you take coumarins (VKORC1 TT).                                                                                                         | Genetic plan (e.6) weak            | VKORC1: high                                  |
| Spirulina<br><b>Normal</b>                             | <b>PAH</b> : no variants                                                      | normal for the tested variants           | No pathogenic variants of PAH among those tested (improbable phenylketonuria).                                                                                               | ClinVar (chip) moderate            | PAH: partial                                  |
| sacha inchi (oil)<br><b>Normal</b>                     | <b>FADS1/FADS2</b> : high activity                                            | good conversion ALA→EPA                  | With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.                                                | FADS × ALA studies moderate        | FADS1/FADS2: high                             |
| chia seeds<br><b>Normal</b>                            | <b>FADS1/FADS2</b> : high activity                                            | good conversion ALA→EPA                  | With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.                                                | FADS × ALA studies moderate        | FADS1/FADS2: high                             |
| ground linseed seeds<br><b>Normal</b>                  | <b>FADS1/FADS2</b> : high activity                                            | good conversion ALA→EPA                  | With high activity FADS, the conversion of ALA/SDA to EPA is one of the best; however, it does not replace the preformed DHA.                                                | FADS × ALA studies moderate        | FADS1/FADS2: high                             |

#### No known genetic relevance (38):

açaí  
baobab (post)  
hypoxis (Hypoxys) (CYP3A4 (inhib, in vitro)) CYP3A4: normal  
Buchu

camu camu  
catuaba  
tart cherry  
chanca piedra (Phyllanthus)

chuchuhuasi  
clove  
damiana  
jackfruit (barley grass)

wheatgrass (wheatgrass)  
stevia  
Yellow leaf (Vernonia)  
Cat claw (CYP3A4 (inhibits, weak)) CYP3A4: normal  
soursop  
hoodia gorgonii  
jatoba  
kanna (Sceletium tortuosum)  
lucuma

maca (Lepidium meyenii)  
black maca  
irvingia (Irvingia gabonensis)  
mangostion  
manuka honey  
Wild blueberry powder  
muira puama  
pau d'archo (lapacho)  
kayenne pepper (CYP1A2) — CYP1A2: normal to intermediate

quinoa / amaranth powder  
hemp seeds  
abstract (Brazilian ginseng)  
supergreen powder (greens) (various)  
sutherlandia  
Cider vinegar (capsules)  
yacon (syrup/FOS)  
castor oil (oral)

3.11 Medicinal mushrooms — 21 supplements

Monitoring: 1 · normal: 3 · without known genetic relevance: 17

| Supplement                                     | Genes (state)                      | Predicted response       | Recommendation (summary) and implication for Sérgio F                                                                   | Source · evidence                | Trust                          |
|------------------------------------------------|------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------|
| reishi (Ganoderma lucidum)<br>Monitoring       | PNPLA3/Pi*MZ (liver): unfavourable | monitoring liver         | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used. | LiverTox weak                    | PNPLA3/Pi*MZ (liver): moderate |
| mushroom king trumpet (ergothionein)<br>Normal | SLC22A4/5 (OCTN1/2): typical       | typical capture          | OCTN1 (SLC22A4) L503 C/C: typical ergothioneine uptake. Maintain.                                                       | Genetic plan (S20) weak          | SLC22A4/5 (OCTN1/2): high      |
| ★ ergothioneine<br>Normal                      | SLC22A4/5 (OCTN1/2): typical       | typical capture          | Current stack (10 mg) — plan S20: KEEP. The same. OCTN1 (SLC22A4) L503 C/C: typical ergothioneine uptake. Maintain.     | Genetic plan (S20) weak          | SLC22A4/5 (OCTN1/2): high      |
| pleurotes (mushroom-ostral)<br>Normal          | SLC01B1: increased function        | natural lovastatin (low) | SLC01B1 *1/*14 (augmented function): without genetic risk with low natural lovastatin.                                  | CPIC statins 2022 (context) weak | SLC01B1: high                  |

No known genetic relevance (17):

|                                      |                                    |                             |
|--------------------------------------|------------------------------------|-----------------------------|
| agaricus blazei (sun mushroom)       | cordyceps militaris                | mesima (Phellinus linteus)  |
| AHCC (cultured shiitake extract)     | cordyceps sinensis (CS-4)          | 7-10 mushroom mixture       |
| yeast beta-glucan (1.3/1.6)          | enoki                              | ear (Auricularia)           |
| peru tail (Trametes, PSK/PSP)        | lion's mantle (Hericium erinaceus) | poria coconuts              |
| chaga (Inonatus obliquus) (oxalates) | lentin                             | shiitake (Lentinula edodes) |
|                                      | mayake (Grifola frondosa)          | trenella                    |

### 3.12 Probiotics, Prebiotics and Fibers — 69 Supplements

dose adjustment: 2 · normal: 20 · without known genetic relevance: 47

| Supplement                                              | Genes (state)                                                                            | Predicted response              | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                 | Source · evidence                     | Trust                                                       |
|---------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------|
| ★ PHGG (partially hydrolyzed guar gum) ⊕<br>Adjust dose | TCF7L2/FTO/GCKR: unfavourable<br>CETP/9p21/APOB: genetic vascular risk<br>FUT2: secretor | adjust dose (plan)              | Current stack (6 g) — plan S35: ADJUST (subir). 10 g/day. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better.                                                                | Genetic plan (S35) strong             | TCF7L2/FTO/GCKR: high<br>CETP/9p21/APOB: high<br>FUT2: high |
| ★ psyllium (ispaghula bark) ⊕<br>Adjust dose            | TCF7L2/FTO/GCKR: unfavourable<br>CETP/9p21/APOB: genetic vascular risk<br>FUT2: secretor | adjust dose (plan)              | Current stack (10 g) — plan S35: ADJUST (subir). 15–20 g/day in 2–3 takes with water; drugs ≥ 1 h before or 2–4 h after. TCF7L2/FTO/MTNR1B/GCKR (postprandial glycosis) and CETP/9p21/APOB (low to LDL). More fiber is always better. | Genetic plan (S35) strong             | TCF7L2/FTO/GCKR: high<br>CETP/9p21/APOB: high<br>FUT2: high |
| 2'-fucosyl-lactose (HMO)<br>Normal                      | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |
| Akkermansia muciniphila (pasteurized)<br>Normal         | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |
| Bifidobacterium animalis lactis BB-12<br>Normal         | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |
| Bifidobacterium bifidum<br>Normal                       | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |
| Bifidobacterium breve<br>Normal                         | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |
| Bifidobacterium lactis HN019<br>Normal                  | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |
| Bifidobacterium longum<br>Normal                        | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |
| Bifidobacterium longum 1714 (psychobiotic)<br>Normal    | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |
| Bifidobacterium longum subsp.<br>Normal                 | FUT2: secretor                                                                           | typical colonization (secretor) | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                                  |

| Supplement                                                        | Genes (state)                                                                          | Predicted response                     | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                 | Source · evidence                     | Trust                                         |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------|
| L. helveticus R0052 + B. longum R0175<br><b>Normal</b>            | <b>FUT2</b> : secretor                                                                 | typical colonization (secretor)        | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                    |
| Lactacaseibacillus casei (fermented milk strain)<br><b>Normal</b> | <b>AOC1 (DAO)</b> : slightly ↓<br><b>HNMT</b> : Thr105Ile het                          | histamine: only if there are symptoms  | Some strains produce histamine; AOC1 and HNMT heterozygotes (light reduction). No restriction, except symptoms.                                                                                                       | Genetic plan (e.6) weak               | AOC1 (DAO): high<br>HNMT: high                |
| Lactobacillus delbrueckii bulgaricus<br><b>Normal</b>             | <b>AOC1 (DAO)</b> : slightly ↓<br><b>HNMT</b> : Thr105Ile het                          | histamine: only if there are symptoms  | Some strains produce histamine; AOC1 and HNMT heterozygotes (light reduction). No restriction, except symptoms.                                                                                                       | Genetic plan (e.6) weak               | AOC1 (DAO): high<br>HNMT: high                |
| Limosylactobacillus reuteri (DSM 17938)<br><b>Normal</b>          | <b>AOC1 (DAO)</b> : slightly ↓<br><b>HNMT</b> : Thr105Ile het                          | histamine: only if there are symptoms  | Some strains produce histamine; AOC1 and HNMT heterozygotes (light reduction). No restriction, except symptoms.                                                                                                       | Genetic plan (e.6) weak               | AOC1 (DAO): high<br>HNMT: high                |
| oat beta-glucan ☺<br><b>Normal</b>                                | <b>CETP/9p21/APOB</b> : genetic vascular risk<br><b>TCF7L2/FTO/GCKR</b> : unfavourable | useful for LDL and glucose             | β-glucan 3 g/day low to LDL (genetic priority) and postprandial glucose.                                                                                                                                              | EFSA (β-glucan) strong                | CETP/9p21/APOB: high<br>TCF7L2/FTO/GCKR: high |
| barley beta-glucan ☺<br><b>Normal</b>                             | <b>CETP/9p21/APOB</b> : genetic vascular risk<br><b>TCF7L2/FTO/GCKR</b> : unfavourable | useful for LDL and glucose             | β-glucan 3 g/day low to LDL (genetic priority) and postprandial glucose.                                                                                                                                              | EFSA (β-glucan) strong                | CETP/9p21/APOB: high<br>TCF7L2/FTO/GCKR: high |
| bovine colostrum<br><b>Normal</b>                                 | <b>LCT</b> : persistent                                                                | Lactose tolerated                      | LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.                                                                                                                                          | Genetic plan (e.6) strong             | LCT: high                                     |
| Wheat bran/wheat fiber<br><b>Normal</b>                           | <b>HLA-DQ2/DQ8</b> : negative                                                          | without celiac HLA                     | Without DQ2.5 or DQ8: very low genetic risk of celiac disease; gluten/cereals tolerated.                                                                                                                              | Genetic plan (e.6) strong             | HLA-DQ2/DQ8: high                             |
| kefir (grains/liophilized)<br><b>Normal</b>                       | <b>LCT</b> : persistent                                                                | Lactose tolerated                      | LCT –13910 C/T: persistence of lactase, without genetic lactose intolerance.                                                                                                                                          | Genetic plan (e.6) strong             | LCT: high                                     |
| high potency probiotic (8 strains formula)<br><b>Normal</b>       | <b>FUT2</b> : secretor                                                                 | typical colonization (secretor)        | FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products.                                                         | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                    |
| ★ multistrain (generic)<br><b>Normal</b>                          | <b>FUT2</b> : secretor                                                                 | <b>typical colonization (secretor)</b> | Current stack (multistrain) — S36 plan: KEEP. The same. FUT2 secretor (W143X heterozygous): normal intestinal fucosylation, typically bifidobacteria-rich microbiota; non-secretors benefit more from these products. | Wacklin 2011; genetic plan (S36) weak | FUT2: high                                    |

#### No known genetic relevance (47):

resistant starch (raw potato fecule) (microbiota)  
larch arabinogalactan (microbiota)  
Bacillus clausii (microbiota)  
Bacillus coagulans (microbiota)  
Bacillus subtilis (spores) (microbiota)  
sodium butyrate/calcium  
Clostridium butyricum (MIYAIRI 588) (microbiota)

E. coli Nissle 1917 (microbiota)  
Enterococcus faecium SF68 (microbiota)  
acacia fiber (arabic gum) (microbiota)  
FOS (fruit-oligosaccharides) (microbiota)  
glucomannan (konjac)  
GOS (galacto-oligosaccharide) (microbiota)  
bovine serum immunoglobulins (SBI)

inulin (microbiota)  
L. rhamnosus GR-1 + L. reuteri RC-14 (microbiota)  
Lactacaseibacillus paracasei (microbiota)  
Lactacaseibacillus rhamnosus HN001 (microbiota)  
Lactiplantibacillus plantarum (other strains) (microbiota)  
Lactobacillus acidophilus (NCFM) (microbiota)  
Lactobacillus crispatus (microbiota)

Lactobacillus gasseri (microbiota)  
Lactobacillus helveticus (microbiota)  
Lactobacillus johnsonii (microbiota)  
Lactobacillus plantarum 299v (microbiota)  
Lactobacillus rhamnosus GG (microbiota)  
Lactococcus lactis (microbiota)  
lactoferrin (iron)  
lactulose (microbiota)  
Levilactobacillus brevis (microbiota)

Ligilactobacillus salivarius (microbiota)  
Limosylactobacillus fermentum (microbiota)  
methylcellulose  
pectin (incl. modified citrus pectin) (microbiota)  
Pediococcus acidilactici (microbiota)  
polydextrose (microbiota)  
probiotic of spores (multistrain) (microbiota)  
Propionibacterium freudenreichii (microbiota)  
potato protein (PI2 satiety)

postbiotic (L. paracasei inactivated)  
chitosan  
Saccharomyces boulardii (microbiota)  
symbiotic (probiotic + prebiotic) (microbiota)  
Streptococcus salivarius K12 (oral) (oral microbiota)  
Streptococcus thermophilus (microbiota)  
★ **tributyryne**  
XOS (xylo-oligosaccharides) (microbiota)

3.13 Enzymes — 23 supplements

indeterminate: 1 · normal: 5 · without known genetic relevance: 17

| Supplement                                  | Genes (state)                                 | Predicted response         | Recommendation (summary) and implication for Sérgio F                                                                 | Source · evidence                    | Trust                          |
|---------------------------------------------|-----------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------|
| amylase<br>Undetermined                     | AMY1: indeterminate                           | non-valuable AMY1 copies   | The number of copies of AMY1 (salivary amylase) is not visible in the chip.                                           | AMY1 Literature not determinable     | AMY1: non-determinable         |
| AN-PEP (prolyl endopeptidase)<br>Normal     | HLA-DQ2/DQ8: negative                         | without genetic indication | Without celiac HLA; these enzymes do not treat celiac disease in any way.                                             | Genetic plan (e.6) strong            | HLA-DQ2/DQ8: high              |
| DAO (diamine oxidase) in capsules<br>Normal | AOC1 (DAO): slightly ↓<br>HNMT: Thr105Ile het | optional if symptoms       | AOC1 Thr16Met/His645Asp and HNMT Thr105Ile heterozygotes: slight reduction; DAO only if there are histamine symptoms. | Genetic plan (e.6); §12 profile weak | AOC1 (DAO): high<br>HNMT: high |
| DPP-IV (gluten enzyme)<br>Normal            | HLA-DQ2/DQ8: negative                         | without genetic indication | Without celiac HLA; these enzymes do not treat celiac disease in any way.                                             | Genetic plan (e.6) strong            | HLA-DQ2/DQ8: high              |
| lactase<br>Normal                           | LCT: persistent                               | not necessary              | Lactase persistence (LCT –13910 C/T): no genetic requirement.                                                         | Genetic plan (e.6) strong            | LCT: high                      |
| melon superoxide dismutase<br>Normal        | SOD2: Ala/Ala                                 | without genetic indication | SOD2 Ala/Ala (efficient mitochondrial import): without deficit to compensate.                                         | Profile §16 weak                     | SOD2: high                     |

No known genetic relevance (17):

|                                |                                        |                             |
|--------------------------------|----------------------------------------|-----------------------------|
| alpha-galactosidase            | digestive enzymes (mixture)            | pancreatine                 |
| bromelain                      | systemic enzymes (proteolytic formula) | papain                      |
| bile of cattle (biliary salts) | invertase/sucrose                      | pepsin                      |
| catalase (CAT) — CAT: normal   | lipase                                 | fungal protease             |
| cellulase/hemicelulase         | lumbrokinase                           | serrapeptase                |
|                                | nattokinase                            | xylose isomerase (fructose) |

### 3.14 Nootropics, stimulants, sleep and mood — 32 supplements

dose adjustment: 10 · Monitoring: 3 · indeterminate: 1 · normal: 4 · without known genetic relevance: 14

| Supplement                                              | Genes (state)                                                                                               | Predicted response                          | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Source · evidence                          | Trust                                                       |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------|
| caffeine + L-theanine<br><b>Adjust dose</b>             | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                             | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high                               |
| anhydrous caffeine<br><b>Adjust dose</b>                | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                             | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high                               |
| prolonged release caffeine<br><b>Adjust dose</b>        | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                                                | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied).                                                                                                                                                                             | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high                               |
| magnesium + B6 (cognition/stress)<br><b>Adjust dose</b> | <b>PDXK/ALPL/NBPF3 (B6):</b> typical/high<br><b>MTHFR:</b> 677CT/1298AC<br><b>DHFR:</b> slow reduction (CC) | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                                                                                                                                                                                                                       | Genetic plan (S3, S6); EFSA UL B6 moderate | PDXK/ALPL/NBPF3 (B6): high<br>MTHFR: high<br>DHFR: moderate |
| melatonin (immediate release)<br><b>Adjust dose</b>     | <b>MTNR1B:</b> C/G (carrier)<br><b>CYP1A2:</b> normal to intermediate                                       | low dose (0.3–1 mg)                         | MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).                                                                                                                                                                                          | Garaulet 2015; genetic plan (S30) moderate | MTNR1B: high<br>CYP1A2: high                                |
| prolonged release melatonin<br><b>Adjust dose</b>       | <b>MTNR1B:</b> C/G (carrier)<br><b>CYP1A2:</b> normal to intermediate                                       | low dose (0.3–1 mg)                         | MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).                                                                                                                                                                                          | Garaulet 2015; genetic plan (S30) moderate | MTNR1B: high<br>CYP1A2: high                                |
| ★ melatonin high dose (≥ 10 mg)<br><b>Adjust dose</b>   | <b>MTNR1B:</b> C/G (carrier)<br><b>CYP1A2:</b> normal to intermediate                                       | <b>low (0.3–1 mg)</b>                       | Current stack (up to 100 mg) — plan S30: ADJUST (desert much). 0.3–1 mg, 30–60 min before bedtime. MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied). Doses ≥ 10 mg (up to 100 mg) have no genetic basis and prolong the effect on insulin. | Garaulet 2015; genetic plan (S30) moderate | MTNR1B: high<br>CYP1A2: high                                |
| melatonin in spray/sublingual<br><b>Adjust dose</b>     | <b>MTNR1B:</b> C/G (carrier)<br><b>CYP1A2:</b> normal to intermediate                                       | low dose (0.3–1 mg)                         | MTNR1B rs10830963 C/G: in carrier G, melatonin reduces insulin secretion (5 mg worsened glucose tolerance in carriers); Garaulet 2015). Normal CYP1A2 extends it (apigenin, with suggested pause, would add up). 0.3–1 mg, 30–60 min before bedtime and ≥ 3 h after the last meal (suggestion, not applied).                                                                                                                                                                                          | Garaulet 2015; genetic plan (S30) moderate | MTNR1B: high<br>CYP1A2: high                                |

| Supplement                                       | Genes (state)                                                                                                                       | Predicted response                          | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                     | Source · evidence                                  | Trust                                                                           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------|
| paraxanthine<br><b>Adjust dose</b>               | <b>CYP1A2</b> : normal to intermediate<br><b>ADORA2A</b> : C/C                                                                      | limit dose and time                         | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). | Cornelis 2006; genetic plan (e.4) moderate         | CYP1A2: high<br>ADORA2A: high                                                   |
| tryptophan + B6 (combined)<br><b>Adjust dose</b> | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                      | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is ≤ 5 mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                                                           | Genetic plan (S3, S6); EFSA UL B6 moderate         | PDXK/ALPL/NBPF3 (B6): high<br>MTHFR: high<br>DHFR: moderate                     |
| CBD (cannabidiol)<br><b>Monitoring</b>           | <b>CYP2C19</b> : NM<br><b>CYP2C9</b> : IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>CYP3A5</b> :expressor (*1/*3) | inhibits CYP; transaminase                  | CBD is metabolized by CYP2C19 (*1/*1, normal) and inhibits CYP2C19/2C9/3A4; increased levels of tacrolimus/everolimus in cases. At a high dose, transaminases (Epidiolex trials) are relevant with genetic liver. Never in the rapamycin window (3 days before 7 days after each dose).                                   | Epidiolex label; clinical cases moderate           | CYP2C19: high<br>CYP2C9: high<br>PNPLA3/Pi*MZ (liver): moderate<br>CYP3A5: high |
| galantamine (winter bell)<br><b>Monitoring</b>   | <b>CYP2D6</b> : Probable NM (without CNV)                                                                                           | CYP2D6 (confirm CNV)                        | Galantamine is a substrate of CYP2D6/3A4; probable normal without CNV.                                                                                                                                                                                                                                                    | FDA label moderate                                 | CYP2D6: moderate                                                                |
| Hemp oil (total spectrum)<br><b>Monitoring</b>   | <b>CYP2C19</b> : NM<br><b>CYP2C9</b> : IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>CYP3A5</b> :expressor (*1/*3) | inhibits CYP; transaminase                  | CBD is metabolized by CYP2C19 (*1/*1, normal) and inhibits CYP2C19/2C9/3A4; increased levels of tacrolimus/everolimus in cases. At a high dose, transaminases (Epidiolex trials) are relevant with genetic liver. Never in the rapamycin window (3 days before 7 days after each dose).                                   | Epidiolex label; clinical cases moderate           | CYP2C19: high<br>CYP2C9: high<br>PNPLA3/Pi*MZ (liver): moderate<br>CYP3A5: high |
| nicotine (chipelets/gums)<br><b>Undetermined</b> | <b>CYP2A6</b> : probable normal                                                                                                     | debugging not fully evaluated               | Nicotine is eliminated by CYP2A6 (*2/*9 absent; non-visible *4 deletions): non-determinable clearance rate. Dependence; no indication.                                                                                                                                                                                    | CYP2A6 Literature not determinable                 | CYP2A6: low                                                                     |
| L-theanine + magnesium (sleep)<br><b>Normal</b>  | <b>TRPM6</b> : typical                                                                                                              | typical absorption                          | TRPM6 rs3750425 C/C and rs2274924 T/T (variants associated with low magnesium absent). Any form serves; glycinate and threonate for sleep, citrate if constipation.                                                                                                                                                       | TRPM6 (Song 2009) studies; genetic plan (S13) weak | TRPM6: high                                                                     |
| methylene blue (low dose)<br><b>Normal</b>       | <b>G6PD</b> : normal                                                                                                                | Normal G6PD                                 | Normal G6PD: no genetic risk of hemolysis. MAO-A inhibitor: not to combine with serotonergic drugs.                                                                                                                                                                                                                       | CPIC G6PD 2022 strong (G6PD)                       | G6PD: high for the tested variants                                              |
| bacopa + ginkgo (combined)<br><b>Normal</b>      | <b>CYP2C19</b> : NM                                                                                                                 | Normal CYP2C19                              | Ginkgo induces CYP2C19 (omeprazole –40%); with CYP2C19 *1/*1 and without drug CYP2C19 in the stack, without practical effect. Light antiplatelet effect (sum of fish oil).                                                                                                                                                | Yin 2004 weak                                      | CYP2C19: high                                                                   |
| citicoline (patented form) ☺<br><b>Normal</b>    | <b>PEMT</b> : V175M het<br><b>FMO3</b> : typical<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                      | Choline preferred form                      | PEMT V175M het + genetic liver give rational to choline; this form generates little TMAO. 250–500 mg of choline/day in total (suggestion, not applied).                                                                                                                                                                   | Genetic plan (S38) weak-moderate                   | PEMT: high<br>FMO3: high<br>PNPLA3/Pi*MZ (liver): moderate                      |

#### No known genetic relevance (14):

saffron (extract, affron)

bacopa + rhodiola (combined)

CBG (cannabigerol) (CYP (inhib, in vitro)

CBN (cannabinol) (CYP2C9/3A4) — CYP2C9: IM (\*1/\*3)  
centrofenoxin  
DMAE (dimethylaminoethanol)  
Catnip

GABA + theanine (combined)  
methylliberine (ADORA2A) — ADORA2A: C/C  
nootropic mixture (commercial stack) (various)

PQQ (pyrroloquinoline quinone)  
teacrine (ADORA2A) — ADORA2A: C/C  
uridine monophosphate  
vinpocetin

### 3.15 Sport and performance — 36 supplements

prevention: 2 · dose adjustment: 4 · Monitoring: 7 · normal: 7 · without known genetic relevance: 16

| Supplement                                                 | Genes (state)                                                               | Predicted response              | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                     | Source · evidence                          | Trust                              |
|------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------|
| DMHA / DMAA (prohibited stimuli)<br><b>Avoid</b>           | <b>CETP/9p21/APOB:</b> genetic vascular risk<br><b>NOS3:</b> −786CT + 298GT | avoid                           | Prohibited sympathomimetics (UE/EUA): Stroke/farce; 9p21 risk homozygous and NOS3 reinforce.                                                                                                                                                                                                                              | FDA 2004; NCCIH strong                     | CETP/9p21/APOB: high<br>NOS3: high |
| prohormones/SARMs (sold as supplement)<br><b>Avoid</b>     | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                   | avoid                           | Synthesis steroids and SARMs: severe cholestatic lesion (LiverTox) and hormonal axis; genetic liver reinforces. They are not legal supplements in UE.                                                                                                                                                                     | LiverTox strong                            | PNPLA3/Pi*MZ (liver): moderate     |
| electrolytes with caffeine<br><b>Adjust dose</b>           | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                | limit dose and time             | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high      |
| energy gel with caffeine<br><b>Adjust dose</b>             | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                | limit dose and time             | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high      |
| pre-training with caffeine<br><b>Adjust dose</b>           | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                | limit dose and time             | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high      |
| tauro-caffeine (energy drink powder)<br><b>Adjust dose</b> | <b>CYP1A2:</b> normal to intermediate<br><b>ADORA2A:</b> C/C                | limit dose and time             | CYP1A2 *1F het + rs2472297 C/C: normal to intermediate caffeine clearance (no slow genotype C/C); ADORA2A C/C: less sensitive to caffeine; the limit comes from sleep (insomnia p80). ≤ 200 mg/day of total caffeine, only until 12h; apigenin (pause suggested; inhibits CYP1A2) prolonged it (suggestion, not applied). | Cornelis 2006; genetic plan (e.4) moderate | CYP1A2: high<br>ADORA2A: high      |
| BCAA + electrolytes<br><b>Monitoring</b>                   | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                        | the genetic plan advises to add | High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).                                                                                                                                         | Genetic plan (b.3) weak                    | TCF7L2/FTO/GCKR: high              |
| waxy corn starch<br><b>Monitoring</b>                      | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                        | prefer intact hydrates          | TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.                                                                                                                                                                                                                 | Genetic plan (e.1) weak                    | TCF7L2/FTO/GCKR: high              |
| gainer (hydrates + protein)<br><b>Monitoring</b>           | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                        | prefer intact hydrates          | TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.                                                                                                                                                                                                                 | Genetic plan (e.1) weak                    | TCF7L2/FTO/GCKR: high              |
| leucine + HMB (combined)<br><b>Monitoring</b>              | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                        | the genetic plan advises to add | High BCAA are associated with insulin resistance; with risk TCF7L2/FTO and genetic levels of leucine/valine tendentially high, prefer complete protein (suggestion, not applied).                                                                                                                                         | Genetic plan (b.3) weak                    | TCF7L2/FTO/GCKR: high              |
| maltodextrin/cyclic dextrin<br><b>Monitoring</b>           | <b>TCF7L2/FTO/GCKR:</b> unfavourable                                        | prefer intact hydrates          | TCF7L2/FTO/GCKR risk + 4 drugs that move on glucose: rapid absorption hydrates only during long training.                                                                                                                                                                                                                 | Genetic plan (e.1) weak                    | TCF7L2/FTO/GCKR: high              |

| Supplement                                         | Genes (state)                                                                     | Predicted response         | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                        | Source · evidence                                          | Trust                                    |
|----------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------|
| synephrine (Orange-yellow)<br><b>Monitoring</b>    | <b>CYP3A5</b> :expressor (*1/*3)<br><b>CETP/9p21/APOB</b> : genetic vascular risk | stimulant + CYP3A4         | Yellow orange furanocoumarins inhibit CYP3A4 (grapefruit effect); synephrine is a cardiovascular stimulant, especially with caffeine; 9p21 homozygous. Never in the rapamycin window (3 days before 7 days after each dose). | NCCIH (yellow orange) weak                                 | CYP3A5: high<br>CETP/9p21/APOB: high     |
| deer horn velvet (IGF-1)<br><b>Monitoring</b>      | <b>TERT/melanoma</b> : unfavourable                                               | IGF-1 axis                 | Sold by IGF-1: the plan avoids the GH/IGF-1 axis (risk TERT alleles).                                                                                                                                                        | Genetic plan (d) weak                                      | TERT/melanoma: high                      |
| D-ribose<br><b>Normal</b>                          | <b>AMPD1</b> : normal                                                             | without genetic indication | AMPD1 Q12X absent: no myoadenylate deaminase deficit (where ribose was studied).                                                                                                                                             | AMPD1 Literature weak                                      | AMPD1: high                              |
| L-arginine + L-ornithine<br><b>Normal</b>          | <b>NOS3</b> : –786CT + 298GT                                                      | rational NO (small effect) | NOS3 –786 T/C + Glu298Asp G/T: a slightly smaller production of NO. Oral arginine has low bioavailability; with tadalafil the vasodilator effect is added (mild).                                                            | NOS3 studies; NIH ODS (arginine) weak                      | NOS3: high                               |
| Nitric oxide booster (mix) ☺<br><b>Normal</b>      | <b>NOS3</b> : –786CT + 298GT                                                      | rational NO                | NOS3 –786 T/C + Glu298Asp: citrulline rises arginine better than arginine itself. With tadalafil, slight vasodilator sum.                                                                                                    | NOS3 studies; citrulline trials weak                       | NOS3: high                               |
| pre-training without stimulants ☺<br><b>Normal</b> | <b>NOS3</b> : –786CT + 298GT                                                      | rational NO                | NOS3 –786 T/C + Glu298Asp: citrulline rises arginine better than arginine itself. With tadalafil, slight vasodilator sum.                                                                                                    | NOS3 studies; citrulline trials weak                       | NOS3: high                               |
| ketone salts (BHB)<br><b>Normal</b>                | <b>APOE</b> : ε3/ε3                                                               | typical response           | APOE ε3/ε3: typical ketone metabolism (the ketogenic advantage described in ε4 does not apply).                                                                                                                              | APOE × ketones studies weak                                | APOE: high                               |
| whey + creatine (combined) ☺<br><b>Normal</b>      | <b>AMPD1</b> : normal<br><b>ACTN3</b> : R/X<br><b>GNMT</b> : TT                   | useful (muscle + save SAM) | Normal AMPD1; ACTN3 R/X; high fragility genetic score. Exogenous creatine saves SAM spent on endogenous synthesis (GAMT), useful with compound MTHFR/GNMT TT. Monohydrate 3–5 g is the form with evidence.                   | Genetic plan (S10); ISSN (muscle) · moderate (methylation) | AMPD1: high<br>ACTN3: high<br>GNMT: high |
| ketone esters (D-BHB)<br><b>Normal</b>             | <b>APOE</b> : ε3/ε3                                                               | typical response           | APOE ε3/ε3: typical ketone metabolism (the ketogenic advantage described in ε4 does not apply).                                                                                                                              | APOE × ketones studies weak                                | APOE: high                               |

### No known genetic relevance (16):

Oral ATP (PEAK ATP)  
 sodium bicarbonate (buffer)  
 capsimax (capsaicinoids)  
 sodium citrate  
 ecdysterone (beta-ecdysterone)  
 fadogia agreste

glycerol (hyperhydration)  
 hordenin (MAO-B)  
 inosine (uric acid)  
 isomaltulose  
 laxogenin

octopamine (adrenergic)  
 plant protein (mixture)  
 tongkat there (Eurycoma longifolia)  
 turksterone  
 phosphatidic acid

3.16 Hormones, precursors and glandulars — 15 supplements

prevention: 2 · without known genetic relevance: 13

| Supplement                           | Genes (state)                                                                      | Predicted response     | Recommendation (summary) and implication for Sérgio F                          | Source · evidence                | Trust                                                     |
|--------------------------------------|------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------|
| desiccated liver (capsules)<br>Avoid | HFE: H63D het<br>BCO1: slightly ↓ conversion<br>PNPLA3/Pi*MZ (liver): unfavourable | avoid (retinol + iron) | Desiccated liver concentrates retinol (isotretinoin) and iron heme (HFE H63D). | Genetic plan (b.3, e.3) moderate | HFE: high<br>BCO1: high<br>PNPLA3/Pi*MZ (liver): moderate |
| organ glandular (mixture)<br>Avoid   | HFE: H63D het<br>BCO1: slightly ↓ conversion<br>PNPLA3/Pi*MZ (liver): unfavourable | avoid (retinol + iron) | Desiccated liver concentrates retinol (isotretinoin) and iron heme (HFE H63D). | Genetic plan (b.3, e.3) moderate | HFE: high<br>BCO1: high<br>PNPLA3/Pi*MZ (liver): moderate |

No known genetic relevance (13):

|                                                                                                                  |                                                                                                                             |                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-keto-DHEA<br>calcium-D-glucarate (β-glucuronidase)<br>DHEA (CYP17/CYP19; SULT2A1)<br>glandular prostate/testis | glandular thymus<br>adrenal glandular<br>yam-savage (diogenin)<br>menopause isoflavones (UGT)<br>hops (8-prenilnaringenina) | ortho + saw palmetto (prostate)<br>pregnenolone (CYP17)<br>progesterone in cream (OTC) (CYP3A4) — CYP3A4: normal<br>desiccated thyroid (glandular) (deiodination) |
|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.17 Longevity, mitochondria and antioxidants — 21 supplements

prevention: 2 · indeterminate: 1 · normal: 8 · without known genetic relevance: 10

| Supplement                                               | Genes (state)                                                                                                                  | Predicted response            | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Source · evidence                                            | Trust                                                     |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------|
| apigenin + magnesium (sleep)<br><b>Avoid</b>             | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP3A5</b> :expressor (*1/*3)<br><b>TRPM6</b> : typical                                      | avoid / pause (by apigenin)   | Magnesium does not raise a problem (typical TRPM6), but for apigenin a test break (UGT1A1 *28) is suggested. CYP3A4/P-gp with sirolimus: pause while ALT is high (current protocol). Prefer magnesium without apigenin (suggestion, not applied).                                                                                                                                                                                                                                                                                                                 | Genetic plan (S13, S28) weak                                 | UGT1A1: high<br>CYP3A5: high<br>TRPM6: high               |
| synolytic combined (fisetin + quercetin)<br><b>Avoid</b> | <b>UGT1A1</b> : IM (*1/*28)<br><b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC<br><b>CYP3A5</b> :expressor (*1/*3) | avoid/pause                   | UGT1A1 *28 (quercetin inhibits UGT1A1), COMT Val/Val + MTHFR compound (catechol: methylation spent SAM and produces homocysteine) and inhibition of CYP3A4/P-gp (sirolimus). Genetic verdict: pause as an isolated supplement, by coherence with curcumin/luteolin; pause while ALT is high (current protocol) (suggestion, not applied). Coordination note: in the current protocol (decided by bot Supplements + Medicines based on analysis), the monthly senolytic pulse is fisetin + quercetin, only with normal ALT; quercetin is not definitively removed. | Genetic plan (S29 + coordination note) weak                  | UGT1A1: high<br>COMT: high<br>MTHFR: high<br>CYP3A5: high |
| carnosine + ALCAR (combined)<br><b>Undetermined</b>      | <b>CNDP1</b> : indeterminate                                                                                                   | indeterminate (carnosinase)   | Serum carnosinase (CNDP1) is regulated by a repetition (CTG) <sub>n</sub> that the chip does not read: the fraction of oral carnosine that arrives intact is not predictable. The plan maintains 2 g (1 g × 2).                                                                                                                                                                                                                                                                                                                                                   | CNDP1 Literature (Janssen 2005) not determinable             | CNDP1: non-determinable                                   |
| L-ergathione + PQQ (combined)<br><b>Normal</b>           | <b>SLC22A4/5</b> ( <b>OCTN1/2</b> ): typical                                                                                   | typical capture               | OCTN1 (SLC22A4) L503 C/C: typical ergothioneine uptake. Maintain.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Genetic plan (S20) weak                                      | SLC22A4/5 (OCTN1/2): high                                 |
| astaxanthin + DHA ☺<br><b>Normal</b>                     | <b>FADS1/FADS2</b> : high activity<br><b>APOE</b> : ε3/ε3                                                                      | EPA/DHA useful preformed      | FADS1/2 of high activity (plus AA from LA) → benefits from preformed EPA/DHA; APOE ε3/ε3 responds well to DHA; PNPLA3 I148M (Omega-3 helps in the liver). Total dose by coordination: 3.2–4 g/day of EPA+DHA (decision based on measurements) (suggestion, not applied).                                                                                                                                                                                                                                                                                          | Genetic plan (S1 + coordination note); FADS studies moderate | FADS1/FADS2: high<br>APOE: high                           |
| coenzyme Q10 (ubiquinone)<br><b>Normal</b>               | <b>NQO1</b> : *1/*1                                                                                                            | normal; no genetic indication | NQO1 *1/*1 (reduces ubiquinone to ubiquinol and activates idebenone normally); COQ2 without V393A. No genetic indication (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                               | Genetic plan (b.3) weak                                      | NQO1: high                                                |
| creatine + glycine (longevity) ☺<br><b>Normal</b>        | <b>CPS1</b> : low glycine<br><b>GSTM1</b> : probable null<br><b>GNMT</b> : TT                                                  | genetically useful glycine    | CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                  | GWAS glycine (CPS1); genetic plan (S8) moderate              | CPS1: high<br>GSTM1: moderate<br>GNMT: high               |
| glycine + NAC + taurine (combined) ☺<br><b>Normal</b>    | <b>CPS1</b> : low glycine<br><b>GSTM1</b> : probable null<br><b>GNMT</b> : TT                                                  | genetically useful glycine    | CPS1 rs715 T/T + rs1047891 C/C: genetically lower plasma glycine; Probable null GSTM1 (glycine is a glutathione substrate); GNMT TT. 5 g/day in total (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                  | GWAS glycine (CPS1); genetic plan (S8) moderate              | CPS1: high<br>GSTM1: moderate<br>GNMT: high               |
| idebenone<br><b>Normal</b>                               | <b>NQO1</b> : *1/*1                                                                                                            | normal; no genetic indication | NQO1 *1/*1 (reduces ubiquinone to ubiquinol and activates idebenone normally); COQ2 without V393A. No genetic indication (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                               | Genetic plan (b.3) weak                                      | NQO1: high                                                |
| ubiquinol<br><b>Normal</b>                               | <b>NQO1</b> : *1/*1                                                                                                            | normal; no genetic indication | NQO1 *1/*1 (reduces ubiquinone to ubiquinol and activates idebenone normally); COQ2 without V393A. No genetic indication (suggestion, not applied).                                                                                                                                                                                                                                                                                                                                                                                                               | Genetic plan (b.3) weak                                      | NQO1: high                                                |
| vitamin E + selenium<br><b>Normal</b>                    | <b>GPX1</b> : Pro198Leu het                                                                                                    | Nutritional dose              | GPX1 Pro198Leu het: activity slightly lower, but without benefit of high doses; selenium > 200 µg/day is associated with 2 type diabetes. The stack already gives 55 µg; without Brazil nuts.                                                                                                                                                                                                                                                                                                                                                                     | SELECT; genetic plan (S12) weak                              | GPX1: high                                                |

No known genetic relevance (10):

★ calcium alpha-ketoglutarate (Ca-AKG)

- sinolitic herbs (piperlongumin)
- fullerene C60 in olive oil
- molecular hydrogen (compressed)

- mitoquinone (CoQ10 directed to mitochondria)
- oxaloacetate
- powder of antioxidant berries (ORAC)

★ R-ALA (R-lipoic acid)

- trehalosis
- alpha-lipoic acid (racemic)

3.18 Articulations, skin, hair and connective tissue — 22 supplements

prevention: 1 · normal: 5 · without known genetic relevance: 16

| Supplement                                                      | Genes (state)                                                                                                                                           | Predicted response          | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                                                                                                                                                                                                   | Source · evidence                                                    | Trust                                                                                   |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| turmeric + boswellia (articular)<br><b>Avoid</b>                | <b>UGT1A1</b> : IM (*1/*28)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable<br><b>HLA-B*35:01</b> : probable negative<br><b>CYP3A5</b> :expressor (*1/*3) | avoid/pause                 | UGT1A1 *1/*28: curcumin is glucuronidated by UGT1A1 and inhibits it (sobe to bilirubin, confuses liver reading). Genetic liver (PNPLA3, MBOAT7, Pi*MZ). It inhibits CYP3A4/P-gp with unpredictable effect on sirolimus (CYP3A5 expressor). In favor: HLA-B*35:01 (allele of hepatic injury by curcuma) probable. It pauses while the ALT is high (current protocol), that is, until the ALT normalizes; never return to the rapamycin window (suggestion, not applied). | Genetic plan (S26); LiverTox (curcuma, HLA-B*35:01)<br>weak-moderate | UGT1A1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>HLA-B*35:01: moderate<br>CYP3A5: high |
| Polypodium leucotomos (oral photoprotection) ☺<br><b>Normal</b> | <b>TERT/melanoma</b> : unfavourable                                                                                                                     | Useful oral photoprotection | P88 + isotretinoin (photosensitivity): Polypodium reduces erythema UV in trials; it complements (not) the sunscreen.                                                                                                                                                                                                                                                                                                                                                    | Polypodium (Nestor 2015) trials<br>weak-moderate                     | TERT/melanoma: high                                                                     |
| wheat ceramides<br><b>Normal</b>                                | <b>HLA-DQ2/DQ8</b> : negative                                                                                                                           | without celiac HLA          | Without DQ2.5 or DQ8: very low genetic risk of celiac disease; gluten/cereals tolerated.                                                                                                                                                                                                                                                                                                                                                                                | Genetic plan (e.6)<br>strong                                         | HLA-DQ2/DQ8: high                                                                       |
| collagen + vitamin C<br><b>Normal</b>                           | <b>COL1A1/COL5A1</b> : normal                                                                                                                           | without risk alleles        | COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).                                                                                                                                                                                                                                                                                                                                                                                   | Genetic plan (S9)<br>weak                                            | COL1A1/COL5A1: high                                                                     |
| collagen type I+III (skin)<br><b>Normal</b>                     | <b>COL1A1/COL5A1</b> : normal                                                                                                                           | without risk alleles        | COL1A1 Sp1 C/C and COL5A1 T/T: without risk alleles; useful source of glycine (CPS1).                                                                                                                                                                                                                                                                                                                                                                                   | Genetic plan (S9)<br>weak                                            | COL1A1/COL5A1: high                                                                     |
| calcium + D3 + K2 (bone)<br><b>Normal</b>                       | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het                                                                           | typical absorption          | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                                                                                                                                                                                                                         | NIH ODS (calcium); genetic plan (S14)<br>weak                        | VDR: high<br>GC (DBP): high<br>HFE: high                                                |

No known genetic relevance (16):

ASU (unaponifiable of avocado-soy)  
high dose biotin (hair)  
shark cartilage  
chondroitin  
hair / skin / nails formula (biotin; several)

glucosamine (chloride)  
glucosamine (sulfate)  
glucosamine + chondroitin + MSM  
L-cysteine + B (hair)  
eggshell membrane (NEM)  
mussel of green lips (post)

keratin  
silicon (MMST)  
bamboo silica  
hyaluronic acid (oral)  
Marine pine oil (skin)

### 3.19 Metabolism, appetite and cardiovascular — 25 supplements

prevention: 5 · Monitoring: 7 · normal: 1 · without known genetic relevance: 12

| Supplement                                                 | Genes (state)                                                                                                                                   | Predicted response                     | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                                            | Source · evidence                                               | Trust                                                                                 |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| caffeine + green tea + garcinia (combined)<br><b>Avoid</b> | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>COMT:</b> Val/Val (high)<br><b>HLA-B*35:01:</b> probable negative<br><b>UGT1A1:</b> IM (*1/*28) | avoid (liver)                          | Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied). | LiverTox; EFSA 2018 (catechins); genetic plan (b.3)<br>moderate | PNPLA3/Pi*MZ (liver): moderate<br>COMT: high<br>HLA-B*35:01: moderate<br>UGT1A1: high |
| thermogenic green tea (extract)<br><b>Avoid</b>            | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>COMT:</b> Val/Val (high)<br><b>HLA-B*35:01:</b> probable negative<br><b>UGT1A1:</b> IM (*1/*28) | avoid (liver)                          | Concentrated green tea extract is a known cause of liver injury (LiverTox, probability B; risk in fasting and > 800 mg EGCG/day). Genetic liver (PNPLA3, MBOAT7, Pi*MZ); EGCG is catechol (COMT Val/Val). In favor: HLA-B*35:01 probable negative. Tea in drink is OK (suggestion, not applied). | LiverTox; EFSA 2018 (catechins); genetic plan (b.3)<br>moderate | PNPLA3/Pi*MZ (liver): moderate<br>COMT: high<br>HLA-B*35:01: moderate<br>UGT1A1: high |
| red rice yeast (monacolone K)<br><b>Avoid</b>              | <b>SLCO1B1:</b> increased function<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable<br><b>CYP3A5:</b> expressor (*1/*3)                             | avoid (unregulated statin)             | Monacolone K = lovastatin with non-standardized dose and possible citrinine. SLCO1B1 *1/*14 is favourable, but it already takes pitavastatin (statin duplication) and there are genetic livers and cases of hepatitis (suggestion, not applied).                                                 | LiverTox; EFSA 2018; genetic plan (b.3)<br>moderate             | SLCO1B1: high<br>PNPLA3/Pi*MZ (liver): moderate<br>CYP3A5: high                       |
| multi-ingredient thermogenic<br><b>Avoid</b>               | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                                       | avoid (liver)                          | Multi-ingredient thermogenic formulas were associated with acute hepatitis (LiverTox); genetic liver.                                                                                                                                                                                            | LiverTox<br>moderate                                            | PNPLA3/Pi*MZ (liver): moderate                                                        |
| Usnic acid (fat burner)<br><b>Avoid</b>                    | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                                       | avoid (liver)                          | Documented cause of liver injury (LiverTox). With PNPLA3 I148M + MBOAT7 + Pi*MZ probable, the genetic liver vulnerability is higher; the protective HSD17B13 attenuates but does not cancel. Avoid (suggestion, not applied).                                                                    | LiverTox<br>moderate                                            | PNPLA3/Pi*MZ (liver): moderate                                                        |
| Garcinia + green coffee (combined)<br><b>Monitoring</b>    | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                                       | monitoring liver                       | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/ Pi*MZ), use only for short periods, if used.                                                                                                                                                                         | LiverTox<br>weak                                                | PNPLA3/Pi*MZ (liver): moderate                                                        |
| sine detox tea<br><b>Monitoring</b>                        | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                                       | monitoring liver                       | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/ Pi*MZ), use only for short periods, if used.                                                                                                                                                                         | LiverTox<br>weak                                                | PNPLA3/Pi*MZ (liver): moderate                                                        |
| chromium + cinnamon (glucose)<br><b>Monitoring</b>         | <b>CYP2A6:</b> probable normal<br><b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                     | hepatotoxic coumarin                   | Cassia cinnamon has coumarin (EFSA fixed a low tolerable daily intake); 7-hydroxylation depends on CYP2A6 (*2/*9 absent; non-visible deletions). To prefer cinnamon-do-ceil.                                                                                                                     | EFSA 2008 (coumarin)<br>weak-moderate                           | CYP2A6: low<br>PNPLA3/Pi*MZ (liver): moderate                                         |
| garcinia cambogia (HCA)<br><b>Monitoring</b>               | <b>PNPLA3/Pi*MZ (liver):</b> unfavourable                                                                                                       | monitoring liver                       | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/ Pi*MZ), use only for short periods, if used.                                                                                                                                                                         | LiverTox<br>weak                                                | PNPLA3/Pi*MZ (liver): moderate                                                        |
| sodium nitrate/ potassium<br><b>Monitoring</b>             | <b>NOS3:</b> -786CT + 298GT                                                                                                                     | avoid concentrated nitrate (tadalafil) | NOS3 gives rational, but concentrated inorganic nitrate is added to the tadalafil (hypotension). Only vegetal nitrates (suggestion, not applied).                                                                                                                                                | Genetic plan (b.3, f.13)<br>weak                                | NOS3: high                                                                            |

| Supplement                                               | Genes (state)                                                                     | Predicted response           | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                        | Source · evidence                         | Trust                                   |
|----------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------|
| black pepper (extract)<br><b>Monitoring</b>              | <b>CYP3A5</b> :expressor (*1/*3)<br><b>UGT1A1</b> : IM (*1/*28)                   | increases drug exposure      | Piperine inhibits CYP3A4, P-gp and glucuronidation: up to levels of drugs and other supplements. Never in the rapamycin window (3 days before 7 days after each dose).                                                       | Bhardwaj 2002<br>moderate (interaction)   | CYP3A5: high<br>UGT1A1: high            |
| thermogenic (caffeine + synephrine)<br><b>Monitoring</b> | <b>CYP3A5</b> :expressor (*1/*3)<br><b>CETP/9p21/APOB</b> : genetic vascular risk | stimulant + CYP3A4           | Yellow orange furanocoumarins inhibit CYP3A4 (grapefruit effect); synephrine is a cardiovascular stimulant, especially with caffeine; 9p21 homozygous. Never in the rapamycin window (3 days before 7 days after each dose). | NCCIH (yellow orange)<br>weak             | CYP3A5: high<br>CETP/9p21/APOB: high    |
| Liquid L-carnitine (thermogenic)<br><b>Normal</b>        | <b>SLC22A4/5 (OCTN1/2)</b> : typical<br><b>FMO3</b> : typical                     | Typical transport; TMAO form | Typical OCTN2. Typical FMO3: oral carnitine generates TMAO by the microbiota; with genetic vascular priority, moderate and non-chronic dose at a high dose.                                                                  | Genetic plan (S19);<br>Koeth 2013<br>weak | SLC22A4/5 (OCTN1/2): high<br>FMO3: high |

**No known genetic relevance (12):**

|                           |                                    |                                  |
|---------------------------|------------------------------------|----------------------------------|
| lipoic + biotin (glucose) | raspberry ketones                  | white bean (phaseolamine)        |
| banaba (corosolic acid)   | artichoke extract + bergamot (LDL) | glucomannan + chromium (society) |
| bergamot (BPF)            | apple peel extract                 | Morosil (Orange-sanguinea)       |
| Caralluma fimbriata       | whiteberry leaf extract (DNJ)      | grifonia seed (5-HTP) (AADC)     |

### 3.20 Others (bee products, algae, formulas, detoxification, several) — 38 supplements

dose adjustment: 4 · Monitoring: 7 · normal: 5 · without known genetic relevance: 22

| Supplement                                                          | Genes (state)                                                                                                  | Predicted response                          | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                               | Source · evidence                                                        | Trust                                                                       |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| energy formula (B + caffeine)<br><b>Adjust dose</b>                 | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC) | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                | Genetic plan (S3, S6); EFSA UL B6 moderate                               | PDXK/ALPL/NBPF3 (B6): high<br>MTHFR: high<br>DHFR: moderate                 |
| homocysteine formula with folic acid<br><b>Adjust dose</b>          | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)                                               | prefer 5-MTHF (form)                        | MTHFR 677CT + 1298AC (activity ~50–60%) and DHFR rs408626 C/C (slow reduction of folic acid; may leave folic acid unmetabolized in circulation). Exchange by 5-MTHF or food folate; no product with folic acid (suggestion, not applied).                                           | Genetic plan (S3, b.3); MTHFR/DHFR literature moderate                   | MTHFR: high<br>DHFR: moderate                                               |
| methylation formula (B12 + folate + B6 + TMG)<br><b>Adjust dose</b> | <b>PDXK/ALPL/NBPF3 (B6)</b> : typical/high<br><b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC) | choose formula (low B6, without folic acid) | Many formulas bring 10–50 mg of B6 (above the limit with high activity PDXK) and folic acid (DHFR CC). Only if B6 is $\leq 5$ mg and folate is 5-MTHF; otherwise, avoid or start the dose (suggestion, not applied).                                                                | Genetic plan (S3, S6); EFSA UL B6 moderate                               | PDXK/ALPL/NBPF3 (B6): high<br>MTHFR: high<br>DHFR: moderate                 |
| lysine + vitamin C (Pauling protocol)<br><b>Adjust dose</b>         | <b>HFE</b> : H63D het<br><b>G6PD</b> : normal<br><b>SLC23A1/2 (vit. C)</b> : typical                           | drinking between meals (hour)               | HFE H63D het: vitamin C with food increases the absorption of non-heme iron (genetic target: not accumulating iron). Taking $\geq 2$ h away from meals. Normal G6PD (high dose and IV without genetic risk of hemolysis); typical SLC23A1/2 transporters (suggestion, not applied). | Genetic plan (S16); CPIC G6PD 2022 (analogy) moderate (iron interaction) | HFE: high<br>G6PD: high for the tested variants<br>SLC23A1/2 (vit. C): high |
| extract of fava (Vicia faba, L-dopa natural)<br><b>Monitoring</b>   | <b>G6PD</b> : normal<br><b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                            | Normal G6PD; L-dopa (COMT)                  | Normal G6PD (probable favism); the L-dopa of the fava is O-methylated by COMT Val/Val and rises homocysteine (compound MTHFR).                                                                                                                                                      | CPIC G6PD 2022; L-dopa literature moderate                               | G6PD: high for the tested variants<br>COMT: high<br>MTHFR: high             |
| anti-stress formula (adaptogen)<br><b>Monitoring</b>                | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                     | monitoring liver                            | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                             | LiverTox weak                                                            | PNPLA3/Pi*MZ (liver): moderate                                              |
| hepatic support formula (detox)<br><b>Monitoring</b>                | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                     | monitoring liver                            | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used.                                                                                                                                                             | LiverTox weak                                                            | PNPLA3/Pi*MZ (liver): moderate                                              |
| hangover formula (DHM, dihydromyricetin)<br><b>Monitoring</b>       | <b>COMT</b> : Val/Val (high)<br><b>MTHFR</b> : 677CT/1298AC                                                    | catechol: prefer low doses                  | Catechols/pyrogallols methylated by COMT (Val/Val, high activity): at a high dose they spend SAM and can rise to homocysteine (MTHFR compound, p92 score). Food doses without a problem.                                                                                            | Genetic plan (f.2) weak                                                  | COMT: high<br>MTHFR: high                                                   |
| hanging formula with NAC + cardo<br><b>Monitoring</b>               | <b>UGT1A1</b> : IM (*1/*28)<br><b>CYP2C9</b> : IM (*1/*3)<br><b>PNPLA3/Pi*MZ (liver)</b> : unfavourable        | without proven gain; inhibits UGT1A1/CYP2C9 | Silibinin is glucuronidated by UGT1A1 and inhibits it (bilirubin can rise in *28) and inhibits CYP2C9 (effect on losartan greater in *1/*1 than in *3; Han 2009). It does not protect the liver in a proven way in PNPLA3/Pi*MZ.                                                    | Han 2009; silymarin NASH trials weak                                     | UGT1A1: high<br>CYP2C9: high<br>PNPLA3/Pi*MZ (liver): moderate              |
| Sleep formula (mixture of herbs)<br><b>Monitoring</b>               | <b>PNPLA3/Pi*MZ (liver)</b> : unfavourable                                                                     | monitoring liver (valerian)                 | Cases of liver injury described (LiverTox); with the genetic liver (PNPLA3/Pi*MZ), use only for short periods, if used. (valerian/scullery in mixtures).                                                                                                                            | LiverTox weak                                                            | PNPLA3/Pi*MZ (liver): moderate                                              |

| Supplement                                                | Genes (state)                                                                 | Predicted response      | Recommendation (summary) and implication for Sérgio F                                                                                                                                                                                                                        | Source · evidence                                 | Trust                                    |
|-----------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------|
| nutritional yeast<br><b>Monitoring</b>                    | <b>MTHFR</b> : 677CT/1298AC<br><b>DHFR</b> : slow reduction (CC)              | check fortification     | Many brands are fortified with synthetic folic acid (DHFR CC): choosing not fortified or with 5-MTHF.                                                                                                                                                                        | Genetic plan (b.3) weak                           | MTHFR: high<br>DHFR: moderate            |
| eggshell calcium<br><b>Normal</b>                         | <b>VDR</b> : typical<br><b>GC (DBP)</b> : favourable<br><b>HFE</b> : H63D het | typical absorption      | VDR and GC typical/favourable. HFE H63D bonus: calcium at meals reduces iron absorption. Dose: complete the food up to ~1000 mg/day, no longer.                                                                                                                              | NIH ODS (calcium); genetic plan (S14) weak        | VDR: high<br>GC (DBP): high<br>HFE: high |
| immunity formula (zinc + C + D) ☺<br><b>Normal</b>        | <b>SLC30A8</b> : C/C (usual zinc)                                             | genetically useful zinc | SLC30A8 rs13266634 C/C + rs11558471 A/A: in patients with these glucose alleles, more zinc is associated with lower fasting glycemia (Kanoni 2011). Total 20-25 mg/day with 1.5-2 mg of copper (Zn:Cu ≈ 10-15:1; DAO depends on copper, AOC1 het) (suggestion, not applied). | Kanoni 2011 (CHARGE); genetic plan (S12) moderate | SLC30A8: high                            |
| formula for the eye (lutein + bileberry)<br><b>Normal</b> | <b>CFH/ARMS2</b> : CFH het<br><b>BCO1</b> : slightly ↓ conversion             | useful (retina)         | CFH Y402H het, ARMS2 G/G; lutein/zeaxanthin do not depend on BCO1. Keeping with fat.                                                                                                                                                                                         | Genetic plan (S24); AREDS2 weak                   | CFH/ARMS2: high<br>BCO1: high            |
| wheat germ<br><b>Normal</b>                               | <b>HLA-DQ2/DQ8</b> : negative                                                 | without celiac HLA      | Without DQ2.5 or DQ8: very low genetic risk of celiac disease; gluten/cereals tolerated.                                                                                                                                                                                     | Genetic plan (e.6) strong                         | HLA-DQ2/DQ8: high                        |
| Wheat germ oil<br><b>Normal</b>                           | <b>CYP4F2</b> : normal                                                        | normal metabolism       | CYP4F2 V433M C/C (normal): vitamin E ω-hydroxylation is typical. Nutritional dose; no genetic benefit for high doses.                                                                                                                                                        | CYP4F2 × tocopherol (Bardowell 2012) studies weak | CYP4F2: high                             |

### No known genetic relevance (22):

kombu/wakame algae powder (iodine)  
 sodium alginate  
 bentonite clay  
 activated carbon  
 kombucha tea powder (microbiota)  
 Oral EDTA (chelation)  
 fucus/goat (iodine)

Prostate formula  
 royal jelly  
 beer yeast  
 monolaurine  
 Sea moss (sea moss, Chondrus) (iodine)  
 bee pollen  
 spray of propolis + zinc (throat)  
 diatomaceous earth

TUDCA (Tauroursodeoxycholic acid) (biliary conjugation)  
 zeolite (clinoptilolite)  
 fulvic acid  
 malic acid  
 alkaline water/alkaline salts  
 isotonic seawater  
 essential oils in capsules (mixture)

## 4. Limitations

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- **Chip + imputation, no sequencing.** Only the variants that the chip measures or that the imputation reconstructs with  $DR2 \geq 0.8$  are seen. Rare and new variants are not seen; "normal" means "no variants tested".
- **Evidence of weaker nutrition than pharmacogenetics.** There are no CPIC guidelines for supplements. Most of the gene-nutrient interactions come from small, observational or mechanistic studies; the column "evidence" indicates the degree. Colors also reflect the context of stack (sirolimus, isotretinoin, telmisartan, tadalafil, glucose drugs), when genetics changes the importance of an interaction.
- **CNVs and non-visible repetitions:** GSTM1 (inferred) and GSTT1, CYP2D6 (\*5, duplications), CYP2A6 (deletions), AMY1, HP (haptoglobin), CNDP1 (CTG)<sub>n</sub> HSD11B2 not tested.
- **Haplotype phase:** MTHFR 677/1298 assumes compound heterozygosity (cis/trans not determined).
- **Only genetics, on purpose.** Doses of vitamin D, omega-3, zinc, iron, B12 and homocysteine should be tuned with measurements; therefore, where there is a decision based on measurements (D3 10 000 IU; omega-3 3.2–4 g; curcumin pause, luteolin, quercetin and apigenin while ALT is high; monthly pulse fisetin + quercetin only with normal ALT), this decision prevails (Note of coordination of the genetic plan; current protocol).
- **Product quality:** adulteration (stimulants, undeclared drugs), contamination (metals, pyrrolizidine alkaloids) and dose variation between brands are not predictable by genetics and explain much of the hepatotoxicity described.
- **List of supplements:** constructed from NIH ODS and NCCIH chips and curation of the usual commercial categories; it does not derive from sales data. It includes chemical forms, strains and combined formulas as distinct items.
- **It is not a diagnosis.** It is support to the decision to discuss with the doctor. All indications are suggestion (not applied).

## 5. Sources and methods

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- NIH Office of Dietary Supplements — list A-Z of supplements (read to 02-10-2026); NCCIH — Herbs at a Glance (67 inputs); LiverTox (NIH/NIDDK) — plant chapters and supplements (probability of hepatotoxicity).
- EFSA — upper limits of ingestion (B6 2023, selenium, zinc), opinions on green tea catechins (2018), monacolins (2018) and coumarin; EMA/HMPC — plant monographs (pyrrolizidine alkaloids, glycirizine, anthraquinones).
- Pharmacogenetics: CPIC/DPWG profile and guidelines of the 632 drug report (same genotype), used by analogy (e.g. tacrolimus–CYP3A5 for sirolimus, G6PD 2022, 2022 statins, warfarin 2017).
- Gene-nutrient interactions cited by line (e.g. Kanoni 2011 zinc–SLC30A8; Garaulet 2015 melatonin–MTNR1B; Cornelis 2006 caffeine–CYP1A2; Olthof 2001 chlorogenic acid–homocysteine; Chow 2010 resveratrol–CYP; Gurley 2008 hydrate; Guo 2012 berberina; Han 2009 silimarina–CYP2C9; Riche 2014 pterostilbene–LDL).
- Genotypes: chip (GRCh38) + statistical imputation (public reference panel of complete genomes to 30x), direct queries to the 02-10-2026 chip (AMPD1, CYP2E1, UGT1A6, EPHX1, MTR) and the genetic plan of 02-10-2026 (including the coordination note).